DEPARTMENT OF THE ARMY
OFFICE OF THE ASSISTANT SECRETARY OF THE ARMY
ACQUISITION LOGISTICS AND TECHNOLOGY
103 ARMY PENTAGON
WASHINGTON DC 20310-0103

APR 17 2025

The Honorable Tom Cole
Chairman
Committee on Appropriations
U.S. House of Representatives
Washington, DC 20515

Dear Mr. Chairman:

I am pleased to submit the enclosed Army Ammunition Plant Modernization Plan, as directed by Section 2834 of the National Defense Authorization Act for Fiscal Year 2022 (Public Law 117-81), and as amended by Section 2888 of the National Defense Authorization Act for Fiscal Year 2024 (Public Law 118-31).

The enclosed Modernization Plan provides an investment strategy focused on achieving multiple modernization strategy objectives and an associated end state. This includes increased manufacturing safety and readiness to meet current and future requirements, isolating energetic mass from people, and ensuring graceful degradation. The report also addresses resilient operations to improve flexibility, maintainability, and sustainability, as well as reducing the cost of operations and secure supply chains.

Thank you for your continued support of Army programs and our Soldiers.

Sincerely,

Patrick H. Mason
Senior Official Performing the Duties of the
Assistant Secretary of the Army
(Acquisition, Logistics and Technology)

Enclosure

DEPARTMENT OF THE ARMY
OFFICE OF THE ASSISTANT SECRETARY OF THE ARMY
ACQUISITION LOGISTICS AND TECHNOLOGY
103 ARMY PENTAGON
WASHINGTON DC 20310-0103

APR 17 2025

The Honorable Rosa L. DeLauro
Ranking Member
Committee on Appropriations
U.S. House of Representatives
Washington, DC 20515

Dear Representative DeLauro:

I am pleased to submit the enclosed Army Ammunition Plant Modernization Plan, as directed by Section 2834 of the National Defense Authorization Act for Fiscal Year 2022 (Public Law 117-81), and as amended by Section 2888 of the National Defense Authorization Act for Fiscal Year 2024 (Public Law 118-31).

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103 ARMY PENTAGON
WASHINGTON DC 20310-0103

APR 17 2025

The Honorable Ken Calvert
Chairman
Subcommittee on Defense
Committee on Appropriations
U.S. House of Representatives
Washington, DC 20515

Dear Mr. Chairman:

I am pleased to submit the enclosed Army Ammunition Plant Modernization Plan, as directed by Section 2834 of the National Defense Authorization Act for Fiscal Year 2022 (Public Law 117-81), and as amended by Section 2888 of the National Defense Authorization Act for Fiscal Year 2024 (Public Law 118-31).

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OFFICE OF THE ASSISTANT SECRETARY OF THE ARMY
ACQUISITION LOGISTICS AND TECHNOLOGY
103 ARMY PENTAGON
WASHINGTON DC 20310-0103

APR 17 2025

The Honorable Betty McCollum
Ranking Member
Subcommittee on Defense
Committee on Appropriations
U.S. House of Representatives
Washington, DC 20515

Dear Representative McCollum:

I am pleased to submit the enclosed Army Ammunition Plant Modernization Plan, as directed by Section 2834 of the National Defense Authorization Act for Fiscal Year 2022 (Public Law 117-81), and as amended by Section 2888 of the National Defense Authorization Act for Fiscal Year 2024 (Public Law 118-31).

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OFFICE OF THE ASSISTANT SECRETARY OF THE ARMY
ACQUISITION LOGISTICS AND TECHNOLOGY
103 ARMY PENTAGON
WASHINGTON DC 20310-0103

APR 17 2025

The Honorable Mike D. Rogers
Chairman
Committee on Armed Services
U.S. House of Representatives
Washington, DC 20515

Dear Mr. Chairman:

I am pleased to submit the enclosed Army Ammunition Plant Modernization Plan, as directed by Section 2834 of the National Defense Authorization Act for Fiscal Year 2022 (Public Law 117-81), and as amended by Section 2888 of the National Defense Authorization Act for Fiscal Year 2024 (Public Law 118-31).

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Assistant Secretary of the Army
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OFFICE OF THE ASSISTANT SECRETARY OF THE ARMY
ACQUISITION LOGISTICS AND TECHNOLOGY
103 ARMY PENTAGON
WASHINGTON DC 20310-0103

APR 17 2025

The Honorable Adam Smith
Ranking Member
Committee on Armed Services
U.S. House of Representatives
Washington, DC 20515

Dear Representative Smith:

I am pleased to submit the enclosed Army Ammunition Plant Modernization Plan, as directed by Section 2834 of the National Defense Authorization Act for Fiscal Year 2022 (Public Law 117-81), and as amended by Section 2888 of the National Defense Authorization Act for Fiscal Year 2024 (Public Law 118-31).

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Sincerely,

Patrick H. Mason
Senior Official Performing the Duties of the
Assistant Secretary of the Army
(Acquisition, Logistics and Technology)

Enclosure

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OFFICE OF THE ASSISTANT SECRETARY OF THE ARMY
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103 ARMY PENTAGON
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APR 17 2025

The Honorable Susan Collins
Chairwoman
Committee on Appropriations
United States Senate
Washington, DC 20510

Dear Madam Chair:

I am pleased to submit the enclosed Army Ammunition Plant Modernization Plan, as directed by Section 2834 of the National Defense Authorization Act for Fiscal Year 2022 (Public Law 117-81), and as amended by Section 2888 of the National Defense Authorization Act for Fiscal Year 2024 (Public Law 118-31).

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Sincerely,

Patrick H. Mason
Senior Official Performing the Duties of the
Assistant Secretary of the Army
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OFFICE OF THE ASSISTANT SECRETARY OF THE ARMY
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103 ARMY PENTAGON
WASHINGTON DC 20310-0103

APR 17 2025

The Honorable Patty Murray
Vice Chairwoman
Committee on Appropriations
United States Senate
Washington, DC 20510

Dear Senator Murray:

I am pleased to submit the enclosed Army Ammunition Plant Modernization Plan, as directed by Section 2834 of the National Defense Authorization Act for Fiscal Year 2022 (Public Law 117-81), and as amended by Section 2888 of the National Defense Authorization Act for Fiscal Year 2024 (Public Law 118-31).

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Senior Official Performing the Duties of the
Assistant Secretary of the Army
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APR 17 2025

The Honorable Mitch McConnell
Chairman
Subcommittee on Defense
Committee on Appropriations
United States Senate
Washington, DC 20510

Dear Mr. Chairman:

I am pleased to submit the enclosed Army Ammunition Plant Modernization Plan, as directed by Section 2834 of the National Defense Authorization Act for Fiscal Year 2022 (Public Law 117-81), and as amended by Section 2888 of the National Defense Authorization Act for Fiscal Year 2024 (Public Law 118-31).

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Senior Official Performing the Duties of the
Assistant Secretary of the Army
(Acquisition, Logistics and Technology)

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OFFICE OF THE ASSISTANT SECRETARY OF THE ARMY
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WASHINGTON DC 20310-0103

APR 17 2025

The Honorable Chris Coons
Vice Chairman
Subcommittee on Defense
Committee on Appropriations
United States Senate
Washington, DC 20510

Dear Senator Coons:

I am pleased to submit the enclosed Army Ammunition Plant Modernization Plan, as directed by Section 2834 of the National Defense Authorization Act for Fiscal Year 2022 (Public Law 117-81), and as amended by Section 2888 of the National Defense Authorization Act for Fiscal Year 2024 (Public Law 118-31).

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APR 17 2025

The Honorable Roger F. Wicker
Chairman
Committee on Armed Services
United States Senate
Washington, DC 20510

Dear Mr. Chairman:

I am pleased to submit the enclosed Army Ammunition Plant Modernization Plan, as directed by Section 2834 of the National Defense Authorization Act for Fiscal Year 2022 (Public Law 117-81), and as amended by Section 2888 of the National Defense Authorization Act for Fiscal Year 2024 (Public Law 118-31).

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103 ARMY PENTAGON
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APR 17 2025

The Honorable Jack Reed
Ranking Member
Committee on Armed Services
United States Senate
Washington, DC 20510

Dear Senator Reed:

I am pleased to submit the enclosed Army Ammunition Plant Modernization Plan, as directed by Section 2834 of the National Defense Authorization Act for Fiscal Year 2022 (Public Law 117-81), and as amended by Section 2888 of the National Defense Authorization Act for Fiscal Year 2024 (Public Law 118-31).

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Patrick H. Mason
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(Acquisition, Logistics and Technology)

Enclosure

HEADQUARTERS, DEPARTMENT OF THE ARMY

REPORT TO CONGRESS

Army Ammunition Plant
Modernization Plan

April 2025

The estimated cost of this report for the Department of Defense (DOD) is approximately $185,120.00.
This includes $1,000.00 in expenses and $184,120.00 in DOD labor.

Table of Contents

LIST OF FIGURES

LIST OF TABLES

Congressional Language

This report is in response to language found in Section 2834 of the National Defense Authorization Act (NDAA) for Fiscal Year (FY) 2022 (Public Law 117-81), Section 2888 of the NDAA for FY 2024 (Public Law 118-31), and the FY24 Senate Appropriations Committee-Defense Report 118-81 which read:

<u>FISCAL YEAR 2022 NDAA, SECTION 2834:</u>

"SEC. 2834. MASTER PLANS AND INVESTMENT STRATEGIES FOR ARMY AMMUNITION PLANTS GUIDING FUTURE INFRASTRUCTURE, FACILITY, AND PRODUCTION EQUIPMENT IMPROVEMENTS.

(a) SUBMISSION OF MASTER PLANS AND INVESTMENT STRATEGIES—Not later than March 31, 2022, the Secretary of the Army shall submit to the congressional defense committees a report containing the following:

(1) The master plan for each of the ammunition organic industrial base production facilities under the jurisdiction of the Secretary of the Army (in this section referred to as an ''ammunition production facility'') that was developed to guide planning and budgeting for future infrastructure construction, facility improvements, and production equipment needs at the ammunition production facility.

(2) An investment strategy to address the facility, major equipment, and infrastructure requirements at each ammunition production facility in order to support the readiness and material availability goals of current and future weapons systems of the Department of Defense.

(b) ELEMENTS OF MASTER PLAN — To satisfy the requirements of subsection (a)(1), the master plan for an ammunition production facility must incorporate the results of a review of industrial processes, logistics streams, and workload distribution required to support production objectives and the facility requirements to support optimized processes and include the following specific elements:

(1) A description of all infrastructure construction and facility improvements planned or being considered for the ammunition production facility and production equipment planned or being considered for installation, modernization, or replacement.

(2) An explanation of how the master plan for the ammunition production facility will promote efficient, effective, resilient, secure, and cost-effective production of ammunition and ammunition components for the Armed Forces.

(3) A description of how development of the master plan for the ammunition production facility included input from the contractor operating the ammunition production facility and how implementation of that master plan will be coordinated with the contractor.

(4) A review of current and projected workload requirements for the manufacturing of energetic materials, including propellants, explosives, pyrotechnics, and the ingredients

for propellants, explosives, and pyrotechnics, to assess efficiencies in the use of existing facilities, including consideration of new weapons characteristics and requirements, obsolescence of facilities, siting of facilities and equipment, and various constrained process flows.

(5) An analysis of life-cycle costs to repair and modernize existing mission-essential facilities versus the cost to consolidate functions into modern, right-sized facilities at each location to meet current and programmed future mission requirements.

(6) A review of the progress made in prioritizing and funding projects that facilitate process efficiencies and consolidate and contribute to availability cost and schedule reductions.

(7) An accounting of the backlog of restoration and modernization projects at the ammunition production facility.

(c) ELEMENTS OF INVESTMENT STRATEGY — To satisfy the requirements of subsection (a)(2), the investment strategy for an ammunition production facility must include the following specific elements:

(1) A description of the funding sources for such infrastructure construction, facility improvements, and production equipment, including authorized military construction projects, appropriations available for operation and maintenance, and appropriations available for procurement of Army ammunition in order to support the readiness and material availability goals of current and future weapons systems of the Department of Defense.

(2) A timeline to complete the investment strategy.

(3) A list of projects and a brief scope of work for each such project.

(4) Cost estimates necessary to complete projects for mission essential facilities.

(d) ANNUAL UPDATES—Not later than March 31, 2023, and each March 31 thereafter through March 31, 2026, the Secretary of the Army shall submit to the congressional defense committees a report containing the following:

(1) A description of any revisions made during the previous year to master plans and investment strategies submitted under subsection (a).

(2) A description of any revisions to be made or being considered to the master plans and investment strategies.

(3) An explanation of the reasons for each revision, whether made, to be made, or being considered.

(4) A description of the progress made in improving infrastructure, facility, and production equipment at each ammunition production facility consistent with the master plans and investment strategies.

(e) DELEGATION AUTHORITY — The Secretary of the Army shall carry out this section acting through the Assistant Secretary of the Army for Acquisition, Logistics, and Technology.

FY24 NDAA, Section 2888:

"SEC. 2888. EXTENSION AND MODIFICATION OF ANNUAL UPDATES TO MASTER PLANS AND INVESTMENT STRATEGIES FOR ARMY AMMUNITION PLANTS.

Section 2834(d) of the Military Construction Authorization Act for Fiscal Year 2022 (division B of Public Law 117-81; 135 Stat. 2201) is amended--

(1) in the matter preceding paragraph (1), by striking "March 31, 2026" and inserting "March 2030" and

(2) by adding at the end the following new paragraph: "(5) A description of any changes to a master plan for an ammunition production facility made in response to global events, including pandemics and armed conflicts."

FY24 SENATE APPROPRIATIONS COMMITTE-DEFENSE REPORT 118-81, Page 108

"The Committee directs the Secretary of the Army to submit an updated Army Ammunition Plant Modernization Plan with the submission of the fiscal year 2025 President's budget request that includes production capacity and scalability along with the current six objectives. Given the scale and severity of infrastructure needs across the facilities, the Committee directs the Secretary to include an analysis of construction or contracting for new, modern facilities as an additional alternative. This analysis shall include cost estimates and proposed schedules."

EXECUTIVE SUMMARY

The Army has developed a comprehensive Army Ammunition Plant (AAP) Modernization Plan. The AAP Modernization Plan addresses the request in the House Armed Services Committee (HASC) Fiscal Year 2022 National Defense Authorization Act (NDAA) report, Section 2834, that directs the Secretary of the Army, through the Assistant Secretary of the Army for Acquisition, Logistics, and Technology (ASA(ALT)), to submit an AAP Modernization Plan to the congressional committees no later than 31 March 2022, with annual updates each subsequent year for five years. Annual update requirements were amended by Section 2888 of the FY 2024 NDAA which extends the annual submission requirement through 31 March 2030. In addition, this update addresses requirements in the FY 2024 Senate Appropriations Committee-Defense Report 118-81, Page 108. The AAP Modernization Plan was submitted in March 2022, as required, and each year since. This document provides the FY 2025 update, incorporating recent changes in project priorities at our facilities. It also considers the ongoing effects of global events, including: the lasting impact of the COVID-19 pandemic; support for international partners; global supply chain disruptions and efforts to onshore production, and rising material costs and inflation. These factors have influenced our project priorities and are reflected in this updated document.

The purpose of the AAP Modernization Plan is to provide a strategy to modernize facility infrastructure and production capabilities by capitalizing on state-of-the-art manufacturing equipment and technologies while maintaining production continuity. The AAP Modernization Plan provides an investment strategy focused on achieving the following objectives and associated end state:

1. Increase manufacturing safety and readiness to meet current and future requirements.
2. Isolate energetic mass from people.
3. Ensure graceful degradation and resilient operations.
4. Improve flexibility, maintainability, and sustainability.
5. Reduce cost of operations.
6. Secure supply chains.

Executing this AAP Modernization Plan will result in improved safety, resiliency, compliance, and operational effectiveness. The AAP Modernization Plan addresses increasing automation to minimize human exposure to hazardous environments and the application of digital manufacturing methods. Modernization efforts discussed within this report address seven Government Owned, Contractor Operated (GOCO) ammunition production facilities, as identified in Table ES-1 below.

TABLE ES-1. GOCO AMMUNITION PRODUCTION FACILITIES

GOCO Facility	Operating Contractor	Location
Holston Army Ammunition Plant (HSAAP)	BAE Systems – Ordnance Systems Inc. (BAE OSI)	Kingsport, TN
Iowa Army Ammunition Plant (IAAAP)	American Ordnance (AO)	Middletown, IA
Lake City Army Ammunition Plant (LCAAP)	Olin Winchester	Independence, MO
Milan Army Ammunition Plant (MLAAP)	AO*	Milan, TN
Quad City Cartridge Case Facility (QCCCF)	U.S. Army Combat Capabilities Development Command (DEVCOM) maintained; no current Facility Use Contract**	Rock Island, IL
Radford Army Ammunition Plant (RFAAP)	BAE OSI**	Radford, VA
Scranton Army Ammunition Plant (SCAAP)	General Dynamics Ordnance and Tactical Systems (GD-OTS)	Scranton, PA

* No active production at MLAAP.

** A solicitation for a competitive operating contract is planned in FY 2025.

The GOCO facilities serve a vital role in protecting our nation's security—keeping them modernized is necessary for enabling continued ammunition production. Sustaining modernization and maintaining warm ammunition production lines at GOCOs is also necessary when requirements reduce. Procurement of Ammunition, Army (PAA) funding has enabled the Army to execute modernization improvements and recapitalization projects at the GOCO AAPs.

The modernization planning process includes characterizing each GOCO AAP's operating conditions and useful life of core processes, infrastructure, and support systems. This characterization data is used to develop “Heat Maps” that provide a visual representation of these assessments and assist in requirements prioritization by indicating the required time frame for modernization to avoid production disruptions or catastrophic failures. This assessment helps prioritize and plan the required design phases that support modernization project execution and budget formulation. Modernization projects are identified and prioritized based on scoring calculated from weighted criteria and considerations such as criticality of the requirement (e.g., safety and/or environmental compliance), a facility’s ability and capacity to execute the work, and the time remaining on a facility contract.

Project ranking is a critical process because the total annual requirements identified for all the facilities exceed the projected budget availability. Projects that fall within the Total Obligational Authority (TOA) are included in the budget submission; projects that fall outside the TOA are planned for the Extended Planning Period (EPP). Table ES-2 provides a summary of the PAA Provision of Industrial Facilities (PIF) funding spent from FY 2003 through FY 2024, requirements planned for FYs 2025–2031 and projected requirements for FY 2032 and beyond.

TABLE ES-2: PRODUCTION FACILITY PAA PIF FUNDING REQUIREMENTS SUMMARY

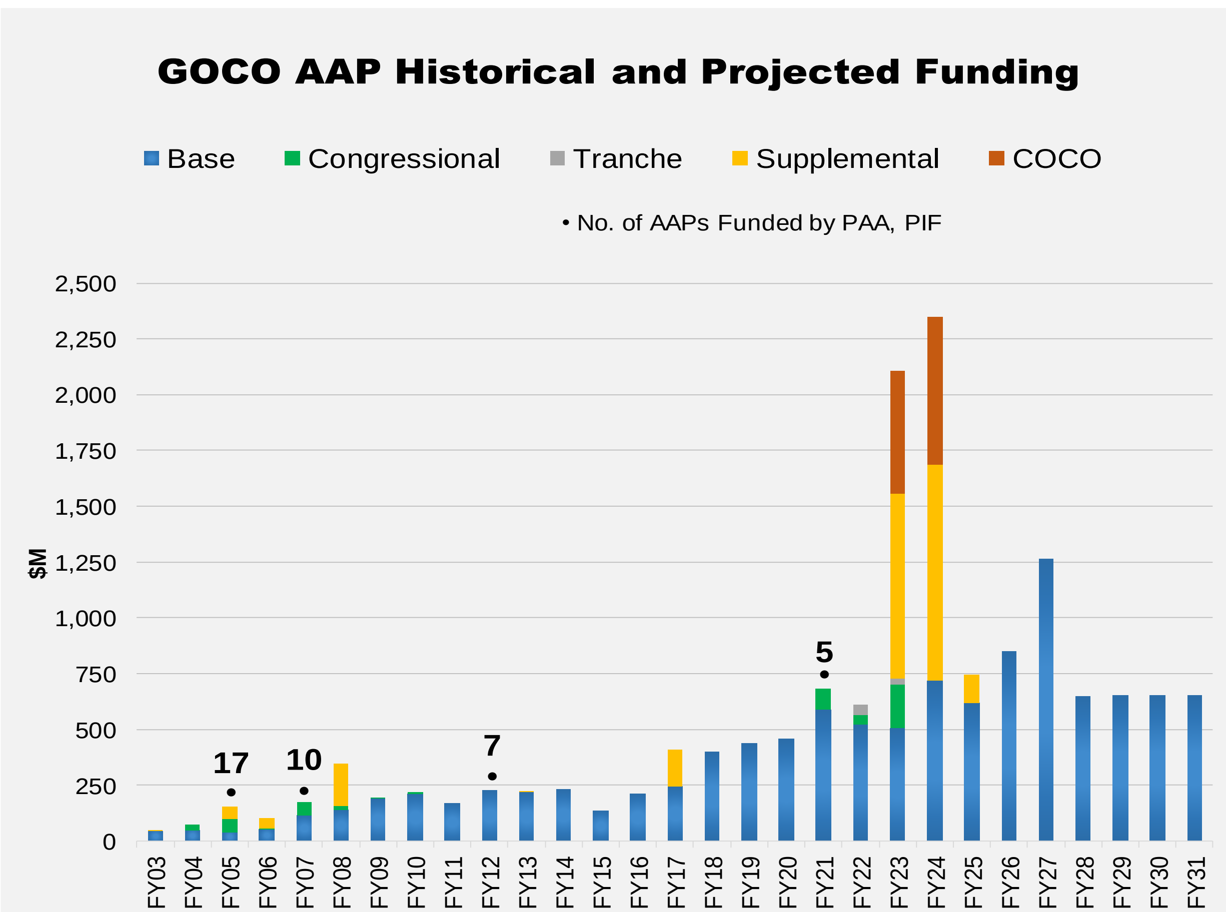
GOCO Ammunition Facility	Core Processes	FYs 2003–2024 ($M)	FYs 2003–2024 COCO ($M)	FYs 2025–2031 ($M)	FYs 2032+ ($M)
HSAAP (TN)	Explosives–HMX, RDX, IMX, TATB	3,007.636	0.000	946.582	897.300
IAAAP (IA)	Load, Assemble, and Pack (LAP) – Mortars, 40mm Cartridges, Tank/Artillery, C-4 Extrusion, Warheads	1,084.804	0.000	657.429	1,070.58
LCAAP (MO)	Small and Medium Caliber	1,594.127	0.000	1,262.401	2,678.14
MLAAP (TN)	Inactive	21.844	0.000	0.000	0.000
QCCCF (IL)	Large Caliber Cartridge Cases	2.205	0.000	11.250	0.000
RFAAP (VA)	Propellant Manufacturing (Rocket, Artillery, Tank, Med Caliber, Nitrocellulose (NC) for all Propellants)	1,842.711	0.000	1,734.573	2,519.18
SCAAP (PA)	Large Caliber Metal Parts–Artillery/Mortars	680.963	0.000	270.176	154.863
All Facilities	Maintenance of Inactive Facilities (MIF), Layaway of Industrial Facilities (LIF), Planning and Design (P&D), Engineering Support, QWE, Emergent Maintenance	715.162	1,212.856	712.645	0.000
	TOTAL	**8,949.452**	**1,212.856**	**8,949.452**	**7,320.06**

Several strategic construction projects that are not currently resourced through FY 2031 could potentially be moved to the left if additional resourcing becomes available.

TABLE ES-3: EXAMPLES OF OUT-YEAR STRATEGIC MODERNIZATION PROJECTS

GOCO Ammunition Facility	Strategic Project Investment by Facility	Rough Order of Magnitude Cost ($M)
HSAAP (TN)	Flexible Explosive Production Facility & Infrastructure CL-20	435-535
HSAAP (TN)	ANSOL Thermal Destruction: Construction	150-250
IAAAP (IA)	Direct Fire Munitions Facility (Tank Line)	400-600
LCAAP (MO)	Modernized Metal Parts Facility	400-500
LCAAP (MO)	Consolidated Chemistry & Energetics Process Laboratory	200-300
RFAAP (VA)	Solventless Propellant Facility: Design & Construction	700-800
RFAAP (VA)	NG-3 (Nitroglycerin) Facility: Construction	600-700
RFAAP (VA)	Utilities Modernization & Permanent Energy Center	500-600

Figure ES-1 graphically depicts historical PAA funding and projected requirements for GOCO AAP modernization. It also shows years where additional funding was provided by Congress in the NDAA or with supplemental appropriations. As indicated in Figure ES-1, significant additional funding was provided in FY 2023 to increase artillery production capacity. FY 2024 Supplemental funding was also used to increase production capacity. FY 2025 Supplemental funding was requested to restore operations from Hurricane Helene damage at RFAAP.



*Supplemental funding in FY 2024 was used to increase capacity and was not used to modernize the GOCOs.

FIGURE ES-1: GOCO AAP HISTORICAL AND PROJECTED FUNDING–PAA, PIF

Figure ES-1 does include Contractor Operated Contractor Owned (COCO) facilities. Although the Army's modernization efforts to date have resulted in an improved industrial base, critical infrastructure and manufacturing process upgrades are still required to support the Warfighter's ammunition requirements, address continued facility deterioration, and ensure a safe working environment. Facilities that can meet ammunition production requirements while ensuring workforce safety, especially in inherently dangerous explosive operations, are critical. Recent years have seen a significant amount of modernization funding aimed at recapitalizing the GOCOs. Many buildings and production lines within our GOCO facilities are facing significant challenges. Key parts are becoming obsolete, unplanned downtime is increasing, and the facilities themselves have exceeded their lifespan. As a result, maintenance and upgrades are no longer sufficient to address these issues. Therefore, constructing new facilities is essential to ensure the continued production of ammunition. It is anticipated that funding levels required to maintain the GOCOs will remain at higher levels based on production requirements and the amount of recapitalization necessary to modernize. In addition, applying transformational solutions such as Digital Engineering (DE), automation, modeling and simulation, and intelligent inspection systems, as well as addressing regulatory and Department of Defense safety and environmental requirements will drive higher modernization costs.

This report describes key modernization objectives and the Army's approach to modernization planning, a summary of the project identification and prioritization process, and a summary of the requirements for modernization activities for each ammunition production facility. Details are provided for AAP projects planned in FYs 2025–2031, and this report also presents strategic investments planned in FY 2032 and beyond. The modernization planning process assumes approximately $700M per year of PAA funding; however, several projects could be moved to the left if additional resourcing becomes available.

1.0 INTRODUCTION

The Army has developed a comprehensive Army Ammunition Plant (AAP) Modernization Plan. The AAP Modernization Plan addresses the request in the House Armed Services Committee Fiscal Year (FY) 2022 National Defense Authorization Act (NDAA) report, Section 2834, that directs the Secretary of the Army, through the Assistant Secretary of the Army for Acquisition, Logistics, and Technology (ASA(ALT)), to submit an AAP master plan and investment strategy to the congressional committees no later than 31 March 2022, with annual updates each subsequent year for five years. Annual update requirements were amended by Section 2888 of the FY 2024 NDAA which extends the annual submission requirement through 31 March 2030. In addition, this update addresses requirements in the FY 2024 Senate Appropriations Committee-Defense Report 118-81, Page 108. The purpose of the AAP Modernization Plan is to outline the strategy for modernizing facility infrastructure and production capabilities which capitalizes on state-of-the-art manufacturing equipment and technologies while maintaining production continuity. The AAP Modernization Plan provides an investment strategy focused on achieving the following objectives and associated end state:

1. Increase manufacturing safety and readiness to meet current and future requirements.
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3. Ensure graceful degradation and resilient operations.
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5. Reduce cost of operations.
6. Secure supply chains.

The AAP Modernization Plan is updated annually, and the investment strategy is continually reassessed to address shifting priorities, project cost variances, program funding adjustments, and national defense initiatives. Executing this AAP Modernization Plan will result in improved safety, resiliency, environmental compliance, and operational effectiveness. The AAP Modernization Plan addresses increasing automation to minimize human exposure to hazardous environments and the application of Digital Engineering (DE). Further use of these emerging technologies, tools, and practices will reduce risk during facility modernization and commissioning, improve production quality, and reduce operational costs.

1.1 2025 AAP MODERNIZATION PLAN UPDATES

Several changes were made to the investment priorities presented in the 2024 AAP Modernization Plan. These updates reflect project priority changes at the facilities, as well as impacts from global events; support to foreign partners and allies; global supply chain issues with materials from foreign suppliers; the strategic desire to onshore production and domestically source raw materials; and increasing material costs and rising inflation. These rapidly changing variables are impacting project costs and schedules, and resulting in shifts to planned funding years. These variables also impact which projects are prioritized among the facilities and when they are planned for funding. The revised project listings are included in Appendices A and B.

1.2 BACKGROUND

As identified in Army Regulation (AR) 700-90, Army Industrial Base Process, the Army's industrial base vision is a complementary and synergistic industrial base (both commercial and government-owned) that has the capability to satisfy the Joint Warfighter's ammunition life-cycle requirements in peacetime, wartime, and during national emergencies.

Ammunition must remain available, reliable, and affordable. To ensure this, ammunition component and end-item production capabilities must also remain available, reliable, and affordable. Procurement of Ammunition, Army (PAA) funding through the Provision of Industrial Facilities (PIF) Budget Line-Item Number (BLIN) has enabled the Army to execute modernization projects that ensure Government Owned, Contractor Operated (GOCO) AAPs avoid catastrophic failures and continue to produce quality ammunition. Modernization efforts are discussed within this report for the following GOCO AAPs (see Table 1).

TABLE 1. GOCO AAPS

GOCO Facility	Core Capabilities	Operating Contractor	Location
Holston Army Ammunition Plant (HSAAP)	Explosives–HMX, RDX, IMX, TATB	BAE Systems–Ordnance Systems, Inc. (BAE OSI)	Kingsport, TN
Iowa Army Ammunition Plant (IAAAP)	Load, Assemble, and Pack (LAP) – Mortars, 40mm Cartridges, Tank/ Artillery, C-4 Extrusion, Warheads	American Ordnance (AO)	Middletown, IA
Lake City Army Ammunition Plant (LCAAP)	Small and Medium Caliber	Olin Winchester	Independence, MO
Milan Army Ammunition Plant (MLAAP)	Inactive	AO	Milan, TN
Quad City Cartridge Case Facility (QCCCF)	Large Caliber Cartridge Cases	U.S. Army Combat Capabilities Development Command (DEVCOM) maintained; no current Facility Use Contract*	Rock Island, IL
Radford Army Ammunition Plant (RFAAP)	Propellant Manufacturing (Rocket, Artillery, Tank, Med Caliber, Nitrocellulose (NC) for all Propellants)	BAE OSI*	Radford, VA
Scranton Army Ammunition Plant (SCAAP)	Large Caliber Metal Parts–Artillery	General Dynamics Ordnance and Tactical Systems (GD-OTS)	Scranton, PA

*A solicitation for a competitive operating contract is planned in FY 2025.

1.3 INDUSTRIAL BASE MANAGEMENT – SINGLE MANAGER FOR CONVENTIONAL AMMUNITION (SMCA)

The Secretary of the Army is designated as the SMCA and assigned authority for Public Law 105-261, Section 806, Procurement of Conventional Ammunition, of the Strom Thurmond NDAA for FY 1999 (herein referred to as Section 806). SMCA authorities are delegated to the Army's Joint Program Executive Office Armaments and Ammunition (JPEO A&A) located at Picatinny Arsenal, NJ. The Army is the primary entity responsible for acquiring ammunition for the Armed Services and has the authority to restrict ammunition procurements to sources within the National Technology and Industrial Base (NTIB). The SMCA's goal is to achieve the highest possible degree of effectiveness and efficiency in the Department of Defense (DOD) operations that are required to obtain top-quality ammunition for United States Armed Forces. By utilizing the Army as the SMCA rather than having the Military Services separately manage their own ammunition, the DOD achieves economies of scale and other managerial efficiencies.

Policies and responsibilities for the SMCA are identified in DOD Instruction 5160.68. The SMCA's responsibilities cover the entire range of the ammunition life cycle, from providing support for Research, Development, Test, and Evaluation (RDT&E) activities to demilitarization (demil) of conventional ammunition. DOD Instruction 5160.68 defines SMCA and Service responsibilities across the following functional areas: RDT&E; Production Base; Acquisition; Supply; Maintenance; Demilitarization and Disposal; Quality Assurance; Technical Data and Configuration Management and Control; Transportation and Handling; Safety; Security; Financial Management and Planning, Programming, Budgeting and Execution; Implementing Regulations and Assessment; Personnel and Unit Training; and Security Assistance. DOD Instruction 5160.68 requires the SMCA to annually identify and prioritize production base deficiencies and then formulate corrective actions to mitigate those deficiencies. The SMCA continually assesses customer satisfaction through an annual customer survey, which is subsequently documented in a published report.

2.0 MODERNIZATION CHALLENGES, OBJECTIVES, AND PLANNING APPROACH

This AAP Modernization Plan is structured to transform the AAPs by utilizing the latest manufacturing technologies and processes to achieve the Army's modernization objectives. It also seeks to overcome a variety of challenges facing these GOCO facilities, such as regulatory compliance, cybersecurity requirements, rising project costs, raw material and equipment sourcing issues, and international conflicts.

2.1 MODERNIZATION CHALLENGES

The GOCO AAPs face several challenges that impact their modernization investment requirements. The useful life of production lines, buildings, and infrastructure presents a planning challenge. Likewise, modernizing while maintaining production requires complex schedule and logistical considerations. One goal of modernization is to prevent catastrophic production failures or significant production disruptions. This includes inserting automation and other emerging technologies for increased efficiency and to reduce operations-related risks. Maintaining environmental compliance, anticipating future regulations, and planning a path for continued regulatory compliance is critical to ensuring production continuity. Establishing cybersecurity strategies and ensuring that GOCO AAP cyber-vulnerabilities are mitigated also pose a new evolving challenge. The proliferation of drones by commercial industry, foreign actors and the public is also of concern across Army facilities. Drones can introduce a physical vulnerability as well as a cyber security threat, assessing these impacts to Army facilities will require additional funding to effectively mitigate potential threats. Accurately estimating the construction cost of unique, high volume manufacturing capabilities is difficult as inflationary pressures, increasing regulatory compliance, defense-unique safety and facility siting requirements, labor shortages, stressed supply chains, and requirements for Continental United States (CONUS) sourced raw materials are significantly driving up construction and material costs. In addition, Government Explosive Safety Standards are not comparable with commercial industry and current Government requirements and standards result in significant increased costs and schedule for projects at Government facilities.

2.1.1 Dangerous and High-Risk Operations

Many ammunition production processes are inherently dangerous due to the use of hazardous chemicals and operations involving energetics. The highest modernization priority is improving safety, especially for the most dangerous operations involving human exposure. Another facet of high-risk operations is ergonomics. AAP production operations involve a lot of manual material handling that requires heavy lifting and repetitive motions. Funds have been aligned to requirements identified as dangerous or high-risk, but a solution is not always readily available. In these situations, analysis and design will be conducted so a path forward can be developed. While pursuit of these solutions will reduce existing operational hazards, no modernization effort will be able to remove all risks associated with the inherently dangerous nature of ammunition production operations.

2.1.2 Automation and Emerging Technology Solutions

This section discusses some elements of transformational modernization that incorporate automation and emerging technology solutions. These types of technologies not only address safety issues and assist with removing personnel from dangerous and high-risk operations but also improve production and operational efficiencies. Although many of these technologies are widely used by commercial chemical and manufacturing industries, actual application is highly process-specific and necessitates specific technology and equipment development. Operations involving energetics introduce another layer of complexity to the design requirements.

Manufacturing Automation

Manufacturing system automation needs data integration and digital connectivity between equipment and operations, thus moving away from traditional islands of automation approaches. To aid this system-level manufacturing automation strategy, the Plan requires digitization of plant processes and operations. Digitized manufacturing processes and systems can rapidly adopt related technologies like Artificial Intelligence (AI) and Machine Learning (ML) in the future. The AAPs can utilize AI to integrate and analyze data to produce meaningful insights into manufacturing processes that can aid decision-making across the AAP and production equipment. ML sorts and analyzes large data sets to identify patterns and trends. ML then uses these patterns and trends to build models that accurately predict scenarios, such as equipment maintenance intervals based on current production outcomes, as well as product quality degradation in connection with those maintenance intervals. AI-driven maintenance prediction will significantly reduce equipment downtime, allowing the Operating Contractor to proactively repair equipment and shift manufacturing flow to limit production disruptions. ML algorithms, coupled with AI-based root cause analysis for anomaly detection within the munitions, will be a key tool to optimize each product’s manufacturing process and provide data for incorporation into future munitions designs.

Modern, automated processes must be introduced in a specific manner to ensure success. First, the new process is inserted at a GOCO facility in parallel with current production to mitigate any loss of production and develop contractor comfort with the new process. Once the new processes are in place and vetted, the legacy system is removed, leaving a fully integrated modern system, which reduces costs and risks.

Flexible and Intelligent Inspection and Manufacturing In-Process Monitoring

Monitoring the manufacturing flow will be vital to ensure traceability, repeatability, and product/process quality assurances. Manufacturing process monitoring has historically used off-line technologies or methods that pause operations for inspection and monitoring. Even though these methods may be automated (e.g., robots that provide off-line inspections), these methods do not provide the required continuity of flow and interoperability between manufacturing equipment, stages, and processes. To ensure flexibility in an automation strategy, the Modernization Plan implements intelligent manufacturing in-process monitoring and inspection that will be integrated with the manufacturing operations. Such measurement includes product parameters specified by the Technical Data Package (TDP) and process parameters that capture how the product is made, and machines are operating. This creates a solidified foundation of the production processes that enables earlier detection of anomalies and troubleshooting in the event of product failures.

Robotics and Mobile Transportation Platform Systems

Stationary robots combined with a mobile transportation platform can replace most human jobs that are dangerous, repetitive, and labor intensive. These technologies can be used for jobs like forklift operations and material handling in hazardous environments.

A major challenge of using robots and mobile platforms is weighing the cost and benefits of such a large investment and its ability to be used in energetic environments.

Modeling and Simulation

Modeling and simulation can be used to evaluate current production system efficiencies and aid in the planning and implementation of new processes. As manufacturing operations become more digitized and networked, modeling and simulation technology will benefit almost every aspect of the manufacturing life cycle including design, process planning, testing, and maintenance. There are several opportunities to include modeling and simulation tools in GOCO modernization and automation strategies. For example, material filling and solidification processes demonstrate how modeling and simulation can play a key role in optimizing processes that have traditionally been run based on experience and operator intuition. One example of planned funding for modeling and simulation is for metal parts fabrication at SCAAP. A simulated production environment was developed to assist with determining an optimized and efficient equipment layout to increase metal parts capacity. This effort included a laser scanning survey of the existing facility to produce a digital, editable model. This model was used to design new production lines, optimize work-in-progress storage and simulate material flow throughout the facility.

Additive Manufacturing Insertion

Additive manufacturing offers many advantages over traditional manufacturing processes. It can significantly reduce material waste by adopting optimized minimum weight product design and reduce on-hand inventory by producing desired parts on demand. Due to its short life cycle time, production with additive manufacturing can be easily scaled up or down based on demands, such as to meet surge capacity requirements. Selected processes can be seamlessly integrated between the existing processes. An ongoing Life Cycle Pilot Process (LCPP) project is evaluating additive manufacturing approaches to produce 155mm metal parts. If successful, this technology could be implemented at SCAAP through a future modernization project.

Digital Engineering

As defined by the Defense Acquisition University, DE is “an integrated digital approach that uses authoritative sources of systems' data and models as a continuum across disciplines to support life-cycle activities from concept through disposal.” This data- and model-centric approach has begun transforming the industrial base, its operation, and the Government’s ability to manage it proactively. Execution of the Plan is ongoing, and significant progress has been made in areas such as the development of digital system models, digital twins of the ammunition portfolio, a digital thread, and a DE Ecosystem (DEE).

The interconnected DEE enables the digital thread, which is an extendable, configurable component of enterprise-level framework that seamlessly connects authoritative technical data, software, information, and knowledge in the enterprise data-information-knowledge systems. It enables decision makers to access, integrate, and transform disparate data into actionable information. To gain value from the DEE and the digital thread, a digital system model of the product and product line is required.

During FY24, these models continued to serve as a digital representation of the manufacturing lines at the facilities and defined specific activities and considerations throughout the system or product life cycle. The process models are created, maintained, and simulated all within the SysML-based Cameo MagicDraw software. These models have allowed the government to regain previously uncaptured manufacturing process knowledge that can be used to simulate what-if scenarios and perform predictive analyses for throughput and bottlenecks. Creation of the process models of each major GOCO is ongoing. However significant progress has been achieved over the last year, where the focus has been on the entire M795 artillery shell production loop from metal parts at SCAAP to Load, Assemble, and Pack (LAP) at Iowa. During this maturation period, simulations to identify bottlenecks, perform throughput analysis, and run what-if scenarios were developed and tested. These capabilities provide Project Lead Joint Services (PL JS) a crucial tool to inform future modernization initiatives and investments. For example, the Iowa Army Ammunition Plant (IAAAP) M795 process model was baselined in June 2024. The executable SysML model focuses on the Line 3A at Iowa which contains every building involved in the LAP process for the M795 shell. The model includes a high-level block diagram that summarizes the activities within each building. The user can select any block within the model and explore the internal block diagram containing the lower-level activities for an increased detail view of the operations. The simulation can perform predictive analysis for any work shift scenario (8hr, 24hr, or even 45 days) and show the net throughput of the shells manufactured over that time. The simulation utilizes a token to represent a shell, cart, or cart string moving through the manufacturing process. During the simulation, a series of tokens are tracked and as bottlenecking occurs, a numeric counter appears on the block signaling a problem area within the manufacturing flow. To address the bottlenecking experienced during the simulation, a what-if scenario may be run in which the configuration of the line may be altered by incorporating additional equipment. By completing this scenario over the desired shift schedule, the user will have the appropriate data to confirm alleviation of the bottleneck(s) as well as the theoretical throughput if the investment of additional equipment is made. This is a key example of how the simulation model is an invaluable tool to inform investment decisions to increase plant efficiencies, verify and validate prospective modernization project designs, as well as inform leadership on the facility impacts for a variety of capacity needs. The status of the ongoing initiatives includes the following:

- Foundational Efforts:
    - After establishing a baseline of all facilities and associated operations through the DE Maturity assessments (reference Figure 1 for the maturity assessment scale and example), reassessment at a defined cadence is necessary. Reassessments will allow the models to remain current and maintain the level of fidelity required to run the simulations to further support modernization and sustainment investment decisions from leadership.

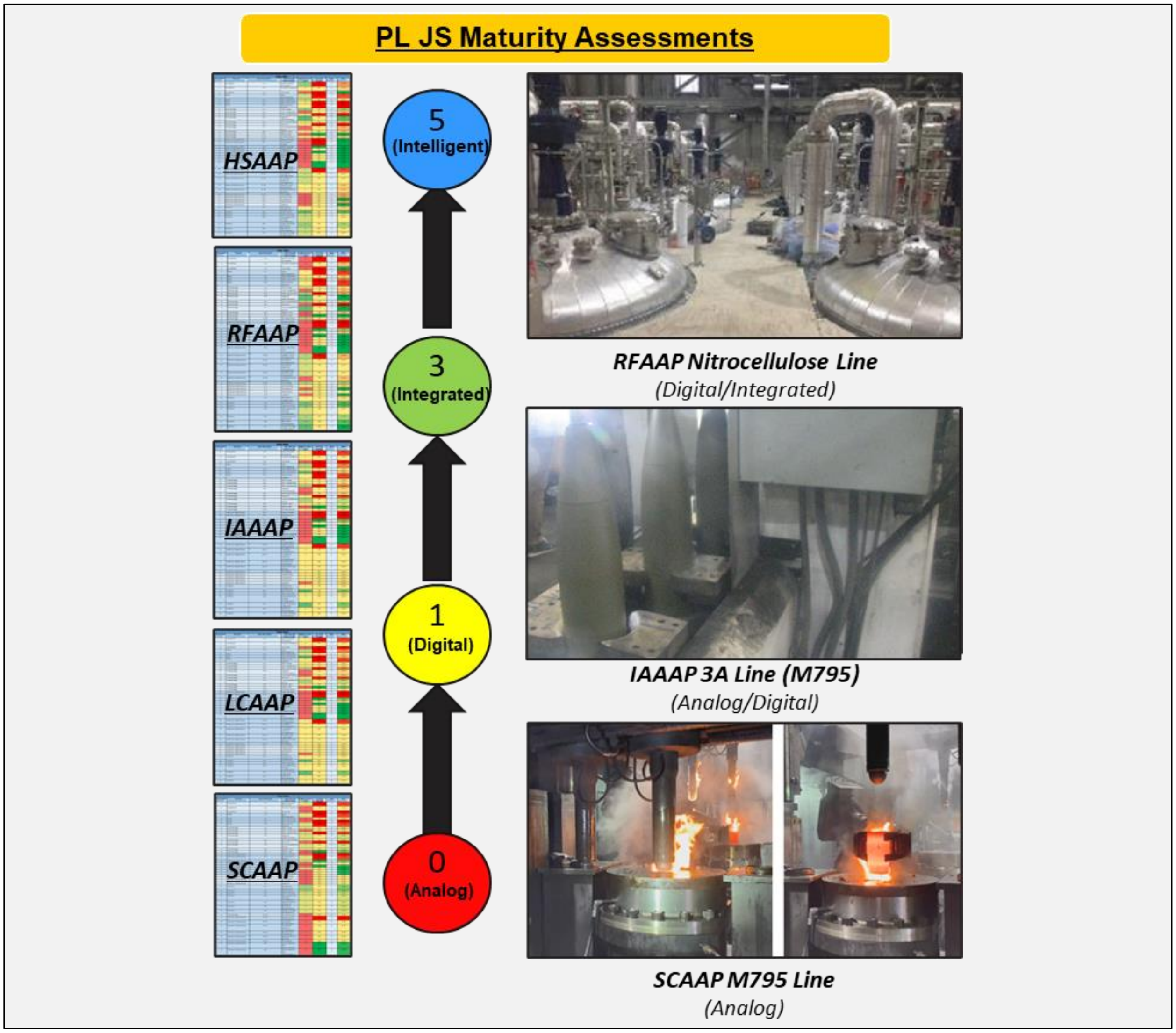


FIGURE 1. DIGITAL ENGINEERING MATURITY SCALE WITH ASSESSMENTS

- The evaluation, implementation, and integration of DEE solutions began in FY 2023 and continued into FY 2024. PL JS continued the exploration of four (4) candidate solutions by the U.S. Army Combat Capabilities Development Command Armaments Center (DEVCOM AC) who performed a robust, hands-on evaluation of the Aras Innovator, Syndeia, SBE Vision, and Interoperability and Integration Framework (IoIF) software solutions. Research is underway for a fifth software evaluation which is estimated to occur in early CY 2025.
- Phase II Small Business Innovation Research for a DEE software solution was awarded in May 2024. This software will be the first ecosystem to be developed specifically utilizing PL JS's System Requirements Document (SRD) which was a byproduct of the System Requirements Review that was completed late FY 2023. The SRD is the authoritative document of the future DE platform which will integrate models and engineering tools throughout the DE platform life cycle.

- The development of a site-specific DE requirements template was completed with the underlying goal to be applied at various levels. First example use case is when recompeting an AAP operating contract, the Government will utilize the set of requirements to establish the objectives for the overall DE implementation at the plant. Another use case is during the planning phase of a modernization investment, the requirements can be used to ensure the project scope is aligned with the DOD DE Strategy to include critical DE components such as user interfaces, data collection, etc.

- Dynamic DE Effort:
    - The first M795 process model for SCAAP was completed in FY 2023 and subsequent versions of the model are underway in FY24. As mentioned above, the M795 LAP process model at IAAAP has been baselined and the simulations are undergoing maturation to include delay, maintenance, and explosive limit logic. LCAAP, RFAAP, and HSAAP modeling were well underway in FY 2024.

2.1.3 Regulatory Compliance

Modernization projects are planned at all the GOCOs which ensure regulatory compliance. One example of a potentially more stringent regulation being considered for the GOCOs is the reduction of open burning and open detonation (OB/OD) practices. OB/OD is permitted and conducted at the GOCO facilities for waste explosives as an exception to the federal ban on open burning practices so long as no other safe method of disposal or treatment exists. There are several projects reflected in the Modernization Plan that address OB/OD and the implementation of alternative technologies. While these projects will reduce OB/OD use, OB/OD must be maintained as capabilities at the GOCOs for safety reasons even after implementation of the alternative technologies.

2.1.4 Cybersecurity

Cybersecurity within a production environment is a complex and challenging issue with immediate impacts to all sectors of the Organic Industrial Base (OIB). Cybersecurity is required to protect information and maintain availability of operational technologies across the enterprise. In an increasingly digital production environment, cybersecurity requirements often evolve and continue to mature. The cybersecurity requirement is further complicated by requirements for automation. Incorporating advanced automation within the facilities, operations, and processes as specified in this Plan will increase the facilities' digitized footprint. Digitizing manufacturing requires integration and connectivity between equipment and operations. This requires increasing cybersecurity processes, controls, and technologies to protect against potential threats and system vulnerabilities. Recognizing the paradigm shift from analog manufacturing systems and processes to digitized and automated manufacturing systems, AAPs are planning for additional cybersecurity requirements. To this effect, the AAPs are in the process of implementing a comprehensive cybersecurity program as required per Department of Defense Instruction (DODI) 8500.01, "Cybersecurity," DODI 8510.01 "Risk Management Framework for DOD Systems," and Army Regulation 25-2, "Army Cybersecurity." The protection and security of information technology (IT), operational technology (OT) and their associated capabilities are of the utmost priority. The strategy outlined below ensures the long-term success and sustainability of the cybersecurity program across the GOCO enterprise.

2.1.4.1 Risk Management Framework

The risk management framework (RMF) defined in the National Institute of Standards and Technology (NIST) Special Publication (SP) 800-37, IAW directive DODI 8510.01, "Risk Management Framework For DOD Systems" is a six-step process: categorize, select, implement, asses, authorize, and monitor (see Figure 2). The RMF process will identify and mitigate potential threats by implementing appropriate cybersecurity controls. Cybersecurity controls need to be carefully designed and implemented through a layered approach to mitigate the risks of both inside and external threats. There should be a long-term strategy to align each AAP with a tailored implementation of the RMF as they finalize their cybersecurity modernization efforts as outlined in Section 2.1.4.2.

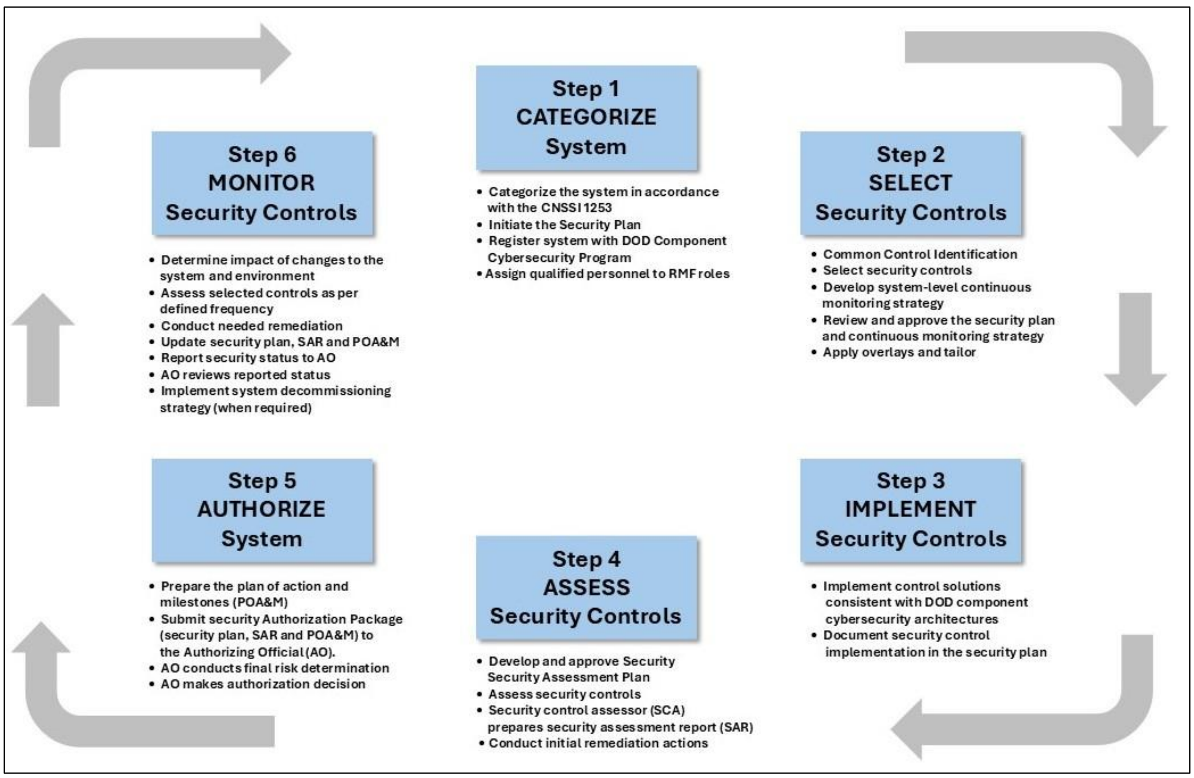


FIGURE 2. DOD IT SYSTEMS RMF PROCESS

2.1.4.2 AAP Cybersecurity Progress

The AAPs are undergoing a three-phased modernization strategy to prepare them for a tailored implementation of the RMF life cycle. The three phases are: (1) asset mapping and identification, (2) design and construction, and (3) active defense/monitoring. Each AAP is at a different point in the timeline (see Figure 3). SCAAP is nearing the end of mapping and inventorying their assets for Phase 1 and has started some preliminary design work for Phase 2. LCAAP was awarded a Phase 2 cybersecurity design project in June 2024 and is actively working on this effort. The Phase 1 deliverables were delayed at IAAAP; however, the project is now on schedule and a proposal for the Phase 2 cybersecurity design is scheduled for 2025. HSAAP completed Phase 2 construction efforts, and Phase 3 is being executed. They will be building out their policy and procedure in further detail as required. RFAAP concluded their phase 3 project by providing some critical documentation deliverables including a system security plan with security control implementation status and detailed network diagrams of each building. Additionally, the DEVCOM AC Cybersecurity team has been conducting independent assessments of each AAP's cybersecurity posture. These assessments have afforded stakeholders an additional lens to view each AAP's progress.

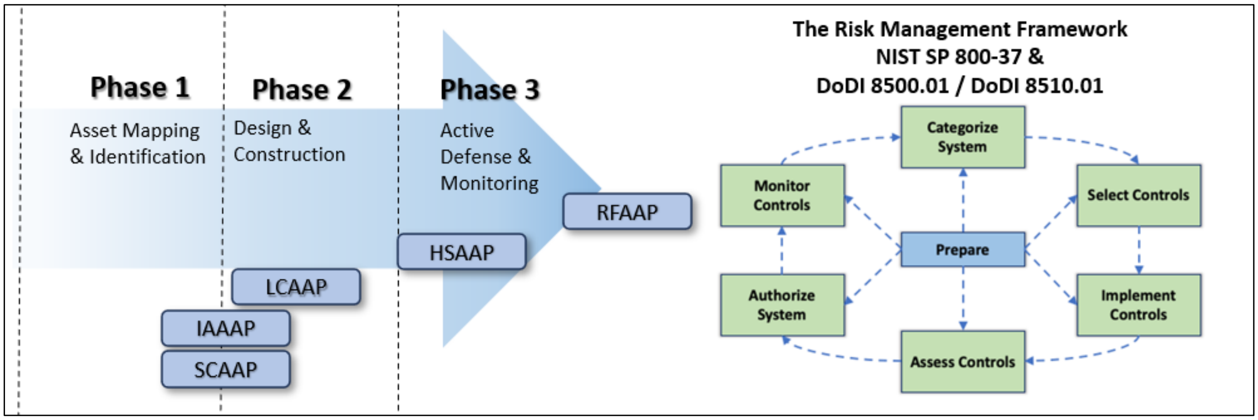


FIGURE 3. GOCO AAP RMF LIFE-CYCLE ROADMAP

2.1.4.3 Continuous Monitoring and Metrics

A continuous monitoring strategy (RMF Step 6) will implement policy and procedure to periodically review and assess cybersecurity controls to validate their effectiveness and verify their implementation. Continuous monitoring is the most critical of all cybersecurity phases. A system cannot remain secure without continuously reviewing and validating the security controls designed to protect it. Simply put, there can be no "set it and forget it" when it comes to cybersecurity. Continuous monitoring includes monitoring audit logs, patching software and operating systems, verifying approved software installations, monitoring privileged access, regular vulnerability scans, and conducting tests of implemented security controls.

2.1.4.4 Collaboration, Awareness, and Training

PL JS has collaborated extensively with Joint Munitions Command (JMC), DEVCOM, U.S. Army Corps of Engineers (USACE), and the Operating Contractors as these cybersecurity efforts have continued to grow and evolve. Fostering professional relationships encourages knowledge sharing and affords expertise where and when it is needed. Systematic cybersecurity training and awareness programs are conducted for all personnel involved in AAP operations to cultivate the philosophy that security is everyone's responsibility. It is critical to champion collaboration and information sharing among AAPs and other Government facilities to stay updated on emerging cyber-threats and industry best practices. This is achieved through participation in industry forums, sharing of lessons learned, and engaging in joint cybersecurity exercises.

2.1.4.5 Incident Response

While a properly funded, implemented, and supported cybersecurity program is essential for operational success, no organization, network, or facility is 100% secure. Negligence or misunderstood training can transform innocent or careless actions into a significant security issue. Implementing a well-defined incident response plan (IRP) ensures the correct steps are taken to mitigate the effects of the incident and restore full mission capability.

This strategy and a program management approach will enhance the cybersecurity posture of the OIB and effectively mitigate the risks associated with cyber-threats. Building, maintaining, and monitoring a risk-managed production base across the AAPs will be critical for decades to come.

2.1.5 Rising Costs

Global events and economic factors such as inflation, support to ongoing foreign conflicts, continued shortages in the touch labor workforce, and supply chain challenges are resulting in 50-200% increased project costs and extended schedules. These cost increases are impacting funding availability, because project budgets that were established in the project planning phases are no longer accurate based on current pricing, inflation, and escalation factors.

2.1.6 Supply Chains and Material Sourcing

The modernization program continues to encounter significant supply chain disruptions that impede the timely delivery of critical equipment and components. These disruptions include large items such as transformers—where delivery lead times can extend up to 18 months, and smaller yet equally crucial parts, such as chips and other electronic controls essential for process automation and manufacturing. Certain high-grade semiconductor chips used in process control systems are now facing delays of up to 12 months due to global shortages and extended lead times. Such components are often produced by limited manufacturers, adding challenges when demand spikes across industries.

Further complicating material sourcing is the reliance on single-source or no-source situations, where specific materials or components are either obtainable only from a limited number of vendors or entirely unavailable. An example is specialized sensors used in automated controls, which are sometimes single sourced from international suppliers, increasing vulnerability to global supply chain fluctuations. This dependency has intensified as the U.S. pushes to onshore more of its raw material sourcing through domestic production. However, for some items, like rare earth metals essential for electronics, there is no current CONUS supply, this forces procurement from overseas increasing lead times and potential risks from geopolitical tensions.

The delays and sourcing challenges are especially impactful in sectors that depend on precision and reliability, such as defense and critical infrastructure, where manufacturing cycles and equipment maintenance are closely tied to the availability of these specialized materials. Supply chain constraints pose ongoing risks and underscore the need for diversified sourcing strategies to mitigate these challenges.

2.1.7 SURGE PRODUCTION CAPACITY CHALLENGES

U.S. support for various international defense interests has also presented a challenge as ammunition requirements, especially for artillery, continue to increase. More recently, an international interest in tank ammunition is impacting GOCO modernization strategies. The required modernization, along with the expedited schedules necessary to meet capacity and production requirements, have resulted in an overall increase in large-scale projects above what had already been planned to meet overall AAP modernization objectives. While these additional projects are critical to meeting ammunition production requirements, the modernization projects already planned that address other critical areas like infrastructure, regulatory compliance, and safety must also be executed. The challenge lies in managing and executing these projects simultaneously.

2.2 MODERNIZATION KEY OBJECTIVES

This AAP modernization plan's objectives are:

(1) Increase manufacturing safety and readiness to meet current and future requirements.
  - Modernize and transform production processes, incorporating best practices from industry.

(2) Isolate energetic mass from people.
  - Remove personnel from energetic operations and replace manual operations with automation.

(3) Ensure graceful degradation and resilient operations.
  - Maintain operability to meet production requirements despite degrading facility conditions and failing equipment, even in the event of catastrophic failure, as well as sustain existing capabilities until new construction is complete.
  - Design production processes to minimize the concentration of energetic materials in any one place.
  - Incorporate redundancy where possible and practicable.

(4) Improve flexibility, maintainability, and sustainability.
  - Integrate state-of-the-art technology and equipment.
  - Implement a proactive preventive maintenance program and utilize predictive maintenance to improve overall system performance and sustainability.
  - Design production lines that can be alternatively used for multiple products with minimum retooling and reconfiguration of the physical system.

(5) Reduce cost of operations.
  - Identify processes and/or equipment that reduce the cost of operating the facilities.
  - Identify efficiencies to reduce labor, material, and energy costs.

(6) Secure supply chains.
  - Mitigate single point failures.
  - Investigate U.S.-based production and vertical integration.

2.3 MODERNIZATION PLANNING APPROACH

The GOCO AAP modernization approach considers critical production capabilities and capacities that support the National Defense Strategy, the National Defense Industrial Strategy, and the SMCA Industrial Base Strategic Plan (IBSP). Due to the sensitive nature of information related to energetic manufacturing (including production of propellants, explosives, pyrotechnics, and associated intermediates) and associated production capacities of the GOCO facilities, specific details are not included herein. However, current, and projected workload requirements (for both current and future weapons) are reviewed and assessed against facility modernization needs, required facility capacity, and project planning. The approach also presents a roadmap for facing the modernization challenges discussed in Section 2.1 above.

Part of the planning process also includes characterizing each GOCO AAP's operating conditions and identifying the useful life of core processes, infrastructure, and support systems. This characterization data is used to develop "Heat Maps" that provide a visual representation of these assessments and assist in requirements prioritization by indicating the required time frame for modernization. This assessment helps prioritize and plan the required design phases that support modernization project execution and budget formulation (refer to Section 2.5). These Heat Maps illustrate current facility condition and are provided for each facility in Section 4.0.

Modernization Integrated Project Teams (IPTs) comprised of members from the AAP Operating Contractor, JPEO A&A, DEVCOM-AC, JMC, DOD, Department of Energy (DOE), and academia identify modernization projects to address production and infrastructure deficiencies. The IPT considers the current and future munitions requirements for the facility. It is a holistic approach focusing on safety, process changes, automation, robotics, and ergonomics that remove facility personnel from direct interface with high-risk and dangerous operations. Interdependencies between projects, facility and equipment obsolescence, Quantity Distance (QD) arcs, process flow constraints, and material flow and intra-facility travel routes are all considerations when selecting and planning modernization projects. In addition, multiple alternatives are assessed against the project requirement to ensure selection of the most effective path forward. This assessment includes analyses of capacity requirements (including upstream and downstream process impacts) and ensuring a right-sized capacity design. The work of these IPTs results in an overarching strategy that identifies both project requirements and execution pathways for each individual GOCO and is documented in a facility-specific Planning and Transformation Strategy (PaTS). While PL JS leverages USACE for execution of select modernization efforts, the necessary expertise for Industrial Base modernization lies within the Modernization IPTs. PL JS manages projects contracted to the AAP operating contractor through their Contracting Officer's Representative (CORs) and relies on DEVCOM as technical experts. PaTS development is an ongoing effort continuously updated and improved in response to additional data and changing requirements.

Identified modernization projects are quantitatively scored against weighted scoring criteria. Table 2 below identifies these criteria and aligns them with the modernization objectives discussed above in Section 2.2.

TABLE 2. MODERNIZATION OBJECTIVES AND PROJECT SCORING ALIGNMENT

OIB Key Objectives	Weighted Scoring Criteria
1. Increase manufacturing safety and readiness to meet current and future requirements. **2.** Isolate energetic mass from people. **3.** Ensure graceful degradation and resilient operations. **4.** Improve flexibility, maintainability, and sustainability. **5.** Reduce cost of operations. **6.** Secure supply chains.	**1. Safety** ➢ Objectives 1 and 2 **2. Readiness/Production Continuity** ➢ Objectives 1, 3, 4, and 6 **3. Regulatory Compliance** ➢ Objectives 1 and 4 **4. Cost Reduction** ➢ Objective 5

While the six objectives provide an overarching end state goal, the defined scoring criteria and weighting provide a more functional basis for project selection. Each scoring criteria are assigned weights, as explained in Figure 4.

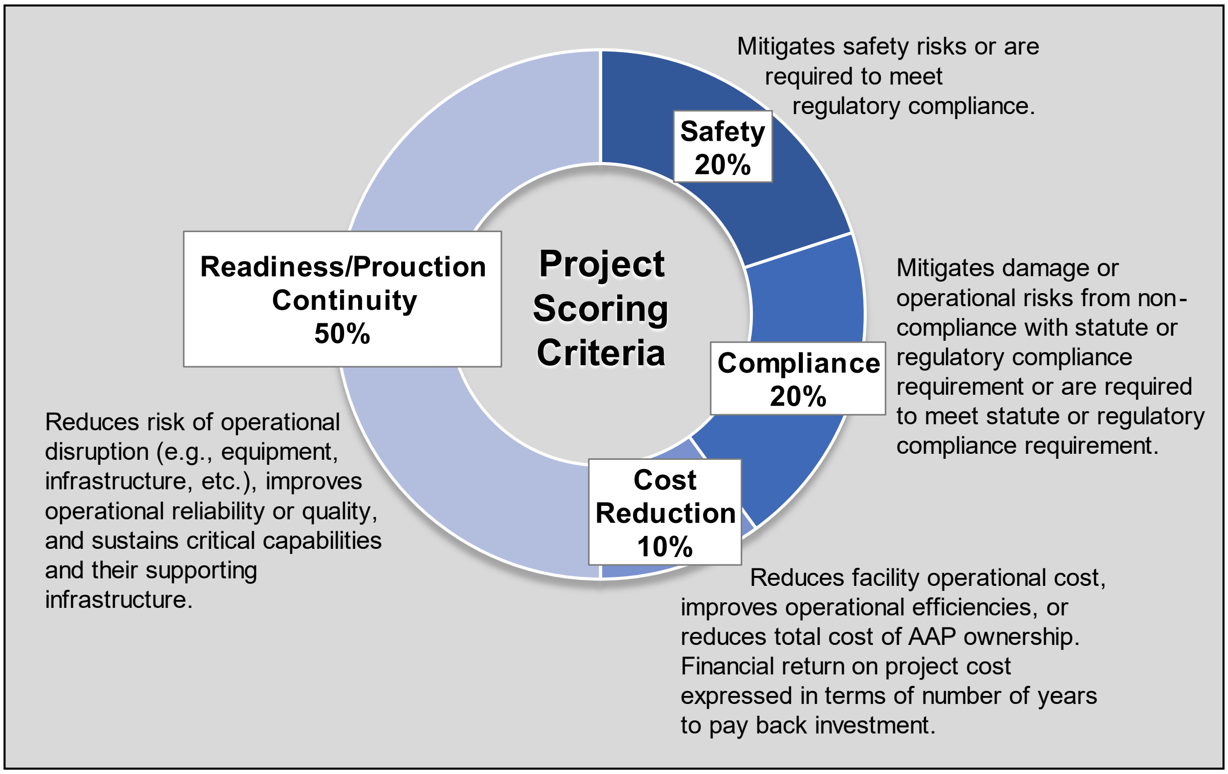
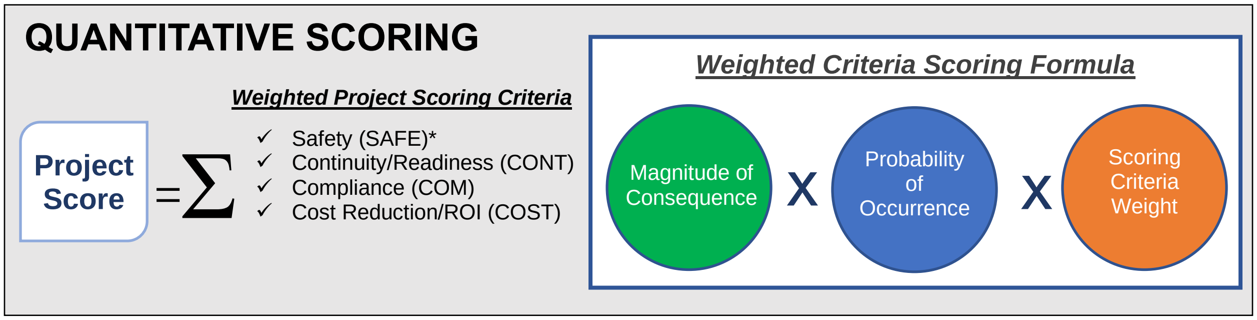


FIGURE 4. PROJECT SCORING CRITERIA

The quantitative scoring uses weighted criteria, and a 0-1-3-9 scoring methodology based on Six Sigma. Each project is measured against two (2) factors: (1) significance of mitigating the deficiency or consequence if the project is not executed (i.e., magnitude of the consequence) and (2) the likelihood (i.e., probability the consequence will occur if the project is not executed). These two metrics are multiplied, along with the corresponding scoring criteria weight, and summed to produce a final score as illustrated in Figure 5.



*Must-fund safety and environmental compliance projects are elevated to the top of the 1-N priority list.

FIGURE 5. QUANTITATIVE SCORING FORMULA

A 1-N list of modernization projects is compiled for each AAP based on this scoring methodology. While not separately scored, Quality of the Work Environment (QWE) initiatives are annually resourced at a level to ensure continuous improvement. Standalone QWE projects would not be prioritized higher than other modernization projects utilizing the above methodology.

When integrating modernization projects across the AAPs, critical projects are considered for prioritization. A project is deemed critical if investment is necessary to address an immediate safety concern, achieve compliance with a specific statue or safety regulation, avoid significant supply disruptions, and/or provide an immediate improvement to operating efficiencies. Projects necessary for safety or regulatory compliance are a top priority in the planning process. They are determined as “must-fund” and elevated to the top of the 1-N list. The 1-N list is further adjusted to account for project sequencing, acquisition strategies, and resource availability for execution. Additional consideration is given to strategic projects that either enhance current capabilities or add new capabilities based on National Defense Strategies and National Defense Industrial Strategies. Projects that fall within the Total Obligational Authority are included in the budget submission; projects falling outside the TOA are planned for the Extended Planning Period.

An annual update on progress of facility improvement projects and details on planned projects within the Service Program Objective Memorandum (POM) is provided to the Office of the Assistant Secretary of Defense for Sustainment (ASD(S)) for approval (Memorandum for Secretaries of the Military Departments, *Delegation of Approval Authority for Industrial Facility Projects*, dated January 15, 2025).

2.4 MODERNIZATION EXECUTION METHODOLOGY

Formulating modernization solutions requires sequencing within the constraints of the current GOCO AAP Facility Use contracts, project phasing requirements and project interconnections. Figure 6 provides timelines for the Facility Use contracts.

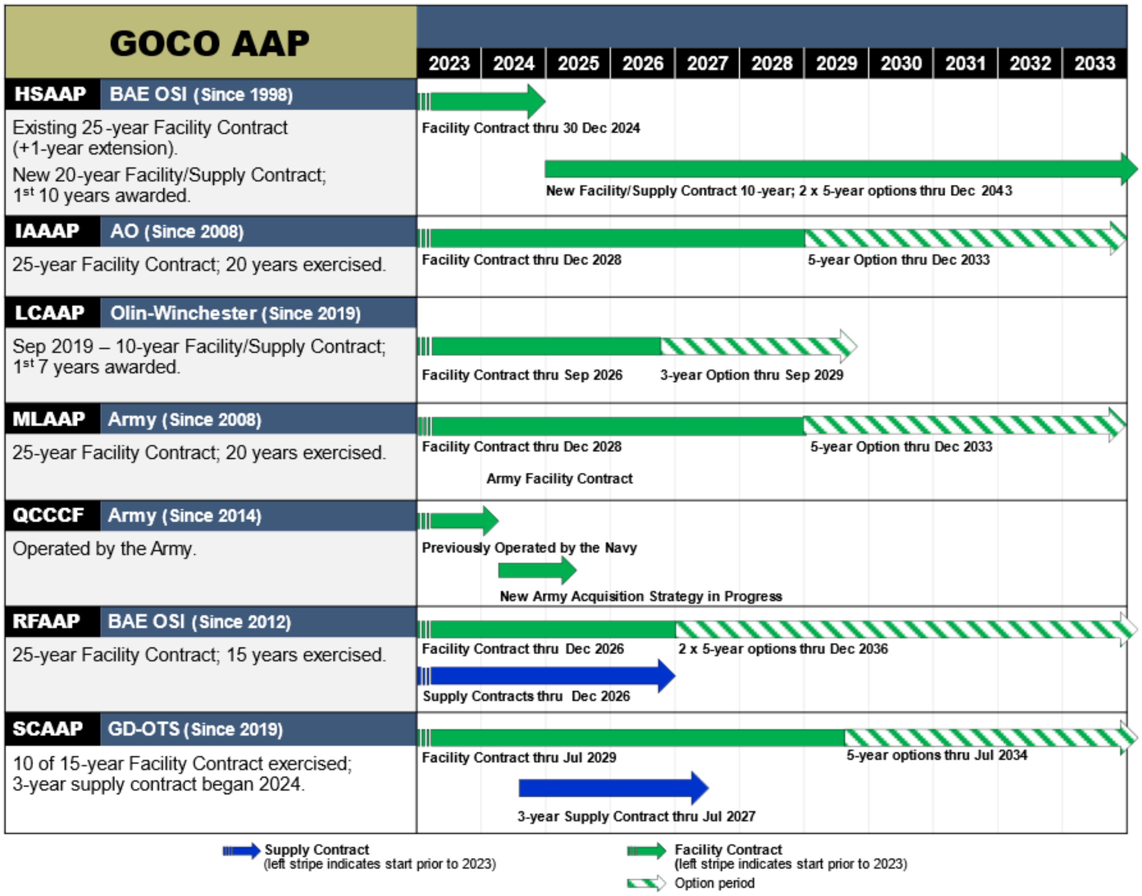


FIGURE 6. GOCO AAP CONTRACT ROADMAP

As indicated in Figure 6 above, the HSAAP Facility Use Contract was successfully recompeted with the award in December 2023. The winner of the competition was the incumbent and no protests were filed. The acquisition and solicitation process for the RFAAP Facility Use Contract recompete kicked off in July 2024.

The solicitation development and competitive source selection process is a significant effort that takes on average, over two (2) years. The timing and execution strategy for projects at HSAAP had been developed considering the timetable for award and the possibility of a change in operating contractor. Likewise, RFAAP's modernization strategy has been adjusted to consider potential impacts from the competition schedule and award.

Significant modernization projects may require three (3) to five (5) years to execute the design, construction, and commissioning. When building the Modernization Plan, projects were planned to minimize encumbering the AAP during a future facility competition. Utilizing project execution strategies other than solely contracting through the Facility Use contracts provides the Army with greater planning flexibility and is a critical strategy for being able to fully obligate funding.

2.5 COST ESTIMATING PROCESS

The GOCO AAP modernization program has faced major challenges with project cost estimating and subsequent budget submissions due to the current inflationary environment and world events. This critical issue is further complicated by the technical complexity and uniqueness of many of the facility projects and the decades-old facilities and infrastructure requiring modernization and/or replacement. To help address these challenges, the modernization program is actively pursuing early design efforts, such as planning charettes, to improve the fidelity of project cost estimates. The following explains design sequencing used for AAP construction projects. This 30/60/90 methodology is standard industry practice and based on widely accepted design practices. The following definitions have been adapted for use by the PBS Program based on established guidelines from other Government organizations (e.g., Veterans Affairs, Los Alamos National Laboratory, USACE), as well as industry partners and contractors.

(1) Define Requirements/Pre-Design (Planning and Design Charrettes)
   a. This initial design phase defines the project scope and includes an evaluation of alternatives.
   b. Primary outputs include project justification and requirements framework; potentially 15% design.

(2) Develop 30% Design
   a. The 30% design phase produces preliminary technical specifications and design basis.
   b. Primary outputs include design to the 30% level, priced equipment list, definition of contracting strategy, preliminary execution plan, overall project schedule, and a cost estimate (+25% / -15%).

(3) Develop 60% Design
   a. The 60% design phase includes complete design drawings, issue for design and specifications, budget quotes for all major equipment, control plan, safety plan, and project schedule.
   b. Primary outputs include a fully designed to spec project. A more detailed cost estimate (+15% / -10%) is also developed at this stage.

(4) Develop 90% Design
   a. The 90% design phase represents a 100% complete design except for the incorporation of final design comments.
   b. Primary outputs include final design drawings and specifications, complete documentation required to support permit application approvals, and a detailed 100% complete cost estimate.

(5) Issued for Construction Design Package and Drawings

Building upon technical knowledge, this process incorporates pre-design and design activities, which allows for the understanding of material requirements and production methods and processes. The process collects data and associated costs early in the project planning process, which provides better and more accurate project budget requests. It also considers life-cycle costs and potential impacts to facility operating costs. These project budgets are used to develop capital investment plans for each AAP, which subsequently provide the budget for each project prioritized in the 1-N list. The detailed list of projects by AAP, including project description and budget, is included in Appendix A. These projects are listed by FY in Appendix B. A discussion of modernization and associated investment plans by AAP is provided in Section 4.0.

3.0 RESOURCING

Since 1970, the Department of Army has utilized the PAA appropriation to fund modernization, construction, and rehabilitation efforts at GOCO AAPs and commercial facilities, as well as preservation of inactive production lines at the AAPs. The PBS program, which is a PAA appropriation, funds both modernization and efforts to maintain inactive capabilities and facilities at the AAPs to retain the capability and capacity for future use. The PBS program comprises the following budget lines:

- Provision of Industrial Facilities–PIF projects may include a new production capability, expansion of current capacity, or modernization of existing facilities to current standards. Modernization efforts should consider improved operating efficiency and reduced operating costs, as well as sustainment and readiness of critical capabilities.
- Maintenance of Inactive Facilities–MIF projects fund efforts to maintain inactive facilities in a laid-away state, ensuring they are available for future use.
- Layaway of Industrial Facilities–LIF projects transition industrial facilities or equipment from production status to long-term storage. LIF projects may also include decontamination, disposal, and/or demolition of excess Government-owned property.

AAP projects include, but are not limited to, construction of acid or propellant facilities or implementation of modernized and next-generation small caliber production lines and are inherently technically complex and unique within the DOD mission. To ensure cost-effective spending of PBS funding, MIF and LIF projects must be identified and prioritized at each facility and across the OIB along with PIF projects.

As discussed in Section 2.3 above, prioritized projects falling within the base TOA are submitted in the budget. Projects that fall outside the base budget are either planned for future years or funded if additional resources become available. Sections 3.1 and 3.2 discuss historical PBS funding levels and current and future resourcing requirements, respectively. Section 3.3 discusses the Life Cycle Pilot Process program (Budget Activity 6, Program Element number 0605805A), which can fund the types of development and demonstration efforts required to apply automation and other emerging technologies to AAP modernization. Section 3.4 provides a summary of the Armament Retooling and Manufacturing Support (ARMS) program and the financial benefits it provides to the Industrial Base.

3.1 HISTORICAL RESOURCING

Most of the OIB installations were designed and built to support the high-volume and sustained production rates of World War II. Since 2003, the Army has invested over $8.5

billion in PBS, Industrial Facilities funding (through FY 2024) to maintain regulatory and safety compliance, ensure and increase readiness and production continuity, and increase operating efficiencies. Figure 7 shows historical and current PBS funding for GOCO AAP modernization. Figure 7 includes COCO funding.

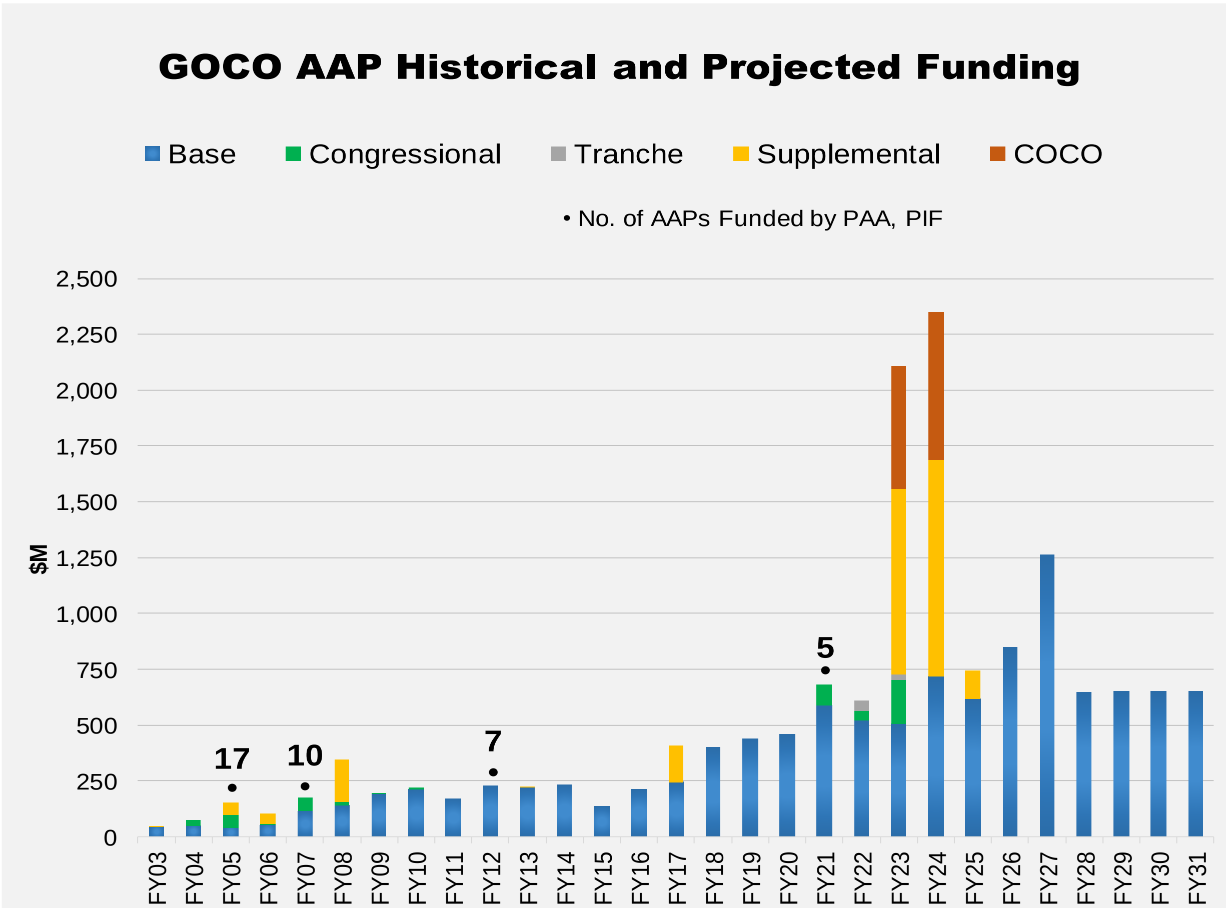


FIGURE 7: GOCO AAP HISTORICAL AND CURRENT FUNDING – PAA, PIF

As observed above, there has been a significant increase in required resourcing (base budget funding) for modernization efforts since FY 2016. This increase is a result of aging facilities and processes—some of which are original to facility construction—increased capacity requirements, new ammunition production, and stricter regulatory and safety regulations. In many cases, simple solutions and upgrades are not enough to fully address infrastructure and/or production issues. These issues require complete replacement or "recapitalization." Future resourcing will continue to address these existing needs, as well as address integration of automation and other emerging technologies into the AAPs to truly modernize their infrastructure and capabilities. The spike in funding in FY 2023 and FY 2024 were a result of production efforts supporting international conflicts and the requirement for increased capacity. FY 2025 Supplemental funding was requested to restore operations from Hurricane Helene damage at RFAAP.

3.2 GOCO AAP Modernization Resourcing

The Army's PIF/MIF/LIF investment requirements for FY 2025 through FY 2031 are presented in Table 3. The investment requirements in Table 3 reflect the base budget. Additional funding in any particular year could potentially move priority requirements to the left.

Table 3. FY25–FY31 Requirements

$M	FY25	FY26	FY27	FY28	FY29	FY30	FY31	Total
PIF (EP1200)	744.791	850.611	1264.77	645.880	651.150	651.036	653.142	5461.379
MIF (EP1500)	7.068	6.634	6.673	6.858	6.929	6.967	6.967	48.096
LIF (EP2000)	12.607	11.793	11.860	12.190	12.319	12.386	12.426	85.851
Congressional Add	-	-	-	-	-	-	-	-
Total Industrial Facilities (EP1000)	764.466	869.038	1,283.302	664.928	670.398	670.389	672.535	5,595.056

Beyond FY 2031, this Modernization Plan considers a list of tactical and strategic requirements in the EPP, totaling over $7B. Several high-value, strategic construction projects, identified in Table 4 below, are not currently resourced through FY 2031 but are planned for in the EPP. These critical projects could be executed earlier if additional funding was made available.

Table 4. Examples of Out-Year Strategic Modernization Projects

GOCO Ammunition Facility	Strategic Project Investment by Facility	Rough Order of Magnitude Cost ($M)
HSAAP (TN)	Flexible Explosive Production Facility & Infrastructure CL-20	435-535
HSAAP (TN)	ANSOL Thermal Destruction: Construction	150-250
IAAAP (IA)	Direct Fire Munitions Facility (Tank Line)	400-600
LCAAP (MO)	Modernized Metal Parts Facility	400-500
LCAAP (MO)	Consolidated Chemistry & Energetics Process Laboratory	200-300
RFAAP (VA)	Solventless Propellant Facility: Design & Construction	700-800
RFAAP (VA)	NG-3 (Nitroglycerin) Facility: Construction	600-700
RFAAP (VA)	Utilities Modernization & Permanent Energy Center	500-600

3.3 Life Cycle Pilot Process Resourcing

LCPP is Budget Activity 6 RDT&E projects that support the implementation of the SMCA Industrial Base Strategic Plan through technology investigations, model-based process controls, pilot prototyping, and industrial assessments. The projects funded through LCPP assess life-cycle production capabilities required for all ammunition families, address design for manufacturability to facilitate economical production, identify industrial and technology requirements, and address the ability of the production base to rapidly and cost effectively produce quality products. LCPP provides a critical risk reduction bridge between science and technology investments in advanced manufacturing technologies and their implementation in GOCOs and ammunition industrial base partner facilities.

LCPP also seeks to improve the efficiency and product quality of existing GOCO manufacturing processes, reduce life-cycle product costs, eliminate single point failures (SPFs), and, in some cases, develop new sources of supply within the National Technology and Industrial Base.

Approved projects under LCPP focus on the end-state of manufacturing technologies and therefore must have a clear transition path to the Organic Industrial Base. LCPP's mission aligns to the GOCO modernization program goals by ensuring that technology assessments and demonstrations are of interest to the Army's Industrial Base. Project execution for the LCPP program includes piloting process technologies from lab-scale environments, technology feasibility studies, and process demonstrations. The LCPP program is particularly focused on executing projects that mitigate single- or no-source supply risk. Research, feasibility studies, and/or demonstrations are conducted to mitigate supply-chain risks and include identifying new or second sources of materials.

LCPP investments often involve synergistic partnerships between JPEO A&A, DEVCOM AC, and OIB partners, which encourage the transition of high return-on-investment technologies from the laboratory to full scale production. By collaborating with industrial base partners to test and validate technologies developed by DEVCOM AC at the pilot scale, LCPP can reduce implementation risks and address critical needs of Program Managers (PMs). This is especially true for ammunition products such as energetic materials which often face supply chain constraints and are produced using traditional, sometimes inefficient manufacturing processes. LCPP also brings the ammunition industrial base and Army laboratory communities together to tackle life-cycle challenges and technology development through modern manufacturing methods. With extensive technology partnering, LCPP increases the likelihood of the Army realizing long term benefits such as better product quality, reduced product costs, automation and a more robust supply chain.

3.3.1 Project Selection

The process for the LCPP Call for Proposals typically unfolds over several months. Initially, manufacturing challenges and technical requirements are compiled and utilized to draft the LCPP Call Letter, which is subsequently released to the technical community. The community is then given 1-2 months to submit technical proposals for the LCPP funding opportunity. Following the submission deadline, a thorough evaluation process takes place to generate a 1-N list of prospective projects. A final evaluation and endorsement by Program Managers/Project Leads takes place prior to final project selection. Figure 8 illustrates the LCPP Project Selection Process.

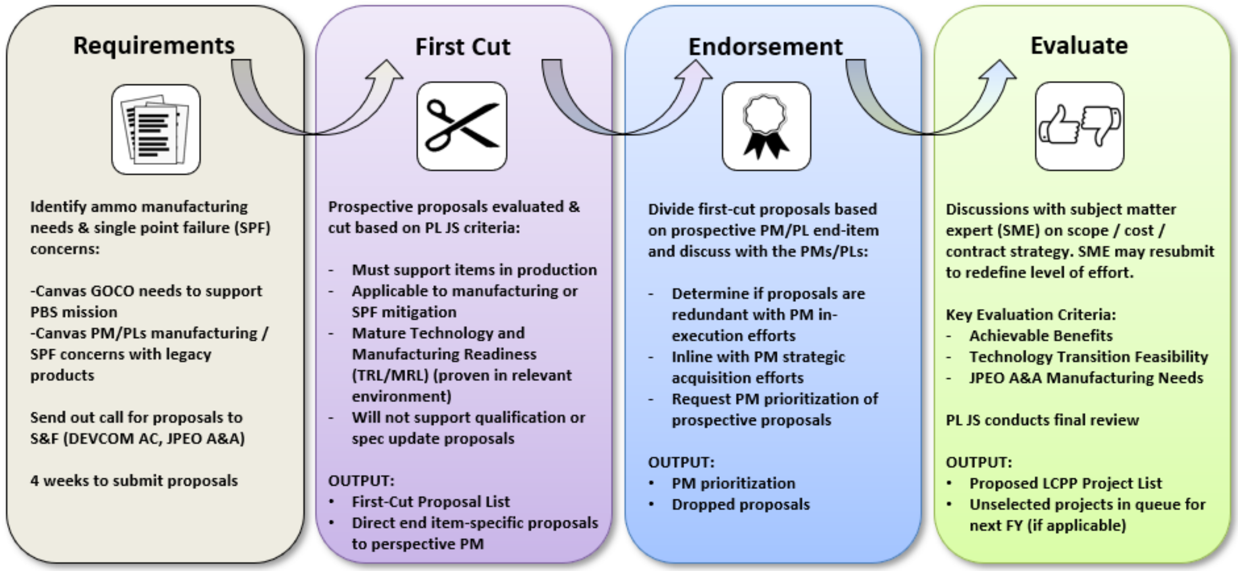


FIGURE 8. THE LCPP PROJECT SELECTION PROCESS

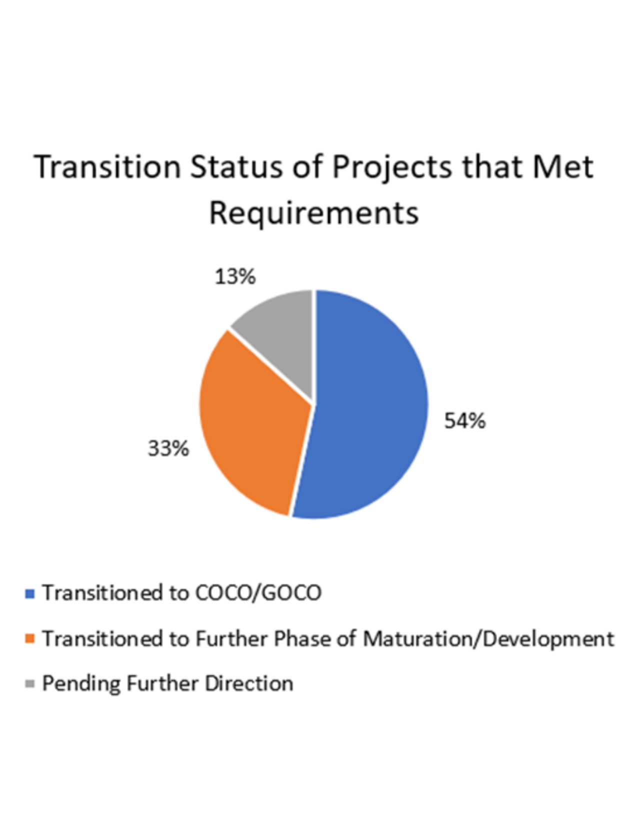
Proposals submitted to the LCPP program are graded on a list of criteria and must meet the following for final selection:

- Must affect in-production JPEO A&A products in line with LCPP mission statement
- PM/PL Endorsement or Technology Transfer Agreement (TTA)
- Mature technology readiness with focus on transition to Industrial Base
- Manufacturability & Producibility Evaluation Weights

3.3.2 PROGRAM STATUS

The acquisition process for weapons systems relies heavily on prototyping, testing, and maturation to guarantee their effectiveness and suitability for intended use. This involves rigorous testing at proving grounds and refinement of deployment methods at training sites. Similarly, manufacturing systems that incorporate new technologies and processes require development, refinement, and testing to ensure their readiness. LCPP plays a crucial role in facilitating this process, enabling the maturation of emerging manufacturing technologies before they are transitioned to the OIB for large-scale implementation.

For the munitions OIB, the collaboration between LCPP and DEVCOM AC has resulted in substantial risk reduction and overall acceleration of efforts to modernize. Many LCPP projects are proposed by DEVCOM AC and are executed in conjunction with both GOCO and other OIB partners, resulting in numerous successful transitions, as documented in prior reports. A program review was recently conducted to evaluate the outcomes of completed projects. A total of 21 projects were reviewed with project start dates since 2015. The following graphic provides an overview of the projects' results. Figure 9 illustrates that most projects successfully met project requirements with eight (8) out of 21 projects successfully transitioning to a COCO or GOCO and another five (5) projects transitioning to a further phase of maturation or development.



Transitioned to COCO/GOCO 54% of Projects	Prototype Riser Crusher for Melt Pour Explosives
	Alternative Anti-Seize Material for Use on Lifting Plugs
	MOFA Liquid Reserve Battery Single Point Failure
	BDNPA/F Prototype Manufacturing Process
	Thermocouple Artillery Projectile for IAAAP
	Automated Dimensional Inspection of Legacy Artillery
	MIC Green Primer Pilot Line
	Manufacturing Automation for Metal Microscale G-Switches
Transitioned to Further Phase of Maturation / Development 33% of Projects	Alternative Dye Synthesis for M18 Grenades
	BDNPA/F Manufacturing Process Scale-Up
	Manufacture KNO3 from HSAAP ANSOL: Phase II
	Twin-Screw Extrusion of Black Powder
	Manufacture Critical Chemicals from HSAAP Waste Streams
Pending Further Direction 13% of Projects	Alternative Smoke Ignition Source Kettle Drying Alternatives for CXMs

FIGURE 9. LCPP PROJECT TRANSITION STATUS

While the goal of the LCPP is to mature technology for and transfer it to the OIB, the partnership with DEVCOM AC provides reach back and organic expertise to mitigate other risks post-transition. LCPP's investments in pilot-scale and process advancements, coupled with the expertise of DEVCOM AC yield significant benefits and enhanced capabilities for the SMCA. The pilot scale capability will support more effective and rapid root cause analysis when product or process failures occur. The Government will retain more process and technical knowledge reducing reliance on single operating contractors. The pilot-scale facilities can be used to investigate proposed changes to production processes, to material, or inspection methods; all in an environment that does not disrupt on-going production.

3.3.3 RESOURCING SUMMARY

Table 5 identifies LCPP funding in FY 2023 and FY 2024 and planned for FY 2025 through FY 2028.

TABLE 5. LCPP FUNDING FYS 2023–2028

	Fiscal Year					
	FY23	FY24	FY25	FY26	FY27	FY28
BASE	5.585	5.838	5.873	5.875	5.938	6.003
Base Total FY23–FY28 ($M)	35.112					
CONGRESSIONAL	13.000	7.500	-	-	-	-
Congressional Total FY23–FY28 ($M)	20.500					
FY Total	18.585	13.338	5.873	5.875	5.938	6.003
FY23–FY28 Total ($M)	55.612					

3.4 ARMAMENT RETOOLING AND MANUFACTURING SUPPORT (ARMS) RESOURCING

The ARMS Act of 1992 authorized the commercial use of Government facilities, and in 2000, the program was codified in 10 U.S.C. The ARMS legislation allows commercial use of Government facilities and equipment at the GOCOs for commercial munitions production. This legislation is critical to supporting commercial activities at the GOCOs which reduces manufacturing costs to the Government and provides a means for keeping manufacturing lines operating when Government requirements decrease.

The ARMS program was established in FY 1993 to encourage commercial firms' use of the Army's ammunition manufacturing facilities at the GOCO AAPs. Through the ARMS Program, the Army invests appropriated funds (PAA funding) and ARMS program revenue to repair, refurbish, and upgrade existing facilities to rent to commercial tenants. Savings include both cash and non-cash benefits that reduce the GOCO AAP cost of operations. As part of the program, sites collect cash revenue from rent and other third-party activities. A portion of this revenue, based on annual revenue growth, is paid to Facility Contractors in the form of Contractor Compensation/Incentive Fees (CIFs). Non-cash savings include overhead absorption (decreasing product prices to the Government by increasing the production base) and tenant services provided in lieu of rent (such as mowing, road repair, training, and environmental remediation). Table 6 summarizes these savings for FY 2022 and FY 2023.

TABLE 6. SUMMARY OF ANNUAL INVESTMENT AND SAVINGS

	FY 2022	FY 2023
Rent Revenue Collected	$3,016,186	$3,158,086
Third-Party Revenue Collected	$3,923,918	$6,060,296
Revenue Paid to CIFs	($709,989)	($789,662)
Total Cash Savings	**$6,230,115**	**$8,428,719**
Overhead Absorption	$75,801,256	$135,202,541
Services Provided in Lieu of Rent	$719,758	$239,758
Total Non-Cash Savings	**$76,521,014**	**$135,442,299**
Total Savings	**$82,751,129**	**$143,871,018**
Investment (PAA)	$2,501,687	$4,989,650

Note: The FY 2024 ARMS report with FY 2024 data is currently in development.

The ARMS program generates additional benefits that may not be considered savings. These benefits include private investment made by Facility Contractors and tenants, economic output generated and proceeds from scrap sales. Table 7 summarizes these benefits for FY 2022 and FY 2023.

TABLE 7. SUMMARY OF ANNUAL BENEFITS

	FY 2022	FY 2023
Private Investment	$6,639,853	$20,035,959
Economic Output Generated	$762,226,562	$728,170,418
Proceeds from Scrap Sales	$610,489	$978,844
Total Monetary Benefit	**$769,476,904**	**$749,185,221**
Direct Employment	1,292	1,270
Indirect Employment	1,476	1,384
Total Employment Benefit	**2,768**	**2,654**

Note: The FY 2024 ARMS report with FY 2024 data is currently in development.

4.0 CAPITAL INVESTMENT PLANS

The GOCO AAP sites discussed in this section consist of five (5) AAPs and the QCCCF. MLAAP is not discussed; it does not have any planned funding in this Modernization Plan. Table 8 presents a summary of these AAPs.

TABLE 8. GOCO FACILITY SUMMARY

Industrial Base Sector	GOCO Ammunition Facility	Core Processes and Products
Explosives	HSAAP (Kingsport, TN)	• Production and development of insensitive munitions explosives • Melt-cast, cast-cured, pressed, and extruded explosives formulation (RDX, HMX, IMX, TATB) • Explosives performance testing • Full-spectrum explosives research and development capability
LAP	IAAAP (Middletown, IA)	• LAP–40mm Cartridges (tactical and training), Tank/Artillery (120mm, 105mm, and 155mm) • C-4 Extrusion (C4 Demo Block and M58 Mine Clearing Line Charge [MICLIC]) • Warhead press loading
	MLAAP *(Milan, TN)*	• *No Active Production*
Small & Medium Caliber Ammunition	LCAAP (Independence, MO)	• 5.56mm, 7.62mm, .50 Caliber LAP, Linking, and Links • Small Caliber Tracer/Incendiary Bullet Manufacturing • Future 6.8mm production • Medium Caliber Links, 20MM Electronic Primers, and 20MM Load Assemble Pack (ARMS)
Propellant	RFAAP (Radford, VA)	• Nitrocellulose (NC)–Grades A, B, C, D, E & F • Solvent Single Base Propellant (M6, PAP7993, AFP001, M64MP/SP) • Solvent Multi Base Propellant (RPD596, CAD PAD) • Solventless Double Base Propellant (AA2, AA6, MICLIC/APOBS, JA-2) • MK 90 Rocket Grain Extrusion & Finishing
Metal Parts	SCAAP (Scranton, PA)	• High volume production of artillery metal parts (M795, M1128, XM1113, XM1210) • Long stroke, high tonnage forging capability • Large volume heat treat
	QCCCF (Rock Island Arsenal, IL)	• 105mm Cartridge Cases • Steel Deep Drawn Cartridge Cases (Navy 5”/54, 105mm Army Direct Fire)

The following sections provide a summary of AAP core capabilities, current and anticipated future state condition, modernization programs and progress, advances in automation, and requirements for FYs 2025–2031.

4.1 HOLSTON ARMY AMMUNITION PLANT

4.1.1 CORE CAPABILITIES

HSAAP is the primary manufacturer of Research Development Explosive (RDX) and High-Melt Explosive (HMX). HSAAP manufactures Insensitive Munitions (IM) Explosive IMX-104, which is a melt-cast explosive used in a variety of mortar and future artillery ammunition. IMX-104 uses materials such as Dinitroanisole (DNAN), Nitrotriazolone (NTO), and RDX to provide equivalent explosive performance to Comp B. The facility also produces a growing number of specialty chemicals, such as 2,3-dimethyl-2,3-dinitrobutane (DMDNB), which is the taggant material used in Composition C-4 explosive.

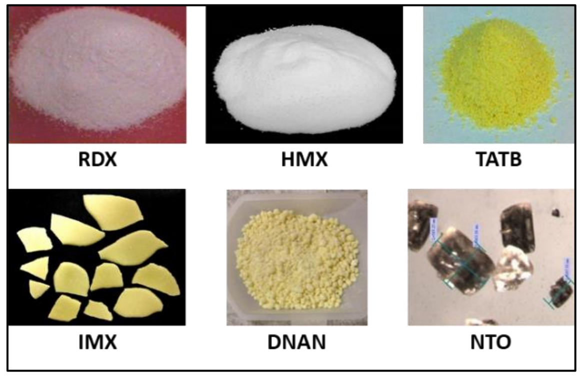


FIGURE 10. HSAAP PRODUCTS

HSAAP, working in conjunction with the Services and DOE, created a triaminotrinitrobenzene (TATB) explosive, which is the main ingredient used in bomb fuzing and booster applications for artillery and mortar applications, as well as other strategic weapons. TATB is one of the most insensitive of all the high-explosive ingredients. Additional research on other critical energetics, like CL-20, is also being conducted at HSAAP and other locations.

Figure 11 provides an updated plant overview based upon facility assessments and engineering studies, illustrating areas of risk and opportunities for investment. They depict the current condition of each of the capabilities and subsystems within the facility. The green/amber/red classifications indicate the condition as a percentage of the whole system. The condition code correlates to the assessed timeframe need for modernization funding for the facility’s core capabilities (i.e., production facilities), infrastructure systems, and support facilities, as shown. For example, if 50% of the rectangle is red it means that 50% of the whole system requires modernization funding in the next 1-5 years to address the deficiency. If the other 50% is green, then the remaining part of the whole system is operating well and will not require funding in the next 10 years.

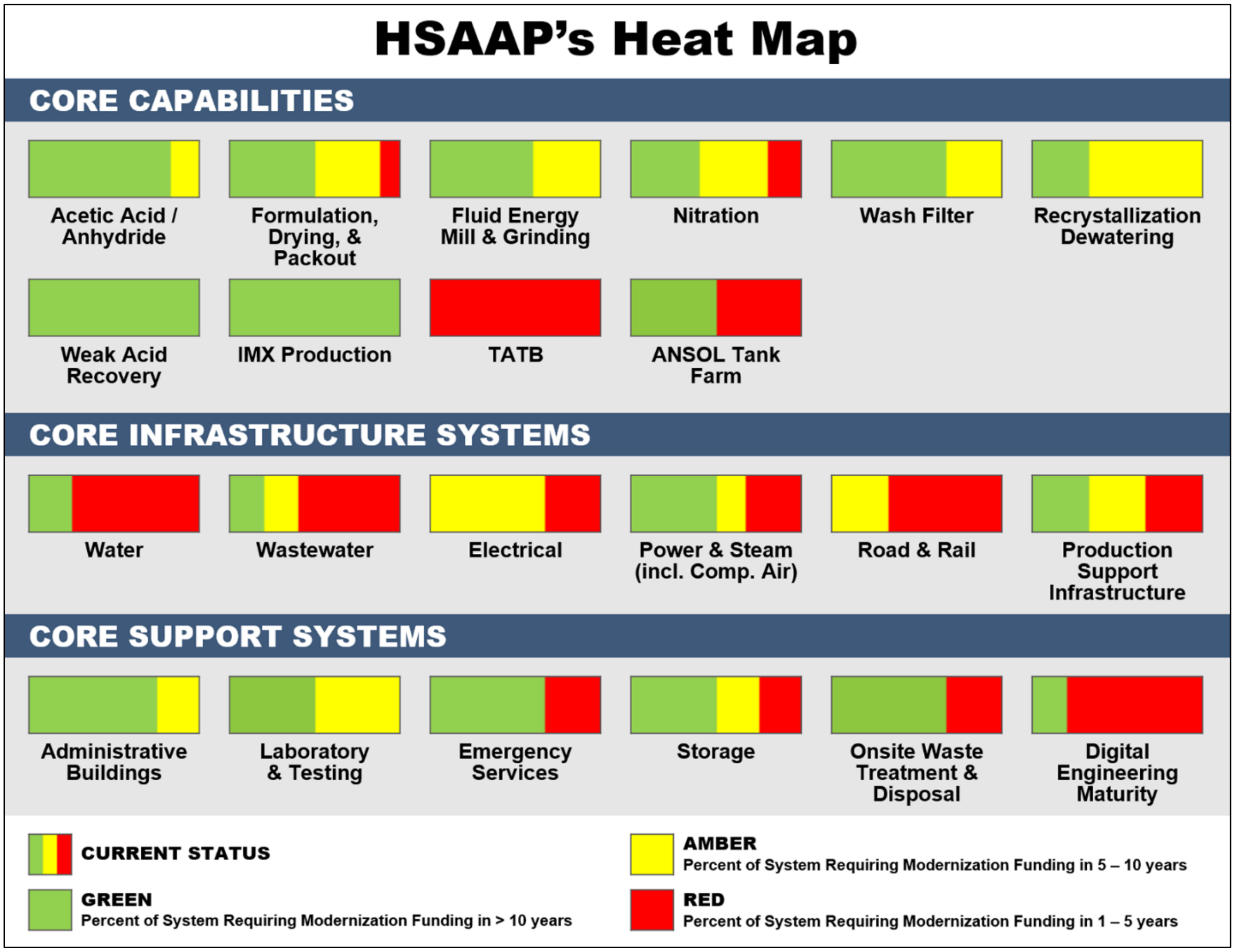


FIGURE 11. HSAAP'S CONDITION AND MODERNIZATION STATUS

4.1.2 MODERNIZATION SUMMARY

Multiple Master Planning efforts have been completed for HSAAP including a Vision Plan in 2017, an Area Development Plan (ADP) in 2021, and an Area Development Execution Plan (ADEP) in 2022. Master Planning efforts for production and supporting infrastructure systems began in 2015. At that time, an RDX production capability gap was identified when the Air Force increased its requirements for explosives to support additional bomb production. Further analysis across all the Service demands over the subsequent five (5) years resulted in the determination of the total capacity increase required to meet current and emerging explosives demand. An AoA was developed and staffed to the Office of the Secretary of Defense, who subsequently recommended a solution to double the RDX capacity at HSAAP that would also reduce dependencies on foreign explosive materials. This recommendation was briefed to the Deputy Secretary of Defense via the Deputy's Management Action Group governance forum in November 2016 and approved. As a result of this decision, the Government worked in collaboration with the HSAAP Operating Contractor to develop an implementation strategy. The strategy included all major constraining-unit operations within the acids area—along with the supporting facilities/infrastructure associated with RDX production—all of which required expansion to meet the new capacity requirement.

Additional manufacturing capacity was also required for facilities dedicated to the production of IM ingredients. The IMX Ingredients facility (by design) is flexible, allowing production of either DNAN or NTO within the facility. DOD demand required a production increase of IMX-104. IMX-104 uses DNAN, NTO, and Fluid Energy Mill RDX as energetic ingredients. Modernization is planned to continue to increase IMX-104 production capacity.

Modernization projects are also ongoing to increase the production capacity of TATB and future projects are planned for additional improvements. A modernization project is planned to transition the production of PAX-3 (used in a grenade and a new tank ammunition round) from the HSAAP pilot plant to a production building to increase production capacity and produce PAX-3 more efficiently. Likewise, additional modernization planning will need to consider other emerging critical energetics, including CL-20, and what the requirements may be for these materials. The flexible energetics production facility, currently included in the EPP, will be designed to produce these products in a semi-flexible manner to better adapt to increases and decreases in capacity demand and HSAAP's overall mission to produce explosives.

The additional IMX expansion efforts will drive modernization efforts at HSAAP similar to how the RDX Expansion program influenced the overall modernization strategy. Besides capacity expansion, another significant driver of modernization requirements is regulatory compliance. Examples of such projects include the Secondary Process River Water Intake Facility construction project, addressing toluene emissions, completing the Flashing Furnace project, and completing the Explosive Decontamination Chamber (EDC).

4.1.3 MODERNIZATION PROGRESS

<u>Facility Designs</u>

The Secondary Process River Water Intake Facility design is about to be completed. The related construction project, once executed, will support Clean Water Act Rule 316b compliance and provide any additional river water source for the filtered water plant and to support production process cooling. The ANSOL Thermal Destruction design has started. While the construction effort is in the EPP, if funded, it would build an on-site treatment technology for the ANSOL that is generated as a by-product of explosives production which would reduce risks associated with reliance on off-site disposal of ANSol and support meeting RDX/HMX capacity requirements.

4.1.4 SUPPLEMENTAL FUNDING IMPACTS

HSAAP received a total of $600M in Supplemental funding in FY 2024 for the eight (8) projects identified in Table 9 below. Collectively, these projects support overall production and capacity expansion efforts, both ongoing and planned, at HSAAP.

TABLE 9. FY 2024 SUPPLEMENTAL APPROPRIATION FOR HSAAP

Holston AAP: $600.0M			
Project Number	**Project**	**Value ($M)**	**Est. Comp. Date**
1	Interim Storage	**$18.0**	Jul 2028
2	NTO Capability	**$220.2**	Jun 2030
3	Third Train Anhydride	**$160.5**	Mar 2029
4	Melt Cast Capability	**$73.3**	Jan 2029
5	Demolition	**$4.7**	Oct 2026
6	Packaging Facility	**$53.6**	Jul 2029
7	NTO Capture Production Buildings	**$69.7**	Apr 2029
	TOTAL ($M):	**$600.0**	

4.1.5 ADVANCES IN AUTOMATION AND EMERGING TECHNOLOGY SOLUTIONS

The G-10 Toluene Emissions and Process Improvements project is planning to add various automated process controls related to the TATB production process and tie them into the Programmable Logic Controller (PLC) system to better control the TATB production process quality and in some cases safety. In addition, the Explosive Decontamination Chamber (EDC) project (currently under construction) will utilize three (3) robots in its operations: one (1) to palletize boxes and two (2) to remove boxes from pallets. Utilization of robotic technology reduces the need for additional operators and repetitive manual handling that can result in operator injury.

HSAAP was assessed using the DE Maturity Model and this baseline assessment resulted in an average value of 1.07 on the scale of 0 to 5, with 0 indicating a fully analog facility and 5 indicating a fully intelligent DEE. This score translates to a semi-digitized/modern facility. A fair degree of automation exists and is complemented with added digital controls and monitors using either Siemens PCS7 or Allen-Bradley control system software.

4.1.6 MODERNIZATION REQUIREMENTS FY 2025 THROUGH FY 2031

The Army has planned nearly $950M in funding for modernization projects at HSAAP during FYs 2025–2031. A complete project list, including project description and funding information, is included in Appendix A.

4.2 IOWA ARMY AMMUNITION PLANT

4.2.1 CORE CAPABILITIES

IAAAP's core capabilities are the explosive Load, Assemble, and Pack (LAP) of medium- and large-caliber ammunition. The plant is a manufacturing center for artillery, missile warheads, tank ammunition, grenades, mortars, mine-clearing charges, and demolition charges. IAAAP is DOD's primary source for:

- High Explosive (HE) Artillery (155mm and 105mm trinitrotoluene (TNT)/IMX filled rounds)
- Tactical Tank Cartridges (105mm and120mm)
- M112 Demo Blocks
- Mine Clearing Line Charge (MICLIC)
- 40mm Grenade Cartridges
- M1147 120mm Advanced Multi-Purpose (AMP) Warhead production

* M1147 120mm AMP production starting 2025

FIGURE 12. IAAAP LAP PRODUCTS

Figure 13 provides an updated plant overview based upon facility assessments and engineering studies, illustrating areas of risk and opportunities for investment. They depict the current condition of each of the capabilities and subsystems within the facility. The green/amber/red classifications indicate the condition as a percentage of the whole system. The condition code correlates to the assessed timeframe need for modernization funding for the facility's core capabilities (i.e., production facilities), infrastructure systems, and support facilities, as shown. For example, if 50% of the rectangle is red it means that 50% of the whole system requires modernization funding in the next 1-5 years to address the deficiency. If the other 50% is green, then the remaining part of the whole system is operating well and will not require funding in the next 10 years.

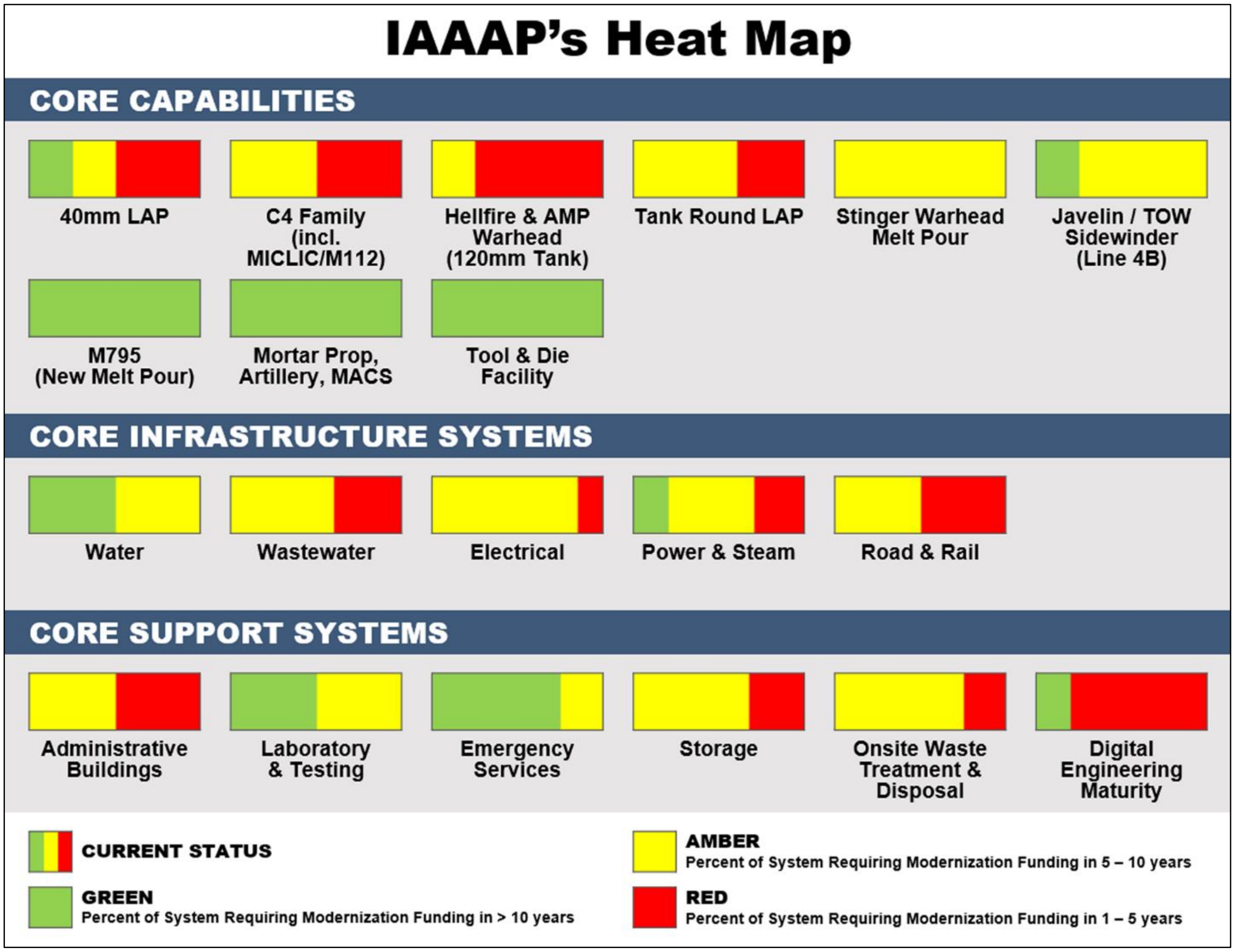


FIGURE 13. IAAAP'S CONDITION AND MODERNIZATION STATUS

4.2.2 MODERNIZATION SUMMARY

Several Master Planning efforts have been completed for IAAAP, including a Vision Plan in 2017, an ADP in 2020 and 2023, Installation Planning Standards in 2021, and an ADEP in 2021 and 2023. The ADEP built upon the Preferred Development concept established in the ADP; the ADEP identified that modernization of the Industrial Core is essential for IAAAP to be able to produce modern munitions, the infrastructure is dated and insufficient to support modern capabilities, flexibility is a key consideration and relocating inappropriately and inefficiently sited functions would improve production efficiency.

As with the other GOCO facilities, capacity increases are needed at IAAAP to meet the increased demand for 155mm projectiles due to world events and the requirement to rebuild the DoD stockpile. The FAC project is being reevaluated due to increased proposal costs and a revised strategy will be implemented in 4th quarter FY 2025.

In 2024 PL JS began the development of a Planning and Transformation Strategy (PaTS) for IAAAP. The overall expectation is the PaTS will be the single source to pull together planning and execution activities as well as provide additional information to help tell

Iowa's story. The PaTS will have the requisite information, data, and documents to inform annual GOCO Army Ammunition Plant updates to Congress as well as assist with POM budgeting.

The first phase of the PaTS effort is expected to conclude in May 2025 and include the following deliverables:

1. IAAAP PaTS Report
2. Process and Procedures
3. Characterizations
4. Project Fact Sheets
5. Execution Plan
6. Unmanned Aerial Vehicle (UAV) mission

Information from the Operating Contractor, Joint Munitions Command (JMC local) staff, and Stakeholders will also be leveraged within the PaTS.

The Operating Contractor annually provides a response to PL JS' Call Letter. The Call Letter provides the opportunity for the Operating Contractor to document issues they identify across plant that may have an adverse effect on their ability to efficiently and effectively produce items to meet mission requirements. The Operating Contractor also produces a P25 Feeder document for each project they identify or are asked to identify by the USG. The Call Letter process also includes the Operating Contractor annually submitting a 1-N priority list of the projects for USG consideration when establishing the execution plan within the PaTS, as well as informing the USG for future POM planning as well as input to future GOCO AAP Modernization Plan updates.

One of IAAAP's core capabilities is the LAP of medium- and large-caliber ammunition. As with the other GOCO facilities, capacity increases are needed at IAAAP to meet the increased demand for 155mm projectiles. The approach to meet this increase in 155mm ammunition requirements is being reevaluated due to increased construction proposal costs.

Buildings throughout IAAAP are from the WWII-era and have reached the end of their design and/or useful life. Because these facilities may be retrofit for production, the production processes are inefficient and not optimized. For example, logistics for the current melt pour facility are challenging due to its distance from other facilities on the installation. It has aging road and rail infrastructure, aging equipment, limited space, high groundwater issues, aging roofs, gutters, and doors, and other structural issues. IAAAP's current production capacity is not sufficient to meet anticipated M795 requirements; as such, a strategy to increase 155mm capacity is needed to meet projected 155mm demand. Likewise, much of the existing Direct Fire facility is at its end-of-life and requires repair parts that are increasingly difficult to acquire. Other deficiencies include a lack of natural gas and the need for a new wastewater treatment facility to meet compliance guidelines.

Project priority at IAAAP has been placed on projects that enable the increased production of munitions (i.e., 155mm artillery rounds). As such, improvements/repairs to the existing melt pour line, and related facility/infrastructure projects are high priority. New infrastructure and improvements to intraplant logistics (i.e., rail) will facilitate improved material handling and transportation at the facility. A phased approach to constructing these facilities is being implemented as a bridging solution to new infrastructure and to minimize risk.

4.2.3 MODERNIZATION PROGRESS

The following are some examples of recently completed modernization projects for IAAAP.

Natural Gas Boilers Phase I

This project was awarded in March 2016 and completed by the Operating Contractor, American Ordnance, in May 2022. This project installed two (2) natural gas boiler systems at artillery melt pour production Line 3A to increase thermal efficiency, provide increased operational control, reduce emissions, and decrease reliance on the main coal plant for operations and personnel comfort. This project eliminated 3.4 of the 12.9 miles of steam distribution lines from the coal-fired heating plant, reducing the amount of potential steam energy lost through distribution. With the installation of package boilers and repairs to steam lines, over 25,000 million British Thermal Units of steam per year will be saved. Based on the successful implementation of Phase I, Phase II was awarded and includes additional centralized boiler systems across the facility at Lines 1, 4A/4B, Roundhouse, and Laundry. Additionally, a coal elimination strategy is in place to remove the coal-fired plant and abandoned steam lines.

K Yard Phase I

The K Yard Phase I project was awarded in March 2023 and was recently completed and accepted. This project increased explosive storage capacity by constructing roads, driveways, and pre-cast dock/ramps at K-yard igloos that were not being used by the operating contractor due to being rail access only. This project also made internal roads commercial truck accessible. Phase II of this project expands upon the work completed in Phase I and was completed in December 2024.

4.2.4 SUPPLEMENTAL FUNDING IMPACTS

IAAAP received a total of $25.0M in Supplemental funding in FY 2024 for one (1) project identified in Table 120 below. This project supports continued and increased M1147 Tank production requirements.

TABLE 10. FY 2024 SUPPLEMENTAL APPROPRIATION FOR IAAAP

Iowa AAP: $25.0M			
Project Number	**Project**	**Value ($M)**	**Est. Comp. Date**
1	M1147 Tank	**$25.0**	Dec 2027
	TOTAL ($M):	**$25.0**	

4.2.5 ADVANCES IN AUTOMATION AND EMERGING TECHNOLOGY SOLUTIONS

Over the next five (5) years, several modernization projects at IAAAP will include automation to improve operator safety and create the opportunity for more efficient production. Graceful degradation will be implemented across future modernization projects. The Direct Fire Munitions Campus (Tank Line) will help remove the operator from handling heavy and dangerous Tank Rounds by automating the process.

The M795 line of IAAAP was assessed using the DE Maturity Model and this baseline assessment resulted in an average value of 0.97 on the scale of 0 to 5 with 0 indicating a fully analog facility and 5 indicating a fully intelligent DEE. This score translates to a mostly analog facility. Although, IAAAP is deficient in modern and efficient inspection methods (go/no-go gauges and manual digital filings). Conversely, IAAAP contains a fair degree of automation since manual/analog processes are complemented with added digital controls or monitors.

4.2.6 MODERNIZATION REQUIREMENTS FY 2025 THROUGH FY 2031

The Army has planned over $650M in funding for modernization projects at IAAAP during FYs 2025–2031. A complete project list, including project description and funding information, is included in Appendix A.

4.3 LAKE CITY ARMY AMMUNITION PLANT

4.3.1 CORE CAPABILITIES

LCAAP produces 5.56mm, 7.62mm, 6.8mm and .50-caliber ammunition, as well as small- and medium-caliber links.

FIGURE 14. LCAAP PRODUCTS

Figure 15 provides an updated plant overview based upon facility assessments and engineering studies, illustrating areas of risk and opportunities for investment. They depict the current condition of each of the capabilities and subsystems within the facility. The green/amber/red classifications indicate the condition as a percentage of the whole system. The condition code correlates to the assessed timeframe need for modernization funding for the facility's core capabilities (i.e., production facilities), infrastructure systems, and support facilities, as shown. For example, if 50% of the rectangle is red it means that 50% of the whole system requires modernization funding in the next 1-5 years to address the deficiency. If the other 50% is green, then the remaining part of the whole system is operating well and will not require funding in the next 10 years.

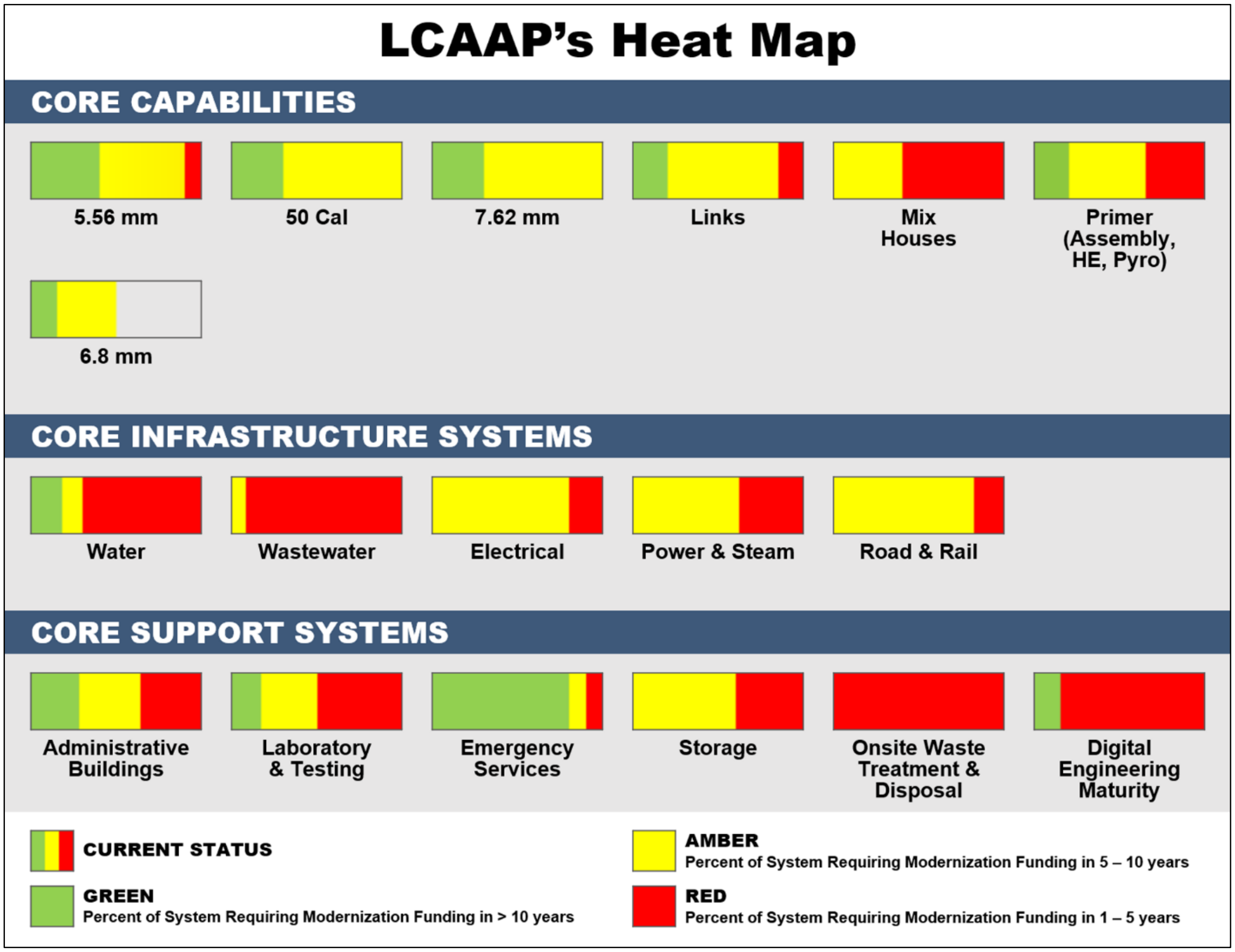


FIGURE 15. LCAAP'S CONDITION AND MODERNIZATION STATUS

4.3.2 MODERNIZATION SUMMARY

A Planning and Transformation Strategy has been documented that incorporates the institutional knowledge from Government Stakeholders at LCAAP, the Operating Contractor, and Subject Matter Experts. The PaTS leverages several planning efforts that have been completed for LCAAP including the Vision Plan (2016), Area Development Plan (2017), Area Development Execution Plan (2021), the Operating Contractor's annual 1-N list based on Call Letter guidance from PL JS, and the accomplishments of the current GOCO Modernization Plan. Process maps within the PaTS provide guidance for the selection, budgeting, and maturation of projects from ideas to awardable construction projects. Another process maps identifies the steps for annual POM Budgeting allowing candidate projects to be identified within required timeframes. In addition, an execution plan has been documented to include the construction award of all viable projects on the 1-N list by 2060.

The PaTS considers that many buildings are beyond their design or useful lifespans. LCAAP requires modernization and transformation for the safe and efficient production of legacy ammunition as well as for new ammunition, such as the Next Generation Squad Weapon-Ammunition (NGSW-A). This major modernization requirement at LCAAP is the construction (awarded in 2024) and commissioning of both the facility and equipment for 6.8mm ammunition production. LCAAP was selected as the site for this high-volume production of NGSW-A, providing approximately 70% of the total required OIB capacity. Additionally, as other small-caliber ammunition items are improved, the processes at LCAAP need to be upgraded and configured to effectively produce these items. Some of these near-term requirements include modifying current processes and developing product-specific equipment to produce the 7.62mm and .50 Cal Reduced Range Ammunition (RRA) and the 5.56mm and 7.62mm One-Way Luminescence (OWL) tracer bullets. LCAAP will also require modernization of current processes to produce Lightweight 7.62mm ammunition.

Ammunition production requirements influence the modernization strategy at LCAAP. The production facilities must be upgraded or replaced with new construction to achieve the required production capacity and operate safely and efficiently. The production requirements also drive the need to evaluate, upgrade, and replace the supporting facilities and infrastructure like waste treatment systems, utility distribution systems, warehousing, access control points, and energetic storage facilities. Many of the existing support facilities and infrastructure have seen limited modernization since their original construction in 1941 and are aged beyond their design and useful life expectancies. Some of the near-term requirements, currently documented in the AAP GOCO Modernization Plan include New Energetic Waste Incinerator, New Fire Water Distribution System, New Five Grain Distribution system, New Zero Grain Water Distribution System, Upgraded Industrial Wastewater Treatment Plant and New Wastewater Collection System, and Expansion of Emergency Services Center. Modernization requirements also contemplate upgrade projects as bridging strategies to align with future year new facility construction in the AAP GOCO Modernization Plan to ensure continued operation.

LCAAP Planning and Transformation activities are being executed in a phased project maturation approach. This process has resulted in identifying data/information to inform or influence project selection to address modernization requirements; when planning and design funds are required to support future construction awards; understanding project interdependencies; and recognizing the need to plan for flexibility given available funding on an annual basis. Strategic leveraging of planning and design funds has created Opportunity Projects for consideration in the EPP for construction award. The following projects are currently designed or in design and available for construction award beginning 4Q FY2024: NGSW-A Utilities Extension and Upgrades, Automated Ammunition Warehouse, Access Control Point, Bridge 5 Replacement, Consolidated Chemistry Laboratory, Industrial Wastewater Treatment Plant Filter Press and Lagoon Closure, Mix Houses and Neutralization, and Powder Delivery at B1.

Project execution strategies are also influenced through understanding siting limitations due to explosive arc requirements. QD arcs are calculated based on net explosive weight (NEW) and can influence how facilities/buildings/structures can be renovated or even govern where new buildings can be sited on the Lake City property in relation to other facilities/buildings/structures. Given LCAAP's limited acreage, there is reduced flexibility to site new facilities given arc influences from existing buildings and operations. Therefore, LCAAP's modernization plan considers the planning and design time to mature each project in preparation for construction award.

Specific outputs from Phase 1 of the Planning and Transformation Strategy deliverables, completed in 2022 include: production infrastructure support facility characterizations, focused unmanned aerial vehicle (UAV) surveys to assess facility condition and assist in the planning and design of selected projects, and processes including incorporating recapitalization, and developmental program needs into the modernization planning process.

Phase 2 outputs, completed in May 2024, documented a Project Execution Plan, Project Fact Sheets, additional characterizations, and additional UAV missions to support project maturation and future planning efforts. Visual aids, derived from data collected during UAV missions, provide support to the Planning and Transformation Strategy. In addition to 2D images captured during UAV missions, the data can be stitched together providing 3D imaging. This data can assist the USG customer and LCAAP stakeholders, as well as design professionals, to visualize existing situations as they mature projects for execution. Thermal imaging capability provides critical data to assist maintenance engineers and staff to identify areas of concern with building/structure envelopes and roof condition. This data assists in scoping areas for execution to repair or replace.

4.3.3 MODERNIZATION PROGRESS

Modernization progress at LCAAP continues with a primary focus on project maturation and Planning and Design activities as the NGSW-A production facility design, equipment design, and construction contract award utilized available funding. Project maturation is the process to mature a project from issue discovery through construction award. Maturation reduces project risk through the execution of charrettes (planning and design) and designs to 100% that iteratively (30/60/90/100) refine project requirements, identify project approvals, and update estimated construction costs. The construction cost estimates follow industry standards with; Class 5 (Rough Order of Magnitude) estimate at the Planning Charrette, Class 4 at the Design Charrette, Class 3 estimate at 30% Design, Class 2 at 60% Design, and Class 1 at 90% and 100% Design. As the design matures the cost estimate's expected accuracy range tightens as there are detailed designs and less unknowns. LCAAP has approximately forty-eight (48) projects active within the maturation process working towards construction award availability.

Whereas there are no recently completed modernization projects, the award of the NGSW-A facility construction contract in 2024 realized a significant milestone in LCAAP's transformation and modernization. A bridging strategy project to provide critical repairs to the stack of the existing Explosive Waste Incinerator (EWI) was also awarded. This project and a follow-on bridging project at the existing EWI facilities will keep the capability viable until the new EWI is awarded (planned for FY 2026), constructed, and commissioned.

4.3.4 SUPPLEMENTAL FUNDING IMPACTS

LCAAP received $172M Supplemental funding in FY 2024, covering $172M in support of the Contaminated Waste Plant Project and a $100M for small caliber ammunition infrastructure upgrades, which is being leveraged for support of the NGSW-A Facility Construction and Equipping Programs. Without this supplement, execution of the associated high-priority projects would have been significantly impacted, if not halted, impacting operational continuity of small caliber production, regulatory compliance, and explosives safety at LCAAP.

TABLE 11. FY 2024 SUPPLEMENTAL APPROPRIATION FOR LCAAP

Lake City AAP: $172M			
Project Number	**Project**	**Value ($M)**	**Est. Comp. Date**
1	Automated Contaminated Waste Plant	**$72.0**	Apr 2027
2	6.8mm Construction and Equipment	**$100**	Mar 2028
	TOTAL ($M):	**$172.0**	

4.3.5 ADVANCES IN AUTOMATION AND EMERGING TECHNOLOGY SOLUTIONS

Recent usage of emerging technologies at LCAAP that support the overall Planning and Transformation (aka Master Planning) efforts include the use of UAVs for building and asset inspection as well as supporting project designs. The UAVs capture high-resolution images of buildings from various angles and heights, providing a comprehensive view of the building's current condition. Additionally, the UAVs have the technology to conduct thermal imaging that can assist with identifying problems, such as heat loss or water damage, allowing for improved maintenance planning. At LCAAP, this technology provided a comprehensive assessment of roofing and HVAC conditions for the targeted buildings, providing valuable information for modernization planning. The recent UAV effort, completed as part of the PaTS Phase 2 deliverables, provided images and video plantwide, to bolster and justify projects for inclusion as well as to inform future GOCO Modernization Plan Updates and POM budget planning.

The LCAAP production/manufacturing processes were assessed using the DE Maturity Model, and this baseline assessment identified an average value of 0.88 on the scale of 0 to 5, with 0 indicating a fully analog facility and 5 indicating a fully intelligent DEE. This score translates to mostly analog production equipment. A fair degree of automation exists and is complemented with added digital controls and monitors. Beneficially, LCAAP collects and retains digital data on critical products and process parameters with many inspections using continuous variable measurements. The maturation of LCAAP, from a digital perspective, should include an understanding of the DE capabilities for facility and infrastructure projects combined with foundational activities associated with digitizing and standardizing any key processes that remain analog for production equipment.

4.3.6 MODERNIZATION REQUIREMENTS FY 2025 THROUGH FY 2031

The Army has planned over $1.3B in funding for modernization projects at LCAAP during FYs 2025–2031. A complete project list, including project description and funding information, is included in Appendix A.

4.4 QUAD CITIES CARTRIDGE CASE FACILITY

4.4.1 CORE CAPABILITIES

QCCCF's core capability is the manufacture of deep draw steel and brass large caliber cartridge cases. Key processes include high-tonnage, long-stroke, deep, cold draw presses; slath-bath heat treating; steel spheroidization process; and advanced machining operations. QCCCF was designed to produce large quantities of cartridge cases including 105mm brass, 105mm steel, Navy 5"54 steel, 40mm brass, 57mm brass and 76mm brass cases.

The QCCCF was established during the execution of BRAC 2005 which directed the relocation of the deep drawn cartridge case line from Riverbank AAP to Rock Island Arsenal (RIA). At the time, there were yearly requirements of approximately 100,000 cases per year. The QCCCF was sized for approximately 6,000-10,000 cases per month given the DOD requirements. The annual requirements for large caliber cartridge cases dropped and the facility was laid away in 2014. In 2017, the facility was re-activated by the Navy. It currently produces low quantities of 5"54 steel cases and is in the initial phases of efforts to produce 105mm tank cases. QCCCF is the **only** known producer of steel deep drawn cartridge cases.

Since the facility was reactivated in 2017, very little investment has been made at the facility. Only general facility maintenance was performed while it was in layaway status. The facility has not undergone a heat map study but will require one going forward to inform its modernization strategy.

4.4.2 MODERNIZATION SUMMARY

As QCCCF represents the only known source for DOD-required deep drawn cartridge cases, it is of strategic value. Since its reactivation by the U.S. Navy, operating sections of the facility have shown that, to be economically viable, the facility will need to be modernized from a designed continuous operation to a batch mode operation. A long-term plan for QCCCF will be developed by 2025, which will identify the production process and personnel changes required to pivot operations from a continuous mode to a batch mode. Specifically, the plan will address modernization of the plating processes, automation, material handling, and production control systems to ensure they are modern and cybersecure. That study will provide the basis for future modernization investment into QCCCF to retain and modernize this key DOD industrial capability. Even though the complete plan has not been defined to date, two requirements have been included in this Modernization Plan to address already defined needs and include the installation of a batch annealing furnace and industrial waste final holding tank.

4.4.3 MODERNIZATION REQUIREMENTS FY 2025 THROUGH FY 2031

The Army has planned approximately $11.25M in funding for modernization projects at QCCCF during FYs 2025–2031. A complete project list, including project description and funding information, is included in Appendix A.

4.5 RADFORD ARMY AMMUNITION PLANT

4.5.1 CORE CAPABILITIES

RFAAP's core capabilities are the production of Nitrocellulose (NC), and solvent and solventless propellants. RFAAP is the only source for production of NC and solventless propellant within the National Technology and Industrial Base. RFAAP also produces Nitroglycerine (NG), and sulfuric and nitric acids, which are required intermediates for NC and NG production.

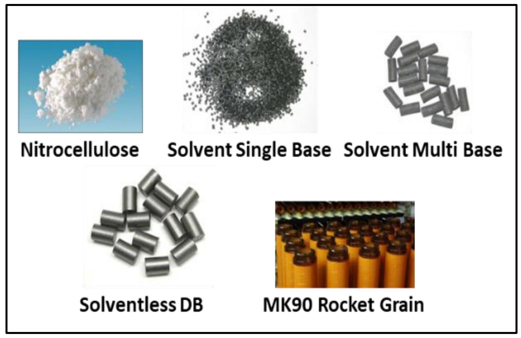


FIGURE 16. RFAAP PRODUCTS

Figure 17 provides an updated plant overview based upon facility assessments and engineering studies, illustrating areas of risk and opportunities for investment. They depict the current condition of each of the capabilities and subsystems within the facility. The green/amber/red classifications indicate the condition as a percentage of the whole system. The condition code correlates to the assessed timeframe need for modernization funding for the facility's core capabilities (i.e., production facilities), infrastructure systems, and support facilities, as shown. For example, if 50% of the rectangle is red it means that 50% of the whole system requires modernization funding in the next 1-5 years to address the deficiency. If the other 50% is green, then the remaining part of the whole system is operating well and will not require funding in the next 10 years.

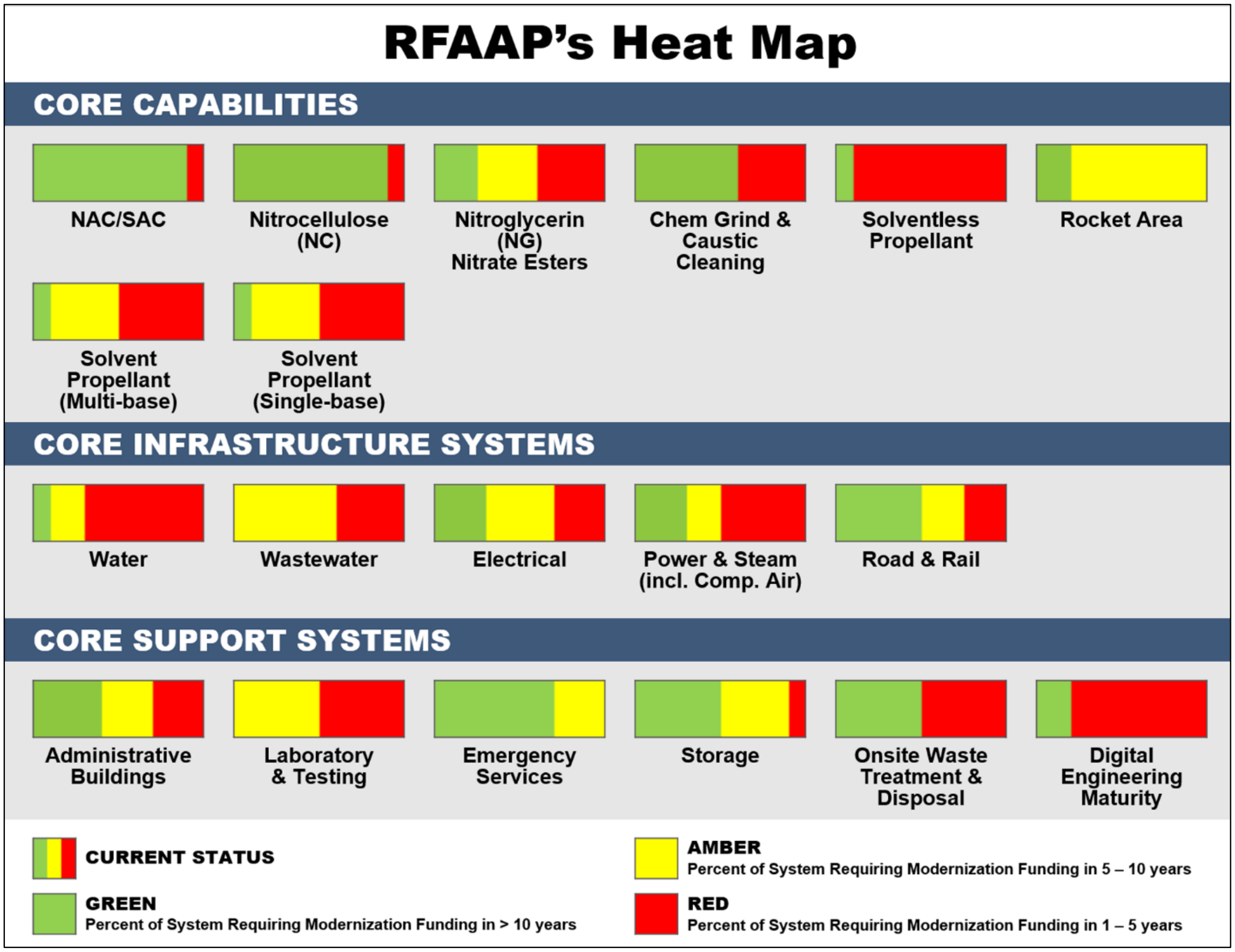


FIGURE 17. RFAAP'S CONDITION AND MODERNIZATION STATUS

4.5.2 MODERNIZATION SUMMARY

Efforts to develop a PaTS for RFAAP have been ongoing for several years. The first phase of the effort, which concluded in late 2020, resulted in an assessment and characterization of RFAAP's production, infrastructure, and support systems. Further efforts included a major operations survey that provided an operational baseline for production capacities and analyzed future production requirements (type and quantity of products). The next phase of the effort used the results from these assessments and surveys to identify critical areas in need of modernization and lay out an approach to sustain legacy facilities until modernized facilities are designed and constructed. An analysis of sitewide QD arcs (for both legacy facilities and modernized facilities based on planned capacities) was also conducted resulting in a siting plan that considered safety, resiliency, construction phasing, and legacy operations. The siting plan aimed not only to minimize the potential impact of adjacent facilities and demonstrate compliance with Army and DOD regulations, but also to identify ways to reduce the facility footprint and the total distance product travels across the site during and after production. Other key considerations for RFAAP's master planning are to improve worker safety, reduce cost of operations (maintenance and utility costs), maintain regulatory compliance, improve the manufacturing process to enhance finished product quality and consistency, and address failing infrastructure and support systems that have exceeded their designed life expectancy and are in poor condition. Recent efforts advancing the PaTS include commencement of comprehensive evaluations of the utility and production support systems including electrical, water, wastewater, natural gas, compressed air, steam, and nitrogen. The results will inform scope development for modernization projects to address identified deficiencies.

4.5.3 MODERNIZATION PROGRESS

The following are some examples of recently completed and in-progress modernization projects for RFAAP.

Solvent Sustainment Phase I

This project was awarded in August 2020 and completed by the Operating Contractor, BAE Systems, in September 2024. The project completed necessary repairs and replacements of equipment and infrastructure to improve safety, reliability, and lifespan of the equipment and facilities in the Solvent Green-lines Area. Equipment and infrastructure upgrades include but not limited to repair/replacement of Modine heaters, heat system spares, roofs, walkways, anti-slip floors, bay walls, etc. All repairs performed with the scope of this project were for purposes of increased reliability and safety. All equipment replaced was like in kind, although the new metal detector, exhaust motor system, RotoClone, and conveyor systems underwent a brief commissioning process to confirm their operability. This project supports multi-year sustainment needs of the legacy Solvent Green-lines to ensure continuous and predictable manufacturing operations manufacturing operations until the new modernized solvent propellant comes online.

Chemical Grind Facility Construction

This project is the continuation of the Chemical Grind Design effort that was completed in 2021. The Chemical Grind Facility will replace the existing facility with better processing equipment, better air quality conditions, and less operator exposure to chemicals. The construction of the new facility is currently underway and is scheduled to be completed by October 2026.

New EWI/CWP Facility Construction

The new EWI/CWP Facility is designed to replace the existing open burning grounds for contaminated energetic waste destruction. The facility will allow contaminated energetic waste to be burned inside of an enclosed chamber allowing the off gases to be treated in an air pollution control system. This will prevent toxic gases from entering the atmosphere and potentially impacting the surrounding community. The construction of this facility has started and is currently estimated to be completed in 2026.

Solvent Propellant Sustainment Phase II and ECP

The solvent propellant facility has not had much modernization work completed in previous years. With the increased demand for solvent propellants, especially propellants supporting 155mm, there has been a renewed interest in increasing the capacity of the Single Base and Multi Base propellant lines. Both production lines have been awarded funding to repair/replace legacy equipment, perform much needed infrastructure upgrades and install additional equipment. This includes repairing a mixer in Building 3508, installing two (2) new mixers in Building 3608, repairing and replacing equipment in the current hydraulic process, repairing old and worn utility systems, replacing old and worn equipment, and bringing old process equipment back online. These tasks will increase the efficiency and capacity of the facility. The main challenge has been scheduling around production due to the increased demands for propellants. This project is estimated to be completed in 2026.

New Nitroglycerine Facility Design

The current nitroglycerine facility at RFAAP needs replacement. The downstream portions of the current facility, NC Premix and Slurry Paste, are operator intensive and have a lot of exposure points to energetics and hazardous chemicals. A design effort has been awarded to design a new facility that would incorporate new equipment, more automation, and reduce operator exposure. In addition, the new facility would be able to produce other nitrate ester products to support new propellant formulations at RFAAP. The current project will provide a 30% design for a new nitroglycerine facility and will be completed in 2026.

4.5.4 SUPPLEMENTAL FUNDING IMPACTS

RFAAP received a total of $93.8M in Supplemental funding in FY 2024 for one (1) project identified in Table 12 below. This project supports continued and increased propellant production requirements, including propellant for 155mm ammunition. RFAAP received a total of $125.1M in Hurricane Helene Supplemental funding in FY 2025 for twelve (12) projects identified in Table 12 below. The total for FY24 and FY25 Supplemental funding at RFAAP was $218.9M.

TABLE 12. FY 2024 SUPPLEMENTAL & FY 2025 HURRICANE HELENE SUPPLEMENTAL APPROPRIATIONS FOR RFAAP

Radford AAP: $218.9M				
Project Number	**Project Year**	**Project**	**Value ($M)**	**Est. Comp. Date**
1	2024	Solvent Propellant Facility Sustainment Phase 2	**$93.8**	May 2026
2	2025	Building 470 - Bio Plant	**$36.7**	Dec 2027
3	2025	Warehouses (River/Butler)	**$5.6**	Jan 2028
4	2025	Infrastructure/T8/Misc	**$3.1**	Apr 2026
5	2025	Open Burning Grounds	**$0.9**	Jun 2027
6	2025	Rolled Powder	**$4.1**	Dec 2026
7	2025	Water Building (407/408/409)	**$10.1**	Feb 2027
8	2025	Building 5511 R&D	**$0.9**	May 2027
9	2025	River Garage	**$3.4**	Apr 2026
10	2025	Supplies and Stored Equipment	**$16.4**	Jun 2027
11	2025	Relocate Warehouses	**$40.4**	Dec 2026
12	2025	Relocate Misc Buildings (T8/River Garage)	**$1.2**	Jan 2028
13	2025	Relocate Building 5511 R&D	**$2.3**	Mar 2027
	Total ($M):	**$218.9**		

4.5.5 ADVANCES IN AUTOMATION AND EMERGING TECHNOLOGY SOLUTIONS

Recent usage of emerging technologies at RFAAP that support the overall Master Planning efforts include 3D LiDAR (Light Detection and Ranging) scanning to generate drawings and models of older facilities where only paper drawings currently exist. Additionally, this technology supports updating drawings post-modernization and sustainment. Scanning and producing digital drawings versus field-measuring and manually creating digital drawings greatly reduces time and cost, as well as improves accuracy. This also supports DE maturity, as discussed below. Specifically, this technology was used to create drawings and models of the current NG-1 Nitration Building for use in the design phases of the NG-3 modernization project as no as-builts were available digitally.

RFAAP was assessed using the DE Maturity Model and this baseline assessment resulted in an average value of 1.48 on the scale of 0 to 5 with 0 indicating a fully analog facility and 5 indicating a fully intelligent DEE. A fair degree of automation exists and is complemented with added digital controls and monitors using either Siemens PCS7 or Allen-Bradley control system software as well as remote accessibility. Beneficially, RFAAP collects and retains digital data on critical product and process parameters; many inspections are using continuous variable measurements and recorded digitally. Lastly, items produced at RFAAP are tracked by batch allowing any data measured to be attributed. The maturation of RFAAP from a digital perspective should include continued modernization to bring all lines to a comparable digital and cybersecurity standard.

Present day cyber practices, such as requiring an "air gap," should be reconsidered as there is a substantial benefit in integration of such systems on a secured closed network.

4.5.6 Modernization Requirements FY 2025 through FY 2031

The Army has planned nearly $1.6B in funding for modernization projects at RFAAP during FYs 2025–2031. A complete project list, including project description and funding information, is included in Appendix A.

While the PaTS discussed in Section 4.5.2 above sets the objectives and priorities for modernization at RFAAP, the execution strategy for these projects is impacted by changing ammunition requirements due to international conflicts, the business space RFAAP currently occupies, and most recently by the competition for the next RFAAP facility contract that commenced in August 2024. The acquisition process will continue through the anticipated award in FY 2027. To accommodate this timeline and the transition from the current contract to the next, modernization projects currently planned for execution over these years needed to be replanned. This resulted in substantial changes in the project list for RFAAP from last year's AAP Modernization Plan Report.

4.6 Scranton Army Ammunition Plant

Core Capabilities

SCAAP's core capability is 155mm artillery metal parts manufacturing. Key processes include long-stroke, high-tonnage, forging; high-volume production; and rotating-band welding.

FIGURE 18. SCAAP PRODUCTS

Figure 19 provides an updated plant overview based upon facility assessments and engineering studies, illustrating areas of risk and opportunities for investment. They depict the current condition of each of the capabilities and subsystems within the facility. The green/amber/red classifications indicate the condition as a percentage of the whole system. The condition code correlates to the assessed timeframe need for modernization funding for the facility's core capabilities (i.e., production facilities), infrastructure systems, and support facilities, as shown. For example, if 50% of the rectangle is red it means that 50% of the whole system requires modernization funding in the next 1-5 years to address the deficiency. If the other 50% is green, then the remaining part of the whole system is operating well and will not require funding in the next 10 years.

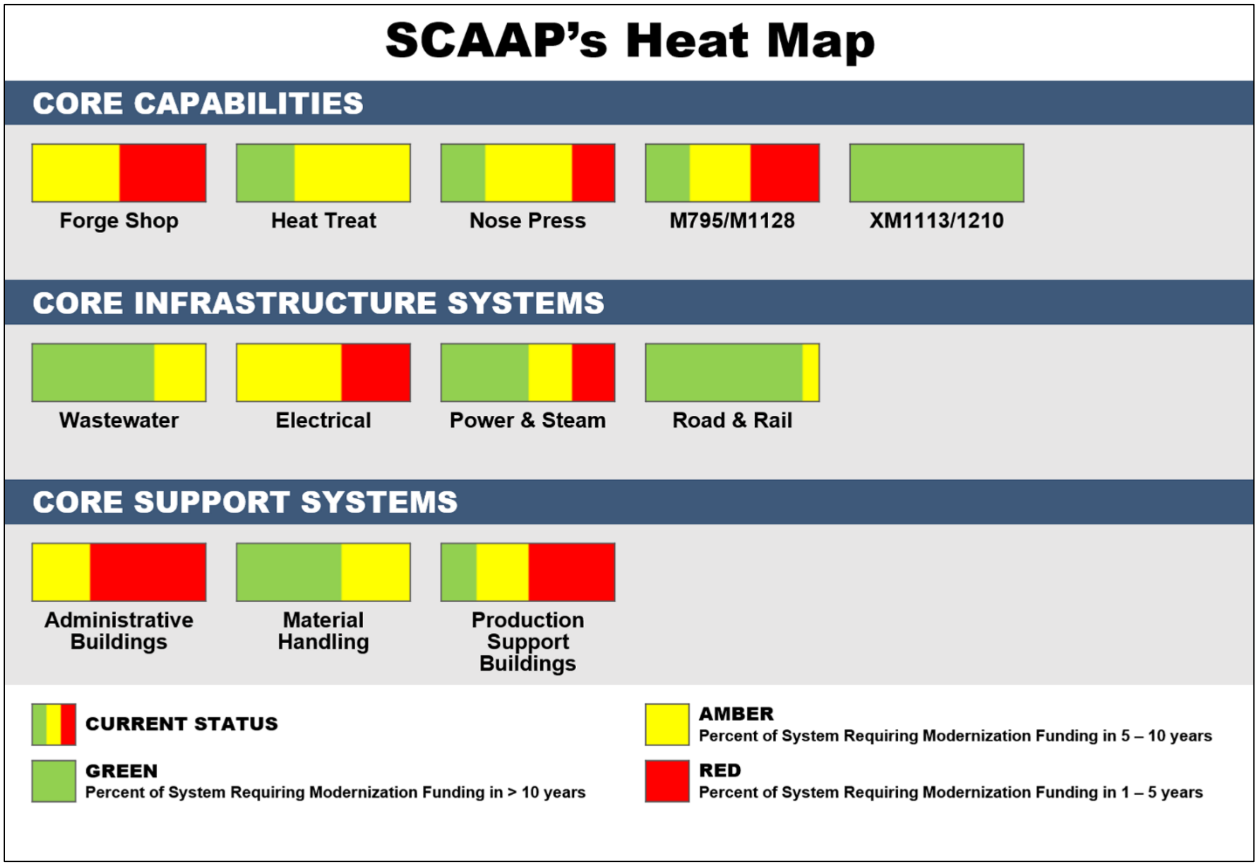


FIGURE 19. SCAAP'S CONDITION AND MODERNIZATION STATUS

4.6.1 MODERNIZATION SUMMARY

Modernization efforts at SCAAP have been driven by the dramatic surge in demand for 155mm artillery metal projectiles. SCAAP has a Vision Plan that was developed in 2017 and calls for facility and equipment optimization and modernization. A March 2023 SCAAP Installation Energy and Water Plan provides an overview of the critical mission facilities at SCAAP, the energy and water security baseline and deficiencies, and strategies to achieve installation goals and address identified deficiencies. Efforts to develop a PaTS for SCAAP began last year. The first phase of the effort, which concluded in July 2024, aimed to document general knowledge and understanding of the facility and its operations to establish a consistent basis of knowledge of production areas, infrastructure, and support systems for future in-depth characterization efforts. It also identified necessary short-range planning, construction, and project efforts that will ensure SCAAP will be positioned to successfully fulfil mission objectives. In addition, a UAV mission was completed to provide data to assist in the development of a solution concerning the retaining walls at the perimeter of the property. The next phase of the PaTS will build upon this foundation and further inform medium- and long-range project development.

4.6.2 Modernization Progress

The following are some examples of recently completed modernization projects for SCAAP.

SCAAP Increased Capacity Conceptual Design

Significant metal parts capacity increases are required at SCAAP to meet the increased demand for 155mm projectiles. In 2023, a concept design was completed to enable a tripling of the production capacity of M795/M1128 shells per month. As part of this effort, a baseline analysis of existing manufacturing processes, equipment, and capabilities was conducted that included a 3D laser scan of the existing facilities and equipment. After the baseline analysis, other technology and equipment options were developed to evaluate options to obtain the desired rate of production at SCAAP. Where new technologies were recommended, machine vendors performed manufacturing simulations and provided time studies that were used to select equipment quantities that would meet the projected production rates and optimize floor space. Although priority was placed on selecting commercial-off-the-shelf (COTS) machines, there were a few instances where a suitable COTS machine was not available, and a custom solution was conceptually designed. This strategy was documented in a digital model of the proposed production process, various concept design deliverables, and a design charrette; it continues to be matured as the design progresses.

SCAAP is in a downtown area and is a relatively small facility with limited parking. From a facility-perspective additional production space, finished goods storage, and personnel accommodations will be required to increase M795/M1128 capacity. From an equipment-perspective, additional rough turn/band weld and finish turn/test capacity is needed to meet both increased requirements as well as enable the transition from M795 to M1128 rounds.

Additional equipment will also need to be procured and planned for, such as new saw lines, forging presses, washing and graphitizing equipment, nosing press, and automated work-in-progress stackers. Required facility and infrastructure improvements (e.g., parking, storage, etc.) are being concurrently planned to accommodate the additional equipment and minimize the potential for production interruption. JPEO A&A and PL JS are moving forward with the selected design option to improve the SCAAP facility and increase M795/M1128 production capacity.

Erie I and Bliss III Press Evaluations

155mm artillery metal part production quality starts with reliable press systems that operate under tremendous stress to shape the raw steel into the initial, rough projectile shape. SCAAP has two press lines (Erie I and Bliss III) capable of producing modern 155mm artillery metal parts each consisting of three individual hydraulic presses that were installed in 1970. This project conducted an in-depth evaluation on each of these presses to identify deficiencies such as misalignment, cracks, and leaks along with remedies that are now being implemented.

Transformer Repair

In September 2022, SCAAP experienced an electrical malfunction that damaged a 5,000 kilovolt-amp transformer along with the induction heater it was powering. The induction heater is used to prepare raw steel for the 155mm artillery metal parts pressing operations. This project was rapidly implemented to fund repairs of the damaged equipment along with temporary solutions to maintain production capacity. This project is an example of the type of unplanned, urgent need experienced in active production facilities.

4.6.3 Supplemental Funding Impacts

SCAAP received a total of $101M in Supplemental funding in FY 2024 for the four (4) projects identified in Table 13. These projects directly provided the required equipment and infrastructure to increase capacity for 155mm ammunition metal parts production. Relocation of M107 and 5"/54 equipment was necessary to make room for the 155mm metal parts production equipment.

Table 13. FY 2024 Supplemental Appropriation for SCAAP

Scranton AAP: $101M			
Project Number	**Project**	**Value ($M)**	**Est. Comp. Date**
1	155mm Metal Parts Expansion at Scranton AAP Base	**$49.9**	Sep 2025
2	155mm Metal Parts Expansion at Scranton AAP – Finished Goods (Option 4)	**$17.7**	Sep 2025
3	155mm Metal Parts Expansion at Scranton AAP – Machine Shop (Option 5)	**$8.4**	Sep 2025
4	155mm Metal Parts Expansion at Hanover Facility – Hanover Press (Option 7)	**$25.0**	Sep 2025
	TOTAL ($M):	**$101.0**	

4.6.4 Advances in Automation and Emerging Technology Solutions

Over the next five (5) years, several key modernization projects at SCAAP will include increased automation to ensure operator safety, minimize workforce shortages, and allow for more efficient production. For example, both the M1128 and the XM1113/1210 production lines will utilize some automated material handling and inspection.

SCAAP was assessed using the DE Maturity Model, and this baseline assessment resulted in an average value of 0.52 on the scale of 0 to 5, with 0 indicating a fully analog facility and 5 indicating a fully intelligent DEE. This score translates to a mostly analog facility. Currently, SCAAP is deficient in digital data capture, product serialization, and modern and efficient inspection methods (go/no-go gauges and manual paper filings or visual indicators). The maturation of SCAAP from a digital perspective should start with foundational activities associated with digitizing and standardizing key processes. With the renewed focus on cannon artillery and M795 in particularly, digital maturation can occur concurrent with any efforts to modernize the facility and increase its capacity. New equipment should be procured with the capability of data measurement, collection, and interoperability as a key consideration. Finally, infrastructure to collect and share that information in a secured environment should be an integral part of any work to upgrade or amend the current facility.

4.6.5 MODERNIZATION REQUIREMENTS FY 2025 THROUGH FY 2031

The Army has planned more than $270M in funding for modernization projects at SCAAP during FYs 2025–2031. A complete project list, including project description and funding information, is included in Appendix A.

5.0 EXTENDED PLANNING PERIOD

This report captures out-year funding requirements totaling over $7.3B with some projects valued between $150M–$800M. Table 4 in Section 3.2 identifies some higher valued projects planned for in the EPP. The consolidated EPP totals by facility is presented in Table 14 in Section 8.0 below. The critical projects, including project description and funding information, are included in Appendix A.

6.0 PRODUCTION CAPACITY AND SCALABILITY

The scalability and maximum capacity data for HSAAP, IAAAP, LCAAP, QCCCF, RFAAP, and SCAAP, are addressed in Appendix D. Production capacity and scalability are planning factors that are considered across the current six objectives. The variability in the maximum capacity of specific capabilities and commodities is designed to balance investments across the multiple GOCO AAPs and maximize production capabilities of all munitions.

7.0 ANALYSIS OF CONSTRUCTION FOR NEW, MODERN FACILITIES AS AN ADDITIONAL ALTERNATIVE

Section 2.3 describes the capability focused modernization planning approach which includes all alternatives, to include new, modern facilities. The detailed process considers critical production capabilities that support the National Defense Strategy, the National Defense Industrial Strategy, and the SMCA IBSP to capture the required capabilities as projects that may result in expansion of current facilities or new, modern facilities.

8.0 SUMMARY

This AAP Modernization Plan describes the methodology and approach to identifying modernization projects for the GOCO AAPs, as well as the execution plan from FYs 2025–2031. Table 3 in Section 3.2 provides a summary of the PAA PIF funding based on current and future requirements. The investment requirements in Table 3 reflect the base budget as well as additional funding. Table 14 also further summarizes individual GOCO funding by fiscal year. The values represent dollars spent during FYs 2003–2024 and planned for FYs 2025–2031. Also provided are the requirements that exceed projected available funding. Additional resourcing could potentially move out-year projects to the left.

TABLE 14. GOCO AAP MODERNIZATION REQUIREMENTS

	Totals ($M)								
GOCO Ammunition Facility	**FYs 2003-2024 ($M)**	**FY25**	**FY26**	**FY27**	**FY28**	**FY29**	**FY30**	**FY31**	**FY32+**
HSAAP (TN)	3,007.636	114.860	184.800	32.844	139.400	220.000	146.678	108.000	897.300
IAAAP (IA)	1,084.804	280.650	22.309	80.350	96.820	27.900	95.230	54.170	1,070.580
LCAAP (MO)	1,594.127	92.000	476.745	135.020	194.916	105.450	141.426	116.844	2,678.144
MLAAP (TN)	21.844	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
QCCCF(IL)	2.205	0.000	0.000	6.500	0.000	4.750	0.000	0.000	0.000
RFAAP (VA)	1,842.711	146.640	98.600	908.025	80.000	210.558	159.700	131.090	2,519.180
SCAAP (PA)	680.963	0.000	4.500	27.956	39.960	5.250	29.365	163.145	154.863
COCO	1,212.856	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Planning & Design, MIF/LIF, Engineering Support, Major Maintenance	715.162	130.356	82.084	92.607	113.832	96.490	97.990	99.286	-
Total	**10,162.308**	**764.466**	**869.038**	**1,283.302**	**664.928**	**670.398**	**670.389**	**672.535**	**7,320.067**

As of January 24, 2025, the GOCO AAP modernization program has 123 projects in execution with a total project cost of over $4.4B. A project listing by GOCO AAP is provided in Appendix C. While this is a substantial number of projects that, upon completion, will have positive impacts, there are projects identified in future years that remain unfunded due to the limited annual budget.

The ammunition production facilities serve a vital role in protecting the security of our nation. Modernized ammunition production installations are necessary to ensure safe, reliable, and effective ammunition operations. Being able to produce at scale serves as a worldwide deterrent against would be adversaries towards the United States and its allies. Many of the facilities within each GOCO AAP have simply aged beyond the point where maintenance or upgrades can address deficiencies, making construction of new facilities essential for continued ammunition production.

The AAP Modernization Plan is an annual investment strategy continually reassessed to address shifting priorities at the AAPs, construction cost variances, and to support national defense initiatives. The Plan is adjusted when additional funding is made available. The strategy assumes approximately $700M per year of PAA funding; however, several projects could be moved to the left if additional resourcing becomes available.

9.0 ABBREVIATION AND ACRONYM LIST

AAP	Army Ammunition Plant
AI	Artificial Intelligence
AMP	Advanced Multi-Purpose
ANSOL	Ammonium Nitrate Solution
AO	American Ordnance, LLC
AoA	Analysis of Alternatives
AR	Army Regulation
ASA(ALT)	Assistant Secretary of the Army for Acquisition, Logistics, and Technology
ASA(FM&C)	Assistant Secretary of the Army for Financial Management and Comptroller
ATP	Army Technical Publication
B	Billion
BAE OSI	BAE Systems Ordnance Systems, Inc.
BES	Budget Estimate Submission
BLIN	Budget Line-Item Number
CAA	Clean Air Act
COTS	Commercial-Off-the-Shelf
CFR	Code of Federal Regulation
DAU	Defense Acquisition University
DDESB	Department of Defense Explosives Safety Board
D&F	Determination and Findings
DE	Digital Engineering
DEE	Digital Engineering Ecosystem
Demil	Demilitarization
DEVCOM AC	U.S. Army Combat Capabilities Development Command Armaments Center
DNAN	Dinitroanisole
DMDNB	2,3-dimethyl-2,3-dinitrobutane
DOD	Department of Defense
DOE	Department of Energy
EA	Economic Analysis
EATOOL	Economic Analysis Tool
EPA	Environmental Protection Agency
EPP	Extended Planning Period
FRP	Full Rate Production
FY	Fiscal Year
GD-OTS	General Dynamics–Ordnance and Tactical Systems
GOCO	Government Owned, Contractor Operated
HASC	House Armed Services Committee
HE	High Explosive
HSAAP	Holston Army Ammunition Plant
HMX	High-Melt Explosive
HVAC	Heating, Ventilation, and Air Conditioning
IAAAP	Iowa Army Ammunition Plant
IBSP	Industrial Base Strategic Plan
IFC	Issued for Construction
IFD	Issue for Design
IM	Insensitive Munitions
IMX	Insensitive Munitions Explosives
IPT(s)	Integrated Project Team(s)
IWWTF/P	Industrial Wastewater Treatment Facility/Plant
JPEO A&A	Joint Program Executive Office Armaments & Ammunition
LAP	Load, Assemble, and Pack
LCAAP	Lake City Army Ammunition Plant
LCPP	Life Cycle Pilot Process
LIF	Layaway of Industrial Facilities
M	Million
MICLIC	Mine Clearing Line Charge
MIF	Maintenance of Inactive Facilities
ML	Machine Learning
MLAAP	Milan Army Ammunition Plant
MMBtu	Million British Thermal Units
NC	Nitrocellulose
NDAA	National Defense Authorization Act
NEPA	Northeastern Pennsylvania
NG	Nitroglycerin
NGSW-A	Next Generation Squad Weapon–Ammunition
NPDES	National Pollutant Discharge Elimination System
NSWC	Naval Surface Warfare Center
NTIB	National Technology and Industrial Base
NTO	Nitrotriazolone
NQ	Nitroguanidine
OB/OD	Open Burning/Open Detonation
OIB	Organic Industrial Base
OSD	Office of the Secretary of Defense
OWL	One-Way Luminescence
PAA	Procurement of Ammunition, Army
PBS	Production Base Support
PL JS	Project Lead Joint Services
PIF	Provision of Industrial Facilities
PLC	Programmable Logic Control
QCCCF	Quad City Cartridge Case Facility
QD	Quantity Distance
QWE	Quality of the Work Environment
RCRA	Resource Conservation and Recovery Act
RDT&E	Research, Development, Test and Evaluation
RDX	Research Development Explosive
RFAAP	Radford Army Ammunition Plant
ROM	Rough Order of Magnitude
RRA	Reduced Range Artillery
SCAAP	Scranton Army Ammunition Plant
SMCA	Single Manager for Conventional Ammunition
Sq. Ft.	Square Foot
TATB	Triaminotrinitrobenzene
TDP	Technical Data Package
TNT	Trinitrotoluene
TOA	Total Obligation Authority
UFC	Unified Facilities Criteria
U.S.	United States
USC	U.S. Code

10.0 REFERENCES AND AMMUNITION AUTHORITIES

- AR 700-90 (Army Industrial Base Process)
- 10 USC § 3406 Task and Delivery Order Contracts: Orders
- 10 USC § 4811 National Security Strategy for National Technology and Industrial Base
- 10 USC § 4862 Requirement to Buy Certain Articles from American Sources: Exceptions
- 10 USC § 4863 Requirement to Buy Strategic Materials Critical to National Security from American Sources: Exceptions
- 10 USC § 4881 (1950, Defense Industrial Reserve Act)
- 10 USC § 3103 Civilian Management of the Defense Acquisition System
- 10 USC § 7542 (The Stratton Amendment)
- 10 USC § 7532 Factories and Arsenals: manufacture at; abolition of (1920, Arsenal Act)
- 10 USC § 7016 Assistant Secretaries of the Army
- DOD Directive 4275.5, Acquisition and Management of Industrial Resources (March 2005)
- DOD Instruction 5160.68, SMCA
- DODI 8500.01
- DODI 8510.01
- NIST SP 800-37
- NIST SP 800-53
- NIST SP 800-53a
- NIST SP 800-82
- Long-range Facilities and Construction Planning at Army Ammunition Plants and Arsenals, February 2015
- Public Law 99-661 § 317 Prohibition of Contracts for the Performance of Certain Army Ammunition Activities
- Public Law 105-261, Section 806, Strom Thurmond FY 1999 National Defense Authorization Act
- A Roadmap for Automation at Army Ammunition Plants, SANDIA, October 2021
- SMCA GOCO AAP 10-Year Modernization Plan, December 2020
- GOCO 15-Year Ammunition Facility Modernization & Technology Implementation, Stevens Institute of Technology, November 2021
- United States Army Ammunition Industrial Base Strategic Plan (IBSP): 2025, SMCA, February 2016

11.0 APPENDICES

11.1 Appendix A: PBS Modernization Requirements FY25-31 and EPP by Facility

The tables in Sections 11.1.1 through 11.1.9 indicate funding types as defined below:

- Base–Refers to the base budget submitted in the Budget Estimate Submission (BES).
- Supp-Refers to Supplemental funding received.

Projects are categorized based on the type of project and their impacts on the facility. The project categories are defined below:

- P–Production
- I–Infrastructure
- E–Environment (Regulatory Compliance)
- S–Safety

11.1.1 HSAAP PBS Modernization Requirements FY25-31

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
1	Drying Energetics Building & Automatic Fire Suppression System: Construction ***Supports: IMX, RDX, HMX Production and Facility Operations***	25	P	Extends a filtered water supply line and replaces existing fire suppression with an automatic fire suppression system that reacts to a rapid rise in heat or a fire without any human intervention.	5.200	Base
2	Consolidated Compressed Air Facility: Construction ***Supports: IMX, RDX, HMX Production and Facility Operations***	25	P	Upgrades legacy compressed air generation facility; replaces aged equipment and infrastructure. Includes smart distribution planning and consolidation, where appropriate.	13.300	Base
3	Secondary Process River Water Intake Facility: Construction ***Supports: IMX, RDX, HMX Production and Facility Operations***	25	I,P	Replaces the current intake facility that is at its end-of-life and not compliant with Rule 316(b) of the Clean Water Act. New facility will be fully compliant with the requirements for screen velocity. It will intake river water and treat the raw water for operations.	88.060	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
4	PAX-3 Production G Building Modernization & Construction ***Supports: PAX-3 Production Operations***	25	P	Consolidates PAX-3 (explosive for grenades, 120mm Tank warhead) into one semi-automated process building and provides production-level volume.	8.300	Base
5	River Water Outfall 40 and Reservoir Upgrade: Construction ***Supports: IMX, RDX, HMX Production and Facility Operations***	26	E,I,P	Redirects the pathway of Outfall 40 from feeding into the Arnott Branch of the Holston River and upgrades the River Water Reservoir system. This project is necessary to maintain compliance with the Clean Water Act.	24.000	Base
6	Wastewater General Upgrades: Construction ***Supports: IMX, RDX, HMX Production and Facility Operations; Production Wastewater Treatment***	26	E,I,P	Project upgrades the IWWTF system and will consider point source treatment, if determined necessary during project planning.	18.000	Base
7	Facility Electrical Upgrades (Sitewide Reliability) ***Supports: IMX, RDX, HMX Production and Facility Operations***	26	I,P,S	General upgrades to plant electrical systems, including electrical distribution. Will mitigate risks of production interruption due to distribution system failures and promote employee safety.	11.400	Base
8	Lacquer Prep Facility Sustainment ***Supports: IMX, RDX, HMX Production and Facility Operations***	26	P,I,E,S	This project will provide necessary improvements and refurbishment to help sustain the existing facility until the new one is constructed.	1.600	Base
9	Sanitary Sewer Infrastructure Upgrades: Construction ***Supports: IMX, RDX, HMX Production and Facility Operations; Production Wastewater Treatment***	26	E,I,P	This project will upgrade the existing sanitary sewer collection system and connections to the IWWTP system.	11.000	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
10	Treatment Technology Alternatives to Open Burning ***Supports: Explosive Production Operations; Production Waste Treatment***	26	E,P	Project will equip the lab that supports Explosive Decomposition Chamber and Flashing Furnace operations. Project will also perform stack testing as required by environmental regulations.	3.000	Base
11	IWWTF (Hydrogen Sulfide Issue) Construction & Modernization ***Supports: IMX, RDX, HMX Production and Facility Operations; Production Wastewater Treatment***	26	S,E,P	Supports construction of new facility to treat and eliminate hydrogen sulfide in Industrial Wastewater treatment facility operations. Improves personnel safety, reduces corrosion to infrastructure, and ensures regulatory compliance.	28.800	Base
12	Consolidated Compressed Air Facility: Construction ***Supports: IMX, RDX, HMX Production and Facility Operations***	26	P,I	Replaces the legacy compressed air generation facility. Includes smart distribution planning and consolidation, where appropriate.	24.000	Base
13	RDX Wastewater—Pump Station Upgrades: Construction ***Supports: RDX, Production Operations; Production Wastewater Treatment***	26	E,I,P	Upgrades aged wastewater treatment pumping systems with new explosive service pumps, new electrical room, new switchgear, new controls, and piping improvements.	55.200	Base
14	G-10 Process Improvements ***Supports: Triaminotrinitrobenzene (TATB) Operations; TATB Capacity Increase***	26	E,S,P	Improve emissions controls to enable additional TATB explosives production for DOD and DOE while reducing safety risks. The project also addresses some of the building infrastructure issues to improve production continuity.	7.800	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
15	Water Distribution Modernization ***Supports: IMX, RDX, HMX Production and Facility Operations***	27	P,I,E	Project will install new main river water and filtered water distribution headers throughout the facility based on the results of the water distribution study (separately funded).	16.200	Base
16	Sewer System Line Replacement Phase III ***Supports: IMX, RDX, HMX Production and Facility Operations***	27	P,E	Supports additional phase to repair sewer lines and upgrade infrastructure that is deteriorating or at end-of-life.	13.644	Base
17	Treatment Technology Alternatives to Open Burning ***Supports: Explosive Production Operations; Production Waste Treatment***	27	E,P	Project will perform stack testing required by regulations and will support the transition to operations.	3.000	Base
18	New Lacquer Prep Facility: Construction ***Supports: IMX, RDX, HMX Production and Facility Operations***	28	P,I,E,S	Project will construct a new Lacquer Preparation Facility to combine solvents and feedstock into dissolver tanks for use in PBX production. The new facility will incorporate automation and will significantly reduce process and product variability. Modernization of the existing facility was not an option as it is critical for production and could not be taken offline to support any upgrade efforts.	97.200	Base
19	G-10 Toluene Emissions and Process Improvements Phase III ***Supports: Triaminotrinitrobenzene (TATB) Operations; TATB Capacity Increase***	28	E,S,P	Supports further modifications to the Trinitrobenzene (TATB) processing equipment and production operations at HSAAP; Phase III represents the maintainability and reliability of previous investments to increase capacity.	16.000	Base
20	New Drumming Packaging Facility: Construction ***Supports: Finished Product (IMX, RDX, HMX) Packaging Operations***	28	P,I	Upgrades legacy packaging capability for finished explosive materials through improvements to process equipment and process optimization.	10.000	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
21	Strong Nitric Acid Tank Farm: Construction ***Supports: Acid Processing and IMX, RDX, HMX Production Operations***	28	P,I,E	Upgrades and improves legacy strong nitric acid capability through replacement of aged tanks and associated support systems.	5.000	Base
22	Dock Deluge (Explosives Packaging/Shipping) Construction ***Supports: Safety Compliance Measure for IMX, RDX, HMX Packaging and Shipping Operations***	28	S,P	Adds a deluge system for improved safety systems in explosive packaging/shipping facilities.	4.700	Base
23	A2B Enclose Platform on Second Level: Construction ***Supports: Safety Compliance Measure for IMX, RDX, HMX Packaging and Shipping Operations***	28	S,P	Encloses a 1,430 sq. ft. area on the second level of the A2B facility, improving safety conditions.	4.400	Base
24	E-4 Boxway Relocation (Utility/Distribution System Protection): Construction ***Supports: G-4 Material Supply Access for RDX, HMX Recrystallization Operations***	28	I,E,P	Relocates the active boxway, which is critical for material supply to G-4, demolishes the building, and returns the site to brown-field status. Building E-4 is unused but active due to a functioning boxway passing through. The building is failing and creating a safety issue.	2.100	Base
25	New Filtered Water Plant: Construction ***Supports: IMX, RDX, HMX Production Operations***	29	I,P	Project will construct a new filtered water plant to support operations at HSAAP.	120.000	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
26	Pump House and River Intake (Bldg. 201) Phase I: Design/Construction ***Supports: IMX, RDX, HMX Production Operations***	29	P,E,S,I	Constructs structural and operational upgrades to the 201 Pump House, including the installation of 10 new pumps, to improve the current, degrading Holston River water system at Area B to increase capacity for RDX expansion. Project addresses structural, safety, capacity, and reliability improvements that can be made without interrupting production operations.	35.000	Base
27	Boxway Upgrades for Spill Prevention: Construction ***Supports: Chemical Supply Transport for IMX, RDX, HMX Production Operations***	29	E,I,P	Project upgrades sections of the boxways, which contain pipelines for transportation of chemicals from one building or area to another, to provide spill prevention (and possible containment) and ensure storm water exclusion.	53.000	Base
28	ANSOL Tank Farm Upgrades ***Supports: IMX, RDX, HMX Production Operations; Production By-Product Management***	29	E,P	Upgrade the tank farm facilities used to store e, a byproduct from the production of RDX and HMX. This project would help support the production capacity expansion of RDX to 15M lbs./year, as well as treatment and/or off-site sales of ANSOL.	12.000	Base
29	G-4 & H-4 (RDX Recrystallization/Dewatering) Consolidated Modernization: Construction ***Supports: Explosive Production Operations; RDX Recrystallization and Dewatering***	30	P,I,E,S	The project will modernize these two related facilities in a single effort. The effort upgrades and improves a variety of systems within the two buildings including restroom renovations, tank farm improvements, steam system infrastructure upgrades, material handling improvements, elevator and lift upgrades, and new process instrumentation and equipment.	30.000	Base
30	Emergency Medical Services and Security Communication System Updates ***Supports: Safety & Security Compliance; IMX, RDX, HMX Production and Facility Operations***	30	S	Enhances safety and security through upgrades to plant communications systems.	6.000	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
31	Barricade: Construction ***Supports: Safety Compliance; IMX, RDX, HMX Production and Facility Operations***	30	S	Repairs HSAAP's wooden explosive protective barricades.	3.500	Base
32	Air/Steam Distribution Replacement: Construction ***Supports: Air & Steam Supply for IMX, RDX, HMX Production and Facility Operations***	30	I,P	The first phase of the project will replace the air and steam distribution systems and add more isolation valves to better isolate sections for maintenance actions and control.	34.678	Base
33	D-5 (Nitration) Operations Consolidation: Construction ***Supports: Nitration Operations; RDX, HMX Production***	30	P,I,E,S	Upgrades legacy nitration facility through replacement of aged equipment and process support systems. This facility is critical to crude RDX/HMX production and will be necessary to supplement the new nitration facility; it is part of RDX expansion efforts.	26.000	Base
34	Increased Filtered Water Storage Capacity ***Supports: Production Water Supply; IMX, RDX, HMX Production***	30	S,I,P	Project will add to the current filtered water storage capacity to support the process filtered water needs of the RDX expansion, but primarily to support fire protection for the existing and planned structures.	1.500	Base
35	G-10 Process Improvements ***Supports: Triaminotrinitrobenzene (TATB) Operations; Agile Production Operations for IMX***	30	E,P	Improve emissions controls to enable additional TATB explosives production for DOD and DOE while reducing safety risks. The project also addresses some of the building infrastructure issues to improve production continuity.	25.000	Base
36	Wastewater Upgrades: Construction ***Supports: IMX, RDX, HMX Production and Facility Operations; Production Wastewater Treatment***	30	I,E,P	Project designs and constructs upgrades the IWWTF system to maintain effluent compliance and reliable operations. Also considers use of point source treatment, where required.	20.000	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
37	HMX Production Facility: Construction ***Supports: HMX Production and Facility Operations***	31	P	Project will maintain and enhance HMX production capabilities, by modernizing and replacing equipment and facilities critical in the HMX production process.	62.000	Base
38	SNA Tank Farm & Pipeline Upgrades Phase II: Construction ***Supports: Strong Nitric Acid Operations; IMX, RDX, HMX Production***	31	P,E,S	Upgrades and improves legacy strong nitric acid (SNA) capability through replacement of aged tanks, pipelines, and associated support systems.	12.000	Base
39	Facility Electrical Upgrades Phase II ***Supports: Electricity Supply for IMX, RDX, HMX Production and Facility Operations***	31	I,P	General upgrades to plant electrical systems, including electrical distribution. Will mitigate risks of production interruption due to distribution system failures and promote employee safety.	12.000	Base
40	Railroad Track Improvements Phase I (Nitric Rail System) ***Supports: Strong Nitric Acid and Glacial Acetic Acid Transport for Explosive Production Operations***	31	S,P	Project will improve safety and correct deficiencies in the 12,850 miles of Mission Critical Essential railroad track that supports Ammonia, SNA, and Glacial Acetic Acid transport within the site.	14.000	Base
41	Critical Spare Parts Phase III ***Supports: IMX, RDX, HMX Production and Facility Operations***	31	P	Critical spares are needed for a number of equipment and infrastructure items since HSAAP does not have enough redundant capability to meet an acceptable level of production in the event of a failure. They are typically long lead time items and/or items that it would be better to replaced instead of repair.	8.000	Base

11.1.2 HSAAP Strategic PBS Modernization Requirements EPP

Line No.	PBS Project Title	Category	Project Description	Project Value ($M)
1	Flexible Explosive Production Facility	P	Design and construction of a new flexible facility that will combine operations that currently take place in four (4) separate buildings into a single	435-535

Line No.	PBS Project Title	Category	Project Description	Project Value ($M)
	Supports: IMX, CL-20, explosives Production Operations		facility (nitration-dewatering), with the current focus on CL-20. The project will also design, and construct waste acid processing/recycling and wastewater treatment facilities required for CL-20 production.	
2	ANSOL Thermal Destruction: Construction ***Supports: RDX, HMX Production Operations; Production By-Product Management***	P,E	Project will construct an onsite thermal treatment system for the destruction of ANSOL that is generated as a by-product of RDX/HMX explosives production. This mitigates the reliance on off-site disposal for ANSOL that is not commercially sold and supports meeting RDX/HMX capacity requirements.	150-250

11.1.3 IAAAP PBS Modernization Requirements FY25-31

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
1	Future Artillery Complex, Phase II Construction ***Supports: 155mm Artillery Production***	25	P	Develops a new, state-of-the-art, flexible production line for artillery. This line will have enhanced safety, automation, flexibility, efficiency, and expandability.	280.65	Base
2	Line 3A Sustainment–Phase III ***Supports: 155mm Artillery Production***	26	P,S	This project sustains the existing Line 3A until the new FAC is constructed and commissioned. Project efforts will include fabricating spare parts for equipment that can no longer be procured externally and addressing structural issues impacting production, like leaking roofs.	4.728	Base
3	Tank Line (Line 2) Sustainment Phase I ***Supports: 120mm Production***	26	P,I	Provides necessary upgrades to sustain the Tank Line. This project specifically will relocate the isostatic press control station and explosive powder pre-heat operations for 120mm AMP currently residing in Building 1-63, enabling compliance with ESSP requirements and removing the need for the current Deviation Approval and Risk Acceptance Document (DARAD).	3.444	Base
4	Coal Elimination ***Supports: Artillery and Tank Production and Facility Operations***	26	E	Project will decommission and demolish the inefficient coal-fired power plant. Demolish the building and steam lines running from the main heating plant to each of the production lines and conduct abatement activities.	14.137	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
5	Yard F Rail Modernization ***Supports: DARAD Compliance; Energetic Material Processing***	27	S,P	Project adds lightning protection system in an area where energetic material is processed and stored; this is required to address the current DARAD in place.	1.850	Base
6	Hellfire & AMP Process Relocation ***Supports: DARAD Compliance & Construction; Hellfire and 120mm Tank AMP Production Operations***	27	S,P	Project addresses explosive site safety issues with the current production operation for Hellfire and 120mm Tank AMP by building a new facility to house these production operations. It will include four (4) bays designed to comply with current UFC and explosive safety siting requirements.	39.000	Base
7	40MM Press Relocation Phase II ***Supports: DARAD Compliance & Construction; 40mm M430A1 Grenade Production Operations***	27	S,P	Project addresses explosive site safety issues with the current production operation for 40mm M430A1 grenades by building a new facility to house these production operations. The new facility will support spit back pressing operations and include new presses, screening table, control systems, and support equipment.	33.000	Base
8	Line 3A Sustainment Phase IV ***Supports: 155mm Artillery Production***	27	P	This project sustains the existing Line 3A until the new FAC is constructed and commissioned.	6.500	Base
9	Stormwater & Erosion Controls for Production Employee Parking ***Supports: 40mm and Direct Fire (120mm AMP) Production Operations***	28	E,S,I,P	Implements erosion controls and modernizes employee parking lots that support production with road surface repair and paving in accordance with UFC-250-01. Will address grading issues and inadequate aggregate thickness and potholes that crease transportation safety issues.	26.880	Base
10	Line 3A Sustainment Phase V ***Supports: 155mm Artillery Production***	28	P	This project sustains the existing Line 3A until the new FAC is constructed and commissioned.	6.940	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
11	Critical Rail Replacement–Yard C ***Supports: TNT, M795 Finished Goods and CAT II Storage Operations***	28	P,I,S	Repairs and replaces the deficient rail infrastructure at Yard C, which is the main storage facility for items such as TNT, M795 finished goods, and CAT II items. This project will enable UFC 40-860-03 compliance with rail lengths and conditions.	31.900	Base
12	Energetics Storage Facility Sustainment Phase I ***Supports: Artillery Production and Storage; Security Compliance***	28	P,S,I	Project components range from replacement of end-of-life dock structures and stairs to design & construction of some of the structures. There is also a need, in some areas, to replace fencing to comply with the most recent security requirements.	16.500	Base
13	Non-Destructive Testing Laboratory Design & Construct ***Supports: Testing Operations for M795, Medium Caliber Ammunition, 120mm Tank Ammunition, and Future Munitions***	28	P,I	Improves non-destructive testing capabilities at IAAAP in support of M795, Medium Caliber, 120mm tank rounds, and other future munitions.	7.790	Base
14	Hellfire Computer Numerical Control (CNC) Equipment Modernization ***Supports: Hellfire Production; Increases Hellfire Capacity***	28	P	Increases Hellfire production capacity through the addition of two (2) new CNC Lathes at IAAAP to allow for higher contract quantities required by future U.S. Army programs.	2.700	Base
15	Indirect Fire Production Quality Control Process Improvements ***Supports: 155mm Artillery Production; Line 3A Quality Control Operations***	28	P	Assists with product quality investigations by installing a camera system covering the entire 3A Indirect Fire production line. The system would include all areas where materials are stored and processed. For Bldg. 3A-05-1, explosion-proof cameras would be required while all other areas would have IP6X (dust tight) cameras. Every building would get at least one NVR (Network Video Recorder) to store and catalog camera data. Camera and NVR are AXIS brand and are approved for use at IAAAP.	4.110	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
16	Telecommunications & Fiber Optic Infrastructure Upgrades ***Supports: Artillery Production and Facility Operations***	29	S,I	Assures facilities telecommunications system meets current and future telephone needs and enhances facility safety and security by providing reliable, expandable communications infrastructure and eliminating antiquated, failing communication system.	15.000	Base
17	Production Road Modernization Phase I ***Supports: Transportation and Material Movement for Production Operations***	29	P,I,S	Project addresses road repairs and improvements necessary for transportation in support of production, material receipt and transfer, and shipping.	10.000	Base
18	Design Rail, Dunnage, & Heavy Equipment Campus ***Supports: Transportation and Material Movement for Production Operations;155mm Artillery***	29	P,I	Project will design a new facility to improve rail/dunnage operations supporting production. The facility's objective is to incorporate a rail system that allows entry from both sides of the building to allow dunnage operations to be conducted indoors and passthrough rail and truck access.	2.900	Base
19	Equipment Maintenance Complex ***Supports: Consolidated Equipment Maintenance for Artillery and Mortar Production Operations***	30	P	Design and construction of a new centrally located maintenance facility for all production activities. Will combine several smaller maintenance areas into a single complex for increased efficiencies.	4.000	Base
20	Access Control Point Upgrades ***Supports: Facility Operations, Artillery and Mortar Production; Security Compliance***	30	S,I	Improves security by consolidating the two (2) site entry points into an updated single point of entry that complies with UFC 4-022-01 and relocates the Welcome Center to within the gate.	14.900	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
21	Critical Rail Replacement–Yard L & F ***Supports: Material Transport; Artillery and Mortar Production Operations***	30	S,P	Repairs and replaces the deficient rail infrastructure in Yards L & F that are crucial to product movements at IAAAP. Complies with UFC 40-860-03 (no rail shorter than 13 feet can be used on Army rail systems). Complies with UFC 40-860-03 section 7-1, Defective Rail and Remedial Actions: Replaces all the existing 90lb. rail with all new 115lb. rail. Replaces all the old wood cross and switch ties with new steel cross and switch ties. Replaces turnout (switches), crossings, culverts.	37.000	Base
22	Tank Line (Line 2) Sustainment Phase III ***Supports: 120mm Production***	30	P	Project sustains the Tank Line to support continuous production.	2.830	Base
23	Operations and Equipment Support Improvement Program Phase 1 ***Supports: Facility Operations, Artillery and Mortar Production***	30	P,E,S	Project is a recurring effort to improve fire protection, material management, and HVAC systems.	6.500	Base
24	Rail Improvement Program Phase 1 ***Supports: Material Transport; Artillery and Mortar Production***	30	P,I	Project addresses critical rail operations that support production.	10.000	Base
25	Munition Process Line Roof Improvement Program Phase 1 ***Supports: Artillery and Mortar Production; Safety and Quality Compliance***	30	P,I	Project addresses roofing repair and replacements for production and support areas.	10.000	Base
26	Munition Production Road Improvement Program Phase 2 ***Supports: Raw Material and Finished Product Transport; Artillery and Mortar Production***	30	P,I	Project addresses road repairs and improvements necessary for transportation in support of production, material receipt and transfer, and shipping.	10.000	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
27	3A Elevator Replacement ***Supports: Line 3A Operations;155mm Artillery Production***	31	P,S	Project replaces two (2) rundown material transport elevators which are critical to production at Line 3A.	3.130	Base
28	Lead Pipe Study & Remediation ***Supports: Artillery, Mortar Production & Facility Operations ; , Health & Safety Compliance***	31	E,S	Project will bring IAAAP into compliance with Maximum Contaminant Level Goals (MCLGs) for lead levels in water by executing a two-phase effort. Phase I is a study to identify areas where lead levels exceed the MCLGs. Phase II will execute the mitigation activities identified during the study.	14.720	Base
29	4B Oil & Water Tertiary Treatment Replacement ***Supports: Artillery, Mortar Production & Facility Operations; Compliance***	31	E,P	Project eliminates risk of oil contamination into the Sanitary Sewer System at Building 4B and further segregates the waste streams for treatment. This project will improve oil/water separation operations and also reduce the amount of water that is currently entering this waste stream.	1.820	Base
30	Steam & Air Supply Support Replacement ***Supports: Steam & Air Supply for Artillery, Mortar Production & Facility Operations***	31	P	Project will replace approximately 529 deteriorating support poles on steam and compressed air supply lines serving multiple production buildings.	9.500	Base
31	Cyber Security Phase II: Construction ***Supports: Artillery, Mortar Production & Facility Operations; Security Compliance***	31	P,I	Project implements planned cybersecurity measures designed to mitigate security risks.	25.000	Base

11.1.4 IAAAP Strategic PBS Modernization Requirements EPP

Line No.	PBS Project Title	Category	Project Description	Project Value ($M)
1	Direct Fire Munitions Facility (Tank Line)	P	Design and construction of a modern, efficient, capable, and flexible Dire Fire Munitions Facility to ensure operational continuity and improved	400-600

Line No.	PBS Project Title	Category	Project Description	Project Value ($M)
	Supports: 120mm and 105mm Production		production for munitions manufacturing for 120mm and 105mm tank rounds.	

11.1.5 LCAAP PBS Modernization Requirements FY25-31

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
1	7.62MM Reduced Range Ammunition Process Modernization & Construction ***Supports: 7.62mm Ammunition Production***	25	P	Supports Bldg. 2 site preparation for new equipment and procurement of new cold heading projectile processing equipment.	57	Base
2	One Way Luminescence Ammo: Facility Construction & Equipment Modernization ***Supports: Small Caliber OWL Production Operations***	25	P	Includes site preparation for Bldg. 3, procurement of tracer charging equipment, and commissioning.	35	Base
3	Energetic Waste Incinerator: Construction ***Supports: Small Caliber Ammunition Production***	26	E,P	Construction of a new complex to maintain the capability to process current scrap from conventional small caliber munitions, as well as handling additional capacity and waste streams resulting from Next Generation materials and near-term technologies transitioning to LCAAP. The use of a modern feed system design will improve the on-stream time and reliability. The new complex will ensure the ability to incinerate explosive waste material at the same high-volume levels as small caliber ammunition deliveries.	239.945	Base
4	NGSW-A Equipping ***Supports: 6.8mm Production***	26	P	Supports the additional equipping and support costs associated with bringing the NGSW-A facility online and producing.	236.800	Base
5	NGSW-A Equipping ***Supports: 6.8mm Production***	27	P	Continues to support additional equipping and support costs associated with bringing the NGSW-A facility into full rate production.	132.420	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
6	B19 Powder Storage Modernization ***Supports: Small Caliber Ammunition Production***	27	S,P	Project replaces failed doors and damaged pavement associated with Building 19 powder storage magazines. This corrects a long-standing challenge with securing the facilities and providing proper bonding for lightning protection.	2.600	Base
7	Five Grain Distribution Replacement: Small Caliber Production & Operations ***Supports: Small Caliber Ammunition Production***	28	I,P	Replaces the legacy five grain (potable water) distribution system, including piping sections, isolation valves, hydrants, and building supplies (inlets and risers) to support continued small caliber manufacturing.	41.446	Base
8	Fire Water System Replacement: Small Caliber Production & Operations ***Supports: Small Caliber Ammunition Production***	28	I,P	Replaces the legacy cast iron underground fire water system, including piping sections, isolation valves, hydrants, and building supplies (inlets and risers) to be compliant with the latest fire and safety standards for small caliber manufacturing.	40.120	Base
9	Zero Grain Distribution Replacement: Small Caliber Production ***Supports: Small Caliber Ammunition Production***	28	I,P	Replaces the legacy cast iron zero grain (production and boiler water) distribution system, including piping sections, isolation valves, hydrants, and building supplies (inlets and risers) to support continued small caliber manufacturing.	4.650	Base
10	NGSW-A Equipping ***Supports: 6.8mm Production***	28	P	Covers additional equipping and support costs associated with maintaining full production capacity for the NGSW-A facility.	108.700	Base
11	Industrial Wastewater Plant Upgrade & Distribution Replacement: Construction ***Supports: Small Caliber Ammunition Production Operations; Production***	29	I,P	Replaces critical underground industrial waste pipelines and rebuilds/expands associated pumping and metering systems.	102.470	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
	Wastewater Treatment					
12	NGSW-A Equipping ***Supports: Small Caliber Ammunition Production; 6.8mm***	29	P	Covers supplementary equipping and support costs associated with maintaining full production capacity for the NGSW-A facility.	2.980	Base
13	Modernize Links Heat Treat Furnace ***Supports: Small Caliber Ammunition Production; 5.56mm, 7.62mm, .50 Cal, 20mm, 25mm links, 6.8mm***	30	P	Replaces three (3) small caliber links heat treat furnaces, two (2) medium caliber furnaces, three (3) endothermic gas generators, and associated supporting subcomponents. This will require installation of a new heat treat system for the processing of 5.56mm, 7.62mm, .50 Cal, 20mm, and 25mm links along with their ancillary conveyor and other support systems. The new process equipment will also be capable of processing 6.8mm (NGSW-A) links.	102.610	Base
14	High Explosive Air and Video Monitoring Upgrades ***Supports: Small Caliber Ammunition Production: Product Quality Compliance***	30	P	Project will replace high-speed camera systems in B8 and B45 to prevent unnecessary retesting, the rejection of acceptable quality lots, and continually increasing maintenance costs.	0.981	Base
15	Automatic Transfer Switch Upgrade ***Supports: Small Caliber Ammunition Production; Safety Compliance***	30	I	Supports a critical safety project to replace Automatic Transfer Switches and Back Up Generator System correcting numerous long-standing deficiencies and in LCAAP backup systems which are critical for safe shutdown and life/health/safety systems during a power outage.	17.853	Base
16	Compressed Air System Improvement (B83A) ***Supports: Compressed Air Supply for Small Caliber Ammunition Production Operations***	30	P,I	Project will replace and upgrade the Air Compressor system in Building 83A to achieve and maintain the desired plant pressure to support operations and eliminate overheating issues currently experienced in summer months.	3.881	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
17	HVAC Instrument and Control Modernization (B33/34 Mix House) ***Supports: Small Caliber Ammunition Production; Product Quality Compliance***	30	E,S,P	Project will replace current HVAC supply system and controls in both Mix Houses (Buildings 33 & 34) to provide environmental conditions conducive to primer production within quality tolerances and to improve personnel safety.	2.789	Base
18	Emergency Services Expansion ***Supports: Small Caliber Ammunition Production; Safety and Security Compliance***	30	S	Project will increase personnel capacity of the Emergency Service Center, B157 by 10 people per shift. The square footage of the facility will increase by approximately 1,750 sq. ft. This will include nine (9) dorm rooms with restroom/shower area. A separate HVAC system will be provided for the new area. All other utility needs will be supported by the existing facility.	6.592	Base
19	NGSW-A Utility Connections ***Supports: 6.8mm Ammunition Production***	30	P,I	Projects make critical utility connections from the NGSW-A facility construction line of demarcation to the nearest reasonably competent connection point.	6.720	Base
20	Roof Replacement: B5, B6, B11, B12A, 19 Series ***Supports: Small Caliber Ammunition Production & Facility Operations***	31	S,P	Replaces roofs on B5, B6, B11, B12A, and the 19-series buildings.	48.749	Base
21	B1: Vacuum Powder Delivery System ***Supports: SCAMP Loading Operations; Small Caliber Ammunition Production***	31	S,P	Installs vacuum powder delivery system for SCAMP loading in B1 eliminating long standing explosives safety hazard and ergonomic risk. Effort is inclusive of all facility upgrades necessary to enable the new equipment installation.	68.095	Base

11.1.6 LCAAP Strategic PBS Modernization Requirements EPP

Line No.	PBS Project Title	Category	Project Description	Project Value ($M)
1	Modernized Metal Parts Facility **_Supports: Small Caliber Ammunition Production & Facility Operations_**	P	Project constructs a centralized metal parts production facility, improving process efficiency versus the decentralized current operations in multiple buildings across the site.	400-500
2	Consolidated Chemistry & Energetics Process Laboratory **_Supports: Small Caliber Ammunition Production & Facility Operations_**		Constructs a new Consolidated Chemistry Lab composed of two buildings (a non-energetic lab and an energetic lab) along with a Prototype Energetics Capabilities Complex. This will consolidate operations versus the current decentralized operations allowing for shared equipment opportunities and improved practices.	200-300

11.1.7 QCCCF PBS Modernization Requirements FY25-31

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
1	Batch Annealing Furnace **_Supports: Deep Draw Cartridge Fabrication Operations; 105mm HE/M1 Artillery & 105 PGU Artillery_**	27	P	Project will support the conversion of production from a continuous to a batch process with installation of a batch annealing furnace to more efficiently and effectively support the DOD's current deep draw cartridge requirements, which are lower than when the facility was originally established. The facility will be able to support both continuous and batch process operations, as required.	6.5	Base
2	Industrial Waste Final Holding Tank **_Supports: Facility Operations; Capacity Increase; Production Wastewater Treatment_**	29	E,P	Project will install a final holding tank at the end of the industrial wastewater treatment plant process to allow for testing of effluent to ensure regulatory compliance with discharge limits prior to discharge. This project is required to support batch operations and ensure discharge compliance.	4.750	Base

11.1.8 RFAAP PBS Modernization Requirements FY25-31

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
1	Water Intake Pump Upgrades, Bldg. 407 Phase 2 ***Supports: Propellant Production and Facility Operations***	25	I,E	Second phase of the project to upgrade and replace water intake pumps and associated infrastructure. Water is a critical utility for RFAAP, and the existing pumps are reaching the end of their useful life. The water intake pumps draw water from the New River and supply the filtered water, fire water, and potable water systems. Without this water supply, critical production would cease.	21.500	Base
2	Hurricane Helene Supplemental ***Supports: Propellant Production and Facility Operations***	25	P	Projects to repair and or replace production support capabilities from Hurricane Helene	125.1	Supp
3	Acid Area Tank Farm Replacement ***Supports: Propellant Production and Facility Operations***	26	P,I,E,S	This project is a full design and scope definition for the replacement of the acid farm. The acid storage tanks are used to store SNA, SSA and Oleum for use in the nitrocellulose nitration process. The spent acid (Pyro) is recovered from the nitrocellulose nitration process and re-concentrated into strong acid by the RFAAP NAC/SAC facility. Currently there are only three fully operationally SNA storage tanks to hold reintroduced acid. The loss of even one of these tanks would result in a significant loss of supply and a possible shutdown of one train of the NAC/SAC until the storage tank(s) are repaired.	19.650	Base
4	NG-3 90% Design ***Supports: Propellant Production and Facility Operations***	26	P,I,E,S	The NG3 Facility Construction is a multi-phased construction approach geared at addressing the four (4) core processes in NG manufacturing (Nitration, Solventless Slurry/Paste, Multi-base Pre-mix, and NC Dehydration). This project will bring the design for the new NG-3 facility to a 90% level.	22.750	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
5	Tie-Line Modernization ***Supports: Propellant Production and Facility Operations***	26	I,P	The objective of this program is to ensure that RFAAP has a stable and dependable connection to commercial electrical grid power. RFAAP is connected to the commercial electrical power grid by a 3.1 mile long electrical 69-kV transmission line commonly referred to as the "Tie-Line" or as the "Electrical Tie-Line". The legacy 69kV tie line is RFAAP's single utility feed, the condition of its poles and conductors are critical to RFAAP electrical reliability and plant operations. This program intends to construct a new transmission line alongside the existing line by utilizing the existing right-of-way and, once complete, the service can be transferred over to the new line with no impact to active operations. The old line and poles will then be demolished and disposed.	28.400	Base
6	Electrical Distribution System Upgrades Phase IV ***Supports: Propellant Production and Facility Operations***	26	I,P	Provides upgrades and modernization of the electrical distribution system supporting the solvent propellant facility. This project will establish a dependable 69 kV electrical distribution loop to support the electrical requirements of RFAAP. This electrical distribution loop (~6 miles long) consist of aerial transmission lines that connect a series of substations which feed all areas of the facility. The substations step the primary loop 69kV power down to 2.4 kV and 12.4kV which then feeds power to transformers at the building and/or process level.	27.800	Base
7	Solvent Propellant Manufacturing Facility: Construction ***Supports: Solvent Propellant Production; Javelin, Hydra 70, & Hellfire***	27	P,I,E,S	This project will finalize the design for the new Solvent propellant manufacturing facility. It will also cover construction, commissioning, and transition to production of the new facility, modernizing the Single Base, Multi Base, and Finishing area operations.	830.725	Base
8	Bioplant Modernization–Phase I 90% Design ***Supports: Propellant Production and Facility Operations***	27	E,P	Project will design a new Bioplant which receives flow from the industrial collection system and treats the energetic-laden wastewater utilizing RBCs before discharging to Outfall 029. The discharged effluent is regularly monitored by VDEQ and prone to compliance issues increasing the likelihood of permit violations. The new plant will also be constructed outside the 100-year flood plain where the existing facility is located.	57.300	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
9	New NC Facility Hammermill Phase II ***Supports: Processing Operation Supporting Triple Base Propellant Production for 155mm Artillery***	27	P,I,E,S	Procures, installs and commissions the new Hammermill. For use in solvent propellants, especially propellants supporting 155mm	20.000	Base
10	Workplace Safety Phase – IV ***Supports: Propellant Operations; Safety Compliance; Fourth-Rolled Powder; Spiral Wrap and Press Bay (Rocket)***	28	S	The project will address major deficiencies in several categories across numerous RFAAP operations that will provide for a safer manufacturing environment for the entire plant. This includes, but is not limited to, new heating units similar to Building 9310-2; ventilation upgrades at all fourth-rolled operations; ventilation System for Spiral Wrap (Rocket); and ventilation System for Press Bay (Rocket).	12.300	Base
11	EWI Legacy Process Improvements: Construction ***Supports: Propellant Production Operations; Production Waste Treatment***	28	E,P	Construction of a new EWI Kiln and Grinder to replace the existing legacy facility.	67.700	Base
12	Wringer Upgrades (NC Support Process) ***Supports: Propellant Production Operations; Javelin, Hydra 70, Hellfire, Small Arms, Cannon Tank, Mortar, & Artillery***	29	P,S	Project will upgrade the wringer machine guarding and the wringer plow automation, as well as replace scales and nip roll label maker/SCR computer in Building 3026, and upgrade Building 3500 controls.	41.930	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
13	Bioplant Modernization Phase II: Construction ***Supports: Propellant Production Operations; Javelin, Hydra 70, Hellfire, Small Arms, Cannon Tank, Mortar, Artillery***	29	E,P	This project will construct the new Bioplant planned for FY26 design. The new treatment facility will have upgraded processes to meet regulatory standards and ensure removal of nitrates from water going to the New River.	69.280	Base
14	Electrical Distribution System Upgrades Phase III ***Supports: Propellant Production Operations; Javelin, Hydra 70, Hellfire, Small Arms, Cannon Tank, Mortar, Artillery***	29	I,P	The current electrical distribution system is unreliable and susceptible to catastrophic failure during adverse weather events. This project will demolish, repair, and replace various circuit components and electrical infrastructure, as well as remove trees from the associated areas to support production continuity and increase safety by ensuring the reliability of critical electrical circuits.	28.000	Base
15	Steam Distribution System Rehab & Repair: Construction Phase II ***Supports: Propellant Production Operations; Javelin, Hydra 70, Hellfire, Small Arms, Cannon Tank, Mortar, Artillery***	29	I,P	Execute a multi-phase project to upgrade and replace the 1940s plantwide steam distribution systems. A critical utility for RFAAP processes, steam is used in propellant production, as well as facility-wide building heating. The steam systems are reaching the end of their useful life with numerous and increasingly frequently failures. The Phase II effort will replace an additional 20% of the 58 miles of the steam distribution system to include piping, valves, traps, and insulation. The replacement will also include overhead piping and ancillary support infrastructures.	29.000	Base
16	Water Distribution System Upgrade Phase I: Metering and Design ***Supports: Propellant Production Operations; Javelin, Hydra 70, Hellfire, Small Arms, Cannon Tank, Mortar, Artillery***	29	I,P	Project develops design for the upgrade and replacement of the facility’s water distribution system.	42.348	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
17	NG-3 Phase I: Construction ***Supports: Nitrate Ester Production; Propellant Production Operations***	30	P,I,E,S	Phase I of the construction of the new NG3 facility will address construction and installation of equipment for Nitration, Storage, and Pre-Mix facilities. The fully completed NG3 Facility will be capable of producing alternative nitrate esters (e.g., DEGN) in addition to the required nitroglycerin. The new facility will include automation to improve production performance and efficiency; address safety, health, and reduce direct labor associated with manufacturing operations.	84.720	Base
18	Caustic Dip Construction & Demo of Legacy Chem Grind ***Supports: Propellant & Facility Operations; Production Process Caustic Cleaning***	30	S,E,P	The Caustic Cleaning operation decontaminates process equipment with propellant and energetic residue from production operations. Cleaning of screens, dies, and other equipment occurs daily in a hot bath of caustic solution that dissolves/neutralizes any energetic contamination so that the equipment can be reused. The legacy caustic cleaning operation has been in service since the 1940s and is in overall poor condition. Since construction, safety, health, and regulations have changed significantly, and the current facility is not compliant with many of the new regulations. The new Caustic Cleaning facility will replace the legacy operation and provide modernized manufacturing processes, safety controls, and improved material management practices.	54.980	Base
19	New Filter Water Treatment Plant: Design ***Supports: Propellant & Facility Operations***	30	I,P	Provides a comprehensive strategy and sign for a New Filter Water Treatment Plant which will replace the existing aged facility. This project will design a new filter water treatment facility and related processes. The new filter water treatment facility will replace the operations at 409 and depending on approach may also include new drinking water facilities, solids handling facilities, etc. The preliminary recommendation of this P-25 is for the design of a new Filtration plant that will replace the legacy filtration operations at Building 409, a new south-side drinking water plant that will replace the legacy drinking water operations at Building 419, and a new solids-handling facility that will replace the legacy waste management (sludge press) operations at Building 451.	20.000	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
20	Lightning Protection, Production Areas Phase III (Design/Build) ***Supports: Propellant & Facility Operations; Safety Compliance***	31	S	Design and installation of a comprehensive lightning protection system to includes the production areas which will provide modernized lightning protection that is in line with existing industry standards and regulations. Currently, several energetics manufacturing buildings lack lightning protection systems or have lightning protection systems that do not meet current standards. AMC regulation AMC-R 385-100 Chapter 8 reads: "It is the policy of AMC to install lightning protection on buildings and structures used in the manufacturing, processing, handling, or storing explosives, ammunition, or explosive ingredients."	14.890	Base
21	Steam Distribution Modernization Phase I (Design) ***Supports: Steam Supply for Propellant & Facility Operations***	31	I,P	This project is to strategically replace steam distribution lines at RFAAP. Replacement as a part of an ongoing multi-phased project will allow for the continued supply of steam for manufacturing propellants and support the proposed new facilities. This program will evaluate the current condition of the system, the overall layout of the system, the balance of energy, a total system redesign, and construction requirements.	40.000	Base
22	New Solvent Tank Farm & Unloading Station ***Supports: Solvent Propellant Production Operations***	31	P	Design of a new solvent tank farm and unloading station to support solvent propellant production. The new tank farm will replace the existing aged tanks and decrease production risks, such as maintenance, repairs and emergency shutdowns that results in production interruptions	66.200	Base
23	Building 474 Relocation (Drinking Water) ***Supports: Facility Operations***	31	S,E,P	Relocation of the existing Building 474 drinking water facility to improve plant efficiency and allocate space for the new solvent, solventless and NG facilities.	10.000	Base

11.1.9 RFAAP Strategic PBS Modernization Requirements EPP

Line No.	PBS Project Title	Category	Project Description	Project Value ($M)
1	Solventless Propellant Facility: Design & Construction ***Supports: Solventless Propellant Production Operations***	P	Design and construction of a new Solventless Propellant facility.	700–800
2	NG-3 Facility: Construction ***Supports: Propellant Production and Facility Operations***	P	Construction of a new NG-3 facility.	600–700
3	Facility Modernization & Permanent Energy Center ***Supports: Propellant Production and Facility Operations***	I,P	Comprehensive modernization of the facility's electrical utilities and construction of a permanent energy center.	500–600

11.1.10 SCAAP PBS Modernization Requirements FY25-31

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
1	Forge Shop-Roof Repairs, Smog Hog Removal & Rainwater Collection Repairs ***Supports: Forge and Production Shop Operations; Artillery, Mortars***	26	S,P	The roof on the Forge Shop has exceeded its life, and its patches are no longer effective. Replaces much of the roof, making the Forge Shop weather-tight, protecting employees and equipment, and lowering the risk of injury due to slips, falls, and electrical hazards.	4.500	Base
2	Boiler #3 Replacement / Upgrade ***Supports: Metal Parts Production Operations; Artillery, Mortars***	27	I,P	Project will replace the existing, obsolete Boiler #3 with a modern boiler to support the capacity increase efforts by providing steam and heat to meet future demands. Project will also upgrade 2-DA tanks and softener system.	1.500	Base
3	Central Coolant System Upgrades ***Supports: Metal Parts Production Operations; Artillery, Mortars***	27	P	Project will upgrade the central coolant system which includes new pumps, holding tank, skimmers, and UV treatment. This is required to support the 35K capacity expansion projects to ensure proper particulate filtration and efficient operation.	4.500	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
4	Infrastructure Repairs ***Supports: Forge and Production Shop Operations; Artillery, Mortars***	27	S,P	This project is to replace the concrete caps on the Forge Shop buildings as they have deteriorated to the point that brick & mortar is falling off the buildings. This project stabilizes and repairs existing structural damage to the Forge and Production Shop buildings. Also, the Arches in the subway material area are deteriorating with concrete falling from the ceiling which poses a safety hazard for people entering these areas.	4.500	Base
5	Cyber Security Phase II ***Supports: Metal Parts Production Operations; Artillery, Mortars***	27	S,P	This project will mitigate any cyber risks identified in Cybersecurity Phase I and bring the existing infrastructure to an acceptable level. Additionally, this is intended to enhance and defend the process control network at SCAAP in order to create a more modern, managed environment.	5.000	Base
6	Employee Health and Safety ***Supports: Critical Safety for Metal Parts Production Operations; Artillery, Mortars***	27	S	Project addresses employee-related health and safety issues to improve QWE, safety, and ensure compliance with all regulations.	3.000	Base
7	Plantwide Electrical Bus System Upgrade ***Supports: Electrical System Supplying Metal Parts Production; Artillery, Mortars***	27	I,P	Replace an estimated 5000 feet of bus ducts, including an estimated 550 disconnects in the Production Shop. This will improve the electrical infrastructure for future growth.	8.000	Base
8	A-Bay Crane Replacement ***Supports: Machine Shop and Metal Parts Production Operations; Artillery, Mortars***	27	P	Remove the existing direct current (DC) Crane and install a new 480V AC 25T Crane in A-Bay to support machine shop and production requirements. This project will include the inspection, repair and alignment of rails, as required.	1.456	Base
9	C-Bay Crane Replacement ***Supports: Metal Parts Production Operations; Artillery, Mortars***	28	P	Remove the existing Crane and install a new 480V 25T Crane in C-Bay to support production requirements. This project will include the inspection, repair, and alignment of rails, as required.	1.580	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
10	Security System and Mass Notification System ***Supports: Artillery & Mortar Production Operations; Safety & Security Compliance***	28	S	Upgrade the security system in SCAAP with state-of-the-art monitoring equipment to protect SCAAP assets; incorporate a mass notification system for safe working environment.	1.080	Base
11	Exhaust (River Street) System Replacement from Subway (QWE) ***Supports: Artillery & Mortar Production Operations; Health & Safety Compliance***	28	S,P	Project replaces four (4) non-functioning exhaust fans and associated ductwork in the subway along River Street with a new, smaller, vertical exhaust system to improve ventilation in the subway. This will improve the work environment for personnel in the area as well as slow deterioration of surrounding infrastructure.	2.300	Base
12	Electrical Upgrades Phase 2 (13KVA Install, 69KV Purchase and Install) ***Supports: Electrical System Supplying Metal Parts Production; Artillery, Mortars***	28	I,P	This project will install the 13.8KVA switch gear purchased under the electrical upgrades phase I effort currently under contract. This project will also purchase a 69KV transformer to replace an existing 69KV transformer on site. The project supports the 35K expansion effort and provides resiliency and better maintainability to the electrical system.	35.000	Base
13	Infrastructure Repairs Phase IV ***Supports: Forge and Production Shop Operations; Artillery, Mortars***	29	I,P	Stabilize and repair existing structural damage to the Administrative, Forge and Production Shop buildings.	5.250	Base
14	Facility-Wide Window Glass Replacement ***Supports: Facility Operations; and Safety Compliance***	30	S,P	Current windows potentially have lead paint that requires remediation, and many windows are broken across the facility. Project will replace all windows facility-wide to eliminate weather infiltration and further building deterioration.	20.000	Base
15	D-Bay Crane Replacement ***Supports: Metal Parts Production; Artillery & Mortars Operations***	30	P	Project removes the existing Crane and installs a new 480V 25T Crane in D-Bay to support production requirements. This project will include the inspection, repair, and alignment of rails, as required.	1.515	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
16	Bliss1 Upgrade (Induction Heater, Feed System, Descaler, Replacement & Conveyors) ***Supports: Metal Parts Production Operations; Artillery & Mortars***	30	P	Modernization of the Bliss 1 Furnace which directing supports the SCAAP mission to produce quality forge and pressed metal parts. Complete upgrades of the induction heater, feed system, descaler replacement and modernization of the conveyors.	4.750	Base
17	Facility Wide - Fire Suppression & Emergency Water Systems Upgrades ***Supports: Artillery & Mortars Operations; Safety Compliance***	30	S	Modernization of the existing fire suppression system plant wide with an automatic fire suppression system that reacts to a rapid rise in heat or a fire without any human intervention. This project will allow the plant to meeting current fire suppression standards.	1.600	Base
18	Replace Compressor 700hp ***Supports: Compressed Air Supply for Artillery & Mortars Operations***	30	P	Replace the existing JOY Compressor with two (2) new Variable Frequent Drive (VFD) Air Compressor to support pressure and volume requirements for entire SCAAP facility. The current JOY compressor requires replacement to ensure SCAAP has the required pressure and volume to prevent production interruption if a compressor fails.	1.500	Base
19	795 Legacy Line Replacement ***Supports: Artillery Production***	31	P	Project replaces the current M795 legacy line with the installation of new equipment to match the recently installed 35K equipment footprint. This will replace and automate items such as the center drill machine, rough turn lathes, welders, stress relief, and finish turn lathes. Replacement will support current and future requirements of M795	64.290	Base
20	Conveyor System Upgrade (Stress/Heat, Treat/Finish Turn) ***Supports: Metal Parts Production Operations; Artillery Production***	31	P	The conveyor systems that move the M795 and new XM rounds between band weld/stress relieve, nosing and to heat treat have been in place for over 30 years and are approaching the end of their useful life. Breakdowns of these conveyors cause delay in products moving to and from heat treat and can impact schedules. This project will replace the conveyor system used in the production shop to transfer 155mm rounds between the Stress Relieve, Heat Treat, and Finish Turn operations.	2.040	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
21	Erie 1 Furnace to Press Material Handling Upgrades ***Supports: Material Handling Operations; Artillery Production***	31	P	Provide a reliable Furnace to press material system to support SCAAP mission to produce quality forge and pressed metal parts. This project will install a new material handling system from the furnace to the press, upgrade the furnace doors and install a new gantry with indexer.	1.800	Base
22	WB Hanover - Phosphate system Replacement ***Supports: Metal Parts Production; Artillery; Compliance***	31	E,P	Project replaces the existing phosphate system at Wilkes-Barre Hanover to support M795 current and future production requirements. The new phosphate line will improve quality and production throughput.	3.500	Base
23	HVAC Replacement (Operations Center) ***Supports: Facility Operations***	31	P,S	Replace existing HVAC system with an energy efficient, zone-controlled, and programmable HVAC system.	7.345	Base
24	Casting Platform Rebuild ***Supports: Casting and Forge Shop Subway Operations; Artillery Production***	31	S,P	Stabilize and repair existing damage to the casting platform; utilize modern concrete mixes and sealers to minimize the possibility of reoccurrences and the amount of deterioration to the casting platform and forge shop subway. Repairing the casting platform will create safe work environment for pedestrians and protect the subway infrastructure.	3.920	Base
25	Bliss 3 Press Line - Rebuild Phase III - Controls & MCC Replacement ***Supports: Artillery Production***	31	P	Provide a reliable line to support SCAAP mission to produce quality forge and pressed metal parts. Complete press rebuilds for Hydraulic, pneumatic, electrical and diagnostic system.	5.750	Base
26	35K Capacity Expansion 1128 Upgrade ***Supports: 155 mm Artillery Production***	31	P	Project upgrades and installs capacity equipment for the 35k expansion of the M1128 project. The M1128 is the next generation projectile, and the capacity expansion is required to meet future mission requirements.	60.000	Base

Line No.	PBS Project Title	FY	Category	Project Description	Project Value ($M)	Funding Type
27	Forge Press Line Capacity and Quality Upgrades ***Supports: Metal Parts Production Operations; 120mm Artillery Production***	31	P	Project replaces the 120mm mortars forge press line system. This includes a new induction heater, two (2) 1000T presses, die lube system, automated material handling, chiller system, and quick-change tooling and other auxiliary equipment. New press line system will improve quality and production throughput. The replacement press line is required to support mortars current and future production requirements.	14.500	Base

11.2 APPENDIX B: PBS MODERNIZATION REQUIREMENTS BY FISCAL YEAR

The tables in Sections 11.2.1 through 11.2.7 indicate funding types as defined below:

- Base – Refers to the base budget submitted in the BES.
- A "✓" in the "2024 Mod. Plan" column indicates the project was included in the March 2024 Army Ammunition Plant Modernization Plan Memo to Congress

11.2.1 FY 2025 PBS MODERNIZATION REQUIREMENTS

Line No.	GOCO AAP	PBS Project Title	Category	Project Value ($M)	Funding Type	2024 Mod. Plan
1	HSAAP	Drying Energetics Building & Automatic Fire Suppression System: Construction ***Supports: IMX, RDX, HMX Production and Facility Operations***	P	5.200	Base	✓
2	HSAAP	Consolidated Compressed Air Facility: Construction ***Supports: IMX, RDX, HMX Production and Facility Operations***	P	13.300	Base	✓
3	HSAAP	Secondary Process River Water Intake Facility: Construction ***Supports: IMX, RDX, HMX Production and Facility Operations***	I	88.060	Base	✓
4	HSAAP	PAX-3 Production G Building Modernization & Construction ***Supports: PAX-3 Production Operations***	P	8.300	Base	✓
5	IAAAP	Future Artillery Complex, Phase II Construction ***Supports: 155mm Artillery Production***	P	280.650	Base	✓
6	LCAAP	7.62mm Reduced Range Ammunition Process Modernization & Construction ***Supports: 7.62mm Ammunition Production***	P	57.000	Base	✓
7	LCAAP	One Way Luminescence Ammo: Facility Construction & Equipment Modernization ***Supports: Small Caliber OWL Production Operations***	P	35.000	Base	✓
8	RFAAP	Water Intake Pump Upgrades, Bldg. 407 Phase 2 ***Supports: Propellant Production and Facility Operations***	I,E	21.500	Base	✓
9	RFAAP	Hurricane Helene Supplemental ***Supports: Propellant Production and Facility Operations***	P	125.1	Supp	

Line No.	GOCO AAP	PBS Project Title	Category	Project Value ($M)	Funding Type	2024 Mod. Plan
10	ALL	Planning & Design, MIF/LIF, Engineering Support, QWE, Emergent Maintenance	-	130.356	Base	✓
			FY25 Total	**764.466**		

11.2.2 FY 2026 PBS Modernization Requirements

Line No.	GOCO AAP	PBS Project Title	Category	Project Value ($M)	Funding Type	2024 Mod. Plan
1	HSAAP	River Water Outfall 40 and Reservoir Upgrade: Construction ***Supports: IMX, RDX, HMX Production and Facility Operations***	E,I,P	24.000	Base	✓
2	HSAAP	Wastewater General Upgrades: Construction ***Supports: IMX, RDX, HMX Production and Facility Operations; Production Wastewater Treatment***	E,I,P	18.000	Base	✓
3	HSAAP	Facility Electrical Upgrades (Sitewide Reliability) ***Supports: IMX, RDX, HMX Production and Facility Operations***	I,P,S	11.400	Base	✓
4	HSAAP	Lacquer Prep Facility Sustainment ***Supports: IMX, RDX, HMX Production and Facility Operations***	P,I,E,S	1.600	Base	✓
5	HSAAP	Sanitary Sewer Infrastructure Upgrades: Construction ***Supports: IMX, RDX, HMX Production and Facility Operations; Production Wastewater Treatment***	E,I,P	11.000	Base	✓
6	HSAAP	Treatment Technology Alternatives to Open Burning ***Supports: Explosive Production Operations; Production Waste Treatment***	E,P	3.000	Base	✓
7	HSAAP	IWWTF (Hydrogen Sulfide Issue) Construction & Modernization ***Supports: IMX, RDX, HMX Production and Facility Operations; Production Wastewater Treatment***	S,E,P	28.800	Base	✓
8	HSAAP	Consolidated Compressed Air Facility: Construction ***Supports: IMX, RDX, HMX Production and Facility Operations***	P,I	24.000	Base	✓
9	HSAAP	RDX Wastewater—Pump Station Upgrades: Construction	E,I,P	55.200	Base	✓

Line No.	GOCO AAP	PBS Project Title	Category	Project Value ($M)	Funding Type	2024 Mod. Plan
		Supports: RDX, Production Operations; Production Wastewater Treatment				
10	HSAAP	G-10 Process Improvements ***Supports: TATB Operations; TATB Capacity Increase***	E,S,P	7.800	Base	
11	IAAAP	Line 3A Sustainment–Phase III ***Supports: 155mm Artillery Production***	P,S	4.728	Base	✓
12	IAAAP	Tank Line (Line 2) Sustainment Phase I ***Supports: 120mm Ammunition Production***	P,I	3.444	Base	✓
13	IAAAP	Coal Elimination ***Supports: Artillery and Tank Production and Facility Operations***	E	14.137	Base	
14	LCAAP	Energetic Waste Incinerator: Construction ***Supports: Small Caliber Ammunition Production***	E,P	239.945	Base	✓
15	LCAAP	NGSW-A Equipping ***Supports: 6.8mm Ammunition Production***	P	236.800	Base	✓
16	RFAAP	Acid Area Tank Farm Replacement ***Supports: Propellant Production and Facility Operations***	P,I,E,S	19.650	Base	✓
17	RFAAP	NG-3 90% Design ***Supports: Propellant Production and Facility Operations***	P,I,E,S	22.750	Base	✓
18	RFAAP	Tie-Line Modernization ***Supports: Propellant Production and Facility Operations***	I,P	28.400	Base	
19	RFAAP	Electrical Distribution System Upgrades Phase IV ***Supports: Propellant Production and Facility Operations***	I,P	27.800	Base	✓
20	SCAAP	Forge Shop-Roof Repairs, Smog Hog Removal & Rainwater Collection Repairs ***Supports: Forge and Production Shop Operations; Artillery, Mortars***	S,P	4.500	Base	✓
21	ALL	Planning & Design, MIF/LIF, Engineering Support, QWE, Emergent Maintenance	-	82.084	Base	
			FY26 Total	**869.038**		

11.2.3 FY 2027 PBS Modernization Requirements

Line No.	GOCO AAP	PBS Project Title	Category	Project Value ($M)	Funding Type	2024 Mod. Plan
1	HSAAP	Water Distribution Modernization ***Supports: IMX, RDX, HMX Production and Facility Operations***	P,I,E	16.200	Base	✓
2	HSAAP	Sewer System Line Replacement Phase III ***Supports: IMX, RDX, HMX Production and Facility Operations***	P,E	13.644	Base	
3	HSAAP	Treatment Technology Alternatives to Open Burning ***Supports: Explosive Production Operations; Production Waste Treatment***	E,P	3.000	Base	
4	IAAAP	Yard F Rail Modernization ***Supports: DARAD Compliance; Energetic Material Processing***	S,P	1.850	Base	
5	IAAAP	Hellfire & AMP Process Relocation ***Supports: DARAD Compliance & Construction; Hellfire and 120mm Tank AMP Production Operations***	S,P	39.000	Base	
6	IAAAP	40mm Press Relocation Phase II ***Supports: DARAD Compliance & Construction; 40mm M430A1 Grenade Production Operations***	S, P	33.000	Base	
7	IAAAP	Line 3A Sustainment Phase IV ***Supports: 155mm Artillery Production***	P	6.500	Base	
8	LCAAP	NGSW-A Equipping ***Supports: 6.8mm Ammunition Production***	P	132.420	Base	✓
9	LCAAP	B19 Powder Storage Modernization ***Supports: Small Caliber Ammunition Production***	S,P	2.600	Base	
10	RFAAP	Solvent Propellant Manufacturing Facility: Construction ***Supports: Solvent Propellant Production; Javelin, Hydra 70, & Hellfire***	P,I,E,S	830.725	Base	✓
11	RFAAP	Bioplant Modernization Phase I: 90% Design ***Supports: Propellant Production and Facility Operations***	E,P	57.300	Base	✓
12	RFAAP	New NC Facility Hammermill Phase II ***Supports: Processing Operation Supporting Triple Base Propellant Production for 105mm Artillery***	P,I,E,S	20.000	Base	
13	QCCCF	Batch Annealing Furnace ***Supports: Deep Draw Cartridge Fabrication Operations; 105mm HE/M1 Artillery & 105 PGU Artillery***	P	6.500	Base	
14	SCAAP	Boiler #3 Replacement / Upgrade ***Supports: Metal Parts Production Operations; Artillery, Mortars***	I, P	1.500	Base	
15	SCAAP	Central Coolant System Upgrades ***Supports: Metal Parts Production Operations; Artillery, Mortars***	P	4.500	Base	
16	SCAAP	Infrastructure Repairs ***Supports: Forge and Production Shop Operations; Artillery, Mortars***	S,P	4.500	Base	✓

Line No.	GOCO AAP	PBS Project Title	Category	Project Value ($M)	Funding Type	2024 Mod. Plan
17	SCAAP	Cyber Security Phase II ***Supports: Metal Parts Production Operations; Artillery, Mortars***	S, P	5.000	Base	
18	SCAAP	Employee Health and Safety ***Supports: Critical Safety for Metal Parts Production Operations; Artillery, Mortars***	S	3.000	Base	
19	SCAAP	Plantwide Electrical Bus System Upgrade ***Supports: Electrical System Supplying Metal Parts Production; Artillery, Mortars***	I,P	8.000	Base	
20	SCAAP	A-Bay Crane Replacement ***Supports: Machine Shop and Metal Parts Production Operations; Artillery, Mortars***	P	1.456	Base	
21	ALL	Planning & Design, MIF/LIF, Engineering Support, QWE, Emergent Maintenance	-	92.607	Base	
			FY27 Total	**1,283.302**		

11.2.4 FY 2028 PBS Modernization Requirements

Line No.	GOCO AAP	PBS Project Title	Category	Project Value ($M)	Funding Type	2024 Mod. Plan
1	HSAAP	New Lacquer Prep Facility: Construction ***Supports: IMX, RDX, HMX Production and Facility Operations***	P,I,E,S	97.200	Base	✓
2	HSAAP	G-10 Toluene Emissions and Process Improvements Phase III ***Supports: TATB Operations; TATB Capacity Increase***	E, S,P	16.000	Base	✓
3	HSAAP	New Drumming Packaging Facility: Construction ***Supports: Finished Product (IMX, RDX, HMX) Packaging Operations***	P,I	10.000	Base	✓
4	HSAAP	Strong Nitric Acid Tank Farm: Construction ***Supports: Acid Processing and IMX, RDX, HMX Production Operations***	P, I,E	5.000	Base	✓
5	HSAAP	Dock Deluge (Explosives Packaging/Shipping): Construction ***Supports: Safety Compliance Measure for IMX, RDX, HMX Packaging and Shipping Operations***	S,P	4.700	Base	✓
6	HSAAP	A2B Enclose Platform on Second Level: Construction ***Supports: Safety Compliance Measure for IMX, RDX, HMX Packaging and Shipping Operations***	S,P	4.400	Base	✓
7	HSAAP	E-4 Boxway Relocation (Utility/Distribution System Protection): Construction ***Supports: G-4 Material Supply Access for RDX, HMX Recrystallization Operations***	I, E,P	2.100	Base	✓
8	IAAAP	Stormwater & Erosion Controls for Production Employee Parking ***Supports: 40mm and Direct Fire (120mm AMP) Production Operations***	E,S,I,P	26.880	Base	✓

Line No.	GOCO AAP	PBS Project Title	Category	Project Value ($M)	Funding Type	2024 Mod. Plan
9	IAAAP	Line 3A Sustainment Phase V ***Supports: 155mm Artillery Production***	P	6.940	Base	
10	IAAAP	Critical Rail Replacement–Yard C ***Supports: TNT, M795 Finished Goods and CAT II Storage Operations***	P,I,S	31.900	Base	✓
11	IAAAP	Energetics Storage Facility Sustainment Phase I ***Supports: Artillery Production and Storage; Security Compliance***	P,S,I	16.500	Base	✓
12	IAAAP	Non-Destructive Testing Laboratory Design & Construct ***Supports: Testing Operations for M795, Medium Caliber Ammunition, 120mm Tank Ammunition, and Future Munitions***	P,I	7.790	Base	✓
13	IAAAP	Hellfire Computer Numerical Control (CNC) Equipment Modernization ***Supports: Hellfire Production; Increases Hellfire Capacity***	P	2.700	Base	✓
14	IAAAP	Indirect Fire Production Quality Control Process Improvements ***Supports: 155mm Artillery Production; Line 3A Quality Control Operations***	P	4.110	Base	✓
15	LCAAP	Five Grain Distribution Replacement: Small Caliber Production & Operations ***Supports: Small Caliber Ammunition Production***	I,P	41.446	Base	✓
16	LCAAP	Fire Water System Replacement: Small Caliber Production & Operations ***Supports: Small Caliber Ammunition Production***	I,P	40.120	Base	✓
17	LCAAP	Zero Grain Distribution Replacement: Small Caliber Production ***Supports: Small Caliber Ammunition Production***	I,P	4.650	Base	✓
18	LCAAP	NGSW-A Equipping ***Supports: 6.8mm Ammunition Production***	P	108.700	Base	✓
19	RFAAP	Workplace Safety Phase – IV ***Supports: Propellant Operations; Safety Compliance; Fourth-Rolled Powder; Spiral Wrap and Press Bay (Rocket)***	S	12.300	Base	✓
20	RFAAP	EWI Legacy Process Improvements: Construction ***Supports: Propellant Production Operations; Production Waste Treatment***	E,P	67.700	Base	✓
21	SCAAP	C-Bay Crane Replacement ***Supports: Metal Parts Production Operations; Artillery, Mortars***	P	1.580	Base	
22	SCAAP	Security System and Mass Notification System ***Supports: Artillery & Mortar Production Operations; Safety & Security Compliance***	S	1.080	Base	
23	SCAAP	Exhaust (River Street) System Replacement from Subway (QWE) ***Supports: Artillery & Mortar Production Operations; Health & Safety Compliance***	S,P	2.300	Base	
24	SCAAP	Electrical Upgrades Phase 2 (13KVA Install, 69KV Purchase and Install)	I,P	35.000	Base	

Line No.	GOCO AAP	PBS Project Title	Category	Project Value ($M)	Funding Type	2024 Mod. Plan
		Supports: Electrical System Supplying Metal Parts Production; Artillery, Mortars				
25	ALL	Planning & Design, MIF/LIF, Engineering Support, QWE, Emergent Maintenance	-	113.832	Base	
			FY28 Total	**664.928**		

11.2.5 FY 2029 PBS Modernization Requirements

Line No.	GOCO AAP	PBS Project Title	Category	Project Value ($M)	Funding Type	2024 Mod. Plan
1	HSAAP	New Filtered Water Plant: Construction ***Supports: IMX, RDX, HMX Production Operations***	I,P	120.000	Base	✓
2	HSAAP	Pump House and River Intake (Bldg. 201) Phase I: Design/Construction ***Supports: IMX, RDX, HMX Production Operations***	P,E,S,I	35.000	Base	✓
3	HSAAP	Boxway Upgrades for Spill Prevention: Construction ***Supports: Chemical Supply Transport for IMX, RDX, HMX Production Operations***	E,I,P	53.000	Base	✓
4	HSAAP	ANSOL Tank Farm ***Upgrades*** ***Supports: IMX, RDX, HMX Production Operations; Production By-Product Management***	E,P	12.000	Base	
5	IAAAP	Telecommunications & Fiber Optic Infrastructure Upgrades ***Supports: Artillery Production and Facility Operations***	S,I	15.000	Base	✓
6	IAAAP	Production Road Modernization Phase I ***Supports: Transportation and Material Movement for Production Operations***	P,I,S	10.000	Base	
7	IAAAP	Design Rail, Dunnage, & Heavy Equipment Campus ***Supports: Transportation and Material Movement for Production Operations;155mm Artillery***	P,I	2.900	Base	
8	LCAAP	Industrial Wastewater Plant Upgrade & Distribution Replacement: Construction ***Supports: Small Caliber Ammunition Production Operations; Production Wastewater Treatment***	I,P	102.470	Base	✓
9	LCAAP	NGSW-A Equipping ***Supports: Small Caliber Ammunition Production; 6.8mm***	P	2.980	Base	✓
10	QCCCF	Industrial Waste Final Holding Tank ***Supports: Facility Operations; Capacity Increase; Production Wastewater Treatment***	E,P	4.750	Base	
11	RFAAP	Wringer Upgrades (NC Support Process) ***Supports: Propellant Production Operations; Javelin, Hydra 70, Hellfire, Small Arms, Cannon Tank, Mortar, & Artillery***	P,S	41.930	Base	✓

Line No.	GOCO AAP	PBS Project Title	Category	Project Value ($M)	Funding Type	2024 Mod. Plan
12	RFAAP	Bioplant Modernization Phase II: Construction ***Supports: Propellant Production Operations; Javelin, Hydra 70, Hellfire, Small Arms, Cannon Tank, Mortar, Artillery***	E,P	69.280	Base	✓
13	RFAAP	Electrical Distribution System Upgrades Phase III ***Supports: Propellant Production Operations; Javelin, Hydra 70, Hellfire, Small Arms, Cannon Tank, Mortar, Artillery***	I,P	28.000	Base	✓
14	RFAAP	Steam Distribution System Rehab & Repair: Construction Phase II ***Supports: Propellant Production Operations; Javelin, Hydra 70, Hellfire, Small Arms, Cannon Tank, Mortar, Artillery***	I,P	29.000	Base	✓
15	RFAAP	Water Distribution System Upgrade - Phase I (Metering and Design) ***Supports: Propellant Production Operations; Javelin, Hydra 70, Hellfire, Small Arms, Cannon Tank, Mortar, Artillery***	I,P	42.348	Base	
16	SCAAP	Infrastructure Repairs Phase IV ***Supports: Forge and Production Shop Operations; Artillery, Mortars***	I,P	5.250	Base	✓
17	ALL	Planning & Design, MIF/LIF, Engineering Support, QWE, Emergent Maintenance	-	96.490	Base	
			FY29 Total	**670.398**		

11.2.6 FY 2030 PBS Modernization Requirements

Line No.	GOCO AAP	PBS Project Title	Category	Project Value ($M)	Funding Type	2024 Mod. Plan
1	HSAAP	G-4 & H-4 (RDX Recrystallization/Dewatering) Consolidated Modernization: Construction ***Supports: Explosive Production Operations; RDX Recrystallization and Dewatering***	P,I,E,S	30.000	Base	✓
2	HSAAP	Emergency Medical Services and Security Communication System Updates ***Supports: Safety & Security Compliance; IMX, RDX, HMX Production and Facility Operations***	S	6.000	Base	✓
3	HSAAP	Barricade: Construction ***Supports: Safety Compliance; IMX, RDX, HMX Production and Facility Operations***	S	3.500	Base	✓
4	HSAAP	Air/Steam Distribution Replacement: Construction ***Supports: Air & Steam Supply for IMX, RDX, HMX Production and Facility Operations***	I,P	34.678	Base	✓
5	HSAAP	D-5 (Nitration) Operations Consolidation : Construction ***Supports: Nitration Operations; RDX, HMX Production***	P,I,E,S	26.000	Base	✓

Line No.	GOCO AAP	PBS Project Title	Category	Project Value ($M)	Funding Type	2024 Mod. Plan
6	HSAAP	Increased Filtered Water Storage Capacity ***Supports: Production Water Supply; IMX, RDX, HMX Production***	S,I,P	1.500	Base	✓
7	HSAAP	G-10 Process Improvements ***Supports: TATB Operations; Agile Production Operations for IMX***	E,P	25.000	Base	✓
8	HSAAP	Wastewater Upgrades: Construction ***Supports: IMX, RDX, HMX Production and Facility Operations; Production Wastewater Treatment***	I,E,P	20.000	Base	
9	IAAAP	Equipment Maintenance Complex ***Supports: Consolidated Equipment Maintenance for Artillery and Mortar Production Operations***	P	4.000	Base	✓
10	IAAAP	Access Control Point Upgrades ***Supports: Facility Operations, Artillery and Mortar Production; Security Compliance***	S,I	14.900	Base	✓
11	IAAAP	Critical Rail Replacement–Yard L & F ***Supports: Material Transport; Artillery and Mortar Production Operations***	S,P	37.000	Base	✓
12	IAAAP	Tank Line (Line 2) Sustainment Phase III ***Supports: 120mm Ammunition Production***	P	2.830	Base	
13	IAAAP	Operations and Equipment Support Improvement Program Phase 1 ***Supports: Facility Operations, Artillery and Mortar Production***	P,E,S	6.500	Base	
14	IAAAP	Rail Improvement Program Phase 1 ***Supports: Material Transport; Artillery and Mortar Production***	P,I	10.000	Base	
15	IAAAP	Munition Process Line Roof Improvement Program Phase 1 ***Supports: Artillery and Mortar Production; Safety and Quality Compliance***	P,I	10.000	Base	
16	IAAAP	Munition Production Road Improvement Program Phase 2 ***Supports: Raw Material and Finished Product Transport; Artillery and Mortar Production***	P,I	10.000	Base	
17	LCAAP	Modernize Links Heat Treat Furnace ***Supports: Small Caliber Ammunition Production; 5.56mm, 7.62mm, .50 Cal, 20mm, 25mm links, 6.8mm***	P	102.610	Base	✓
18	LCAAP	High Explosive Air and Video Monitoring Upgrades ***Supports: Small Caliber Ammunition Production: Product Quality Compliance***	P	0.981	Base	✓
19	LCAAP	Automatic Transfer Switch Upgrade ***Supports: Small Caliber Ammunition Production; Safety Compliance***	I	17.853	Base	✓
20	LCAAP	Compressed Air System Improvement (B83A) ***Supports: Compressed Air Supply for Small Caliber Ammunition Production Operations***	P,I	3.881	Base	✓
21	LCAAP	HVAC Instrument and Control Modernization (B33/34 Mix House)	E,S,P	2.789	Base	✓

Line No.	GOCO AAP	PBS Project Title	Category	Project Value ($M)	Funding Type	2024 Mod. Plan
		Supports: Small Caliber Ammunition Production; Product Quality Compliance				
22	LCAAP	Emergency Services Expansion ***Supports: Small Caliber Ammunition Production; Safety and Security Compliance***	S	6.592	Base	✓
23	LCAAP	NGSW-A Utility Connections ***Supports: 6.8mm Ammunition Production***	P,I	6.720	Base	
24	RFAAP	NG-3 Phase I: Construction ***Supports: Nitrate Ester Production; Propellant Production Operations***	P,I,E,S	84.720	Base	✓
25	RFAAP	Caustic Dip Construction & Demo of Legacy Chem Grind ***Supports: Propellant & Facility Operations; Production Process Caustic Cleaning***	S,E,P	54.980	Base	✓
26	RFAAP	New Filter Water Treatment Plant: Design ***Supports: Propellant & Facility Operations***	I,P	20.000	Base	
27	SCAAP	Facility-Wide Window Glass Replacement ***Supports: Facility Operations; and Safety Compliance***	S,P	20.000	Base	✓
28	SCAAP	D-Bay Crane Replacement ***Supports: Metal Parts Production; Artillery & Mortars Operations***	P	1.515	Base	
29	SCAAP	Bliss1 Upgrade (Induction Heater, Feed System, Descaler, Replacement & Conveyors) ***Supports: Metal Parts Production Operations; Artillery & Mortars***	P	4.750	Base	
30	SCAAP	Facility Wide - Fire Suppression & Emergency Water Systems Upgrades ***Supports: Artillery & Mortars Operations; Safety Compliance***	S	1.600	Base	
31	SCAAP	Replace Compressor 700hp ***Supports: Compressed Air Supply for Artillery & Mortars Operations***	P	1.500	Base	
32	ALL	Planning & Design, MIF/LIF, Engineering Support, QWE, Emergent Maintenance	-	97.990	Base	
			FY30 Total	**670.389**		

11.2.7 FY 2031 PBS Modernization Requirements

Line No.	GOCO AAP	PBS Project Title	Category	Project Value ($M)	Funding Type	2024 Mod. Plan
1	HSAAP	HMX Production Facility: Construction ***Supports: HMX Production and Facility Operations***	P	62.000	Base	
2	HSAAP	SNA Tank Farm & Pipeline Upgrades Phase II: Construction ***Supports: Strong Nitric Acid Operations; IMX, RDX, HMX Production***	P,E,S	12.000	Base	
3	HSAAP	Facility Electrical Upgrades Phase II	I,P	12.000	Base	

Line No.	GOCO AAP	PBS Project Title	Category	Project Value ($M)	Funding Type	2024 Mod. Plan
		Supports: Electricity Supply for IMX, RDX, HMX Production and Facility Operations				
4	HSAAP	Railroad Track Improvements Phase I (Nitric Rail System) ***Supports: Strong Nitric Acid and Glacial Acetic Acid Transport for Explosive Production Operations***	S,P	14.000	Base	
5	HSAAP	Critical Spare Parts Phase III ***Supports: IMX, RDX, HMX Production and Facility Operations***	P	8.000	Base	
6	IAAAP	3A Elevator Replacement ***Supports: Line 3A Operations;155mm Artillery Production***	P,S	3.130	Base	
7	IAAAP	Lead Pipe Study & Remediation ***Supports: Artillery, Mortar Production & Facility Operations; Health & Safety Compliance***	E,S	14.720	Base	
8	IAAAP	4B Oil & Water Tertiary Treatment Replacement ***Supports: Artillery, Mortar Production & Facility Operations; Regulatory Compliance***	E,P	1.820	Base	
9	IAAAP	Steam & Air Supply Support Replacement ***Supports: Steam & Air Supply for Artillery, Mortar Production & Facility Operations***	P	9.500	Base	
10	IAAAP	Cyber Security Phase II: Construction ***Supports: Artillery, Mortar Production & Facility Operations; Security Compliance***	P,I	25.000	Base	
11	LCAAP	Roof Replacement: B5, B6, B11, B12A, 19 Series ***Supports: Small Caliber Ammunition Production & Facility Operations***	S,P	48.749	Base	
12	LCAAP	B1: Vacuum Powder Delivery System ***Supports: SCAMP Loading Operations; Small Caliber Ammunition Production***	S,P	68.095	Base	
13	RFAAP	Lightning Protection, Production Areas Phase III (Design/Build) ***Supports: Propellant & Facility Operations; Safety Compliance***	S	14.890	Base	
14	RFAAP	Steam Distribution Modernization Phase I (Design) ***Supports: Steam Supply for Propellant & Facility Operations***	I,P	40.000	Base	
15	RFAAP	New Solvent Tank Farm & Unloading Station ***Supports: Solvent Propellant Production Operations***	P	66.200	Base	
16	RFAAP	Building 474 Relocation (Drinking Water) ***Supports: Facility Operations***	S,E,P	10.000	Base	
17	SCAAP	795 Legacy Line Replacement ***Supports: Artillery Production***	P	64.290	Base	
18	SCAAP	Conveyor System Upgrade (Stress/Heat, Treat/Finish Turn) ***Supports: Metal Parts Production Operations; Artillery Production***	P	2.040	Base	
19	SCAAP	Erie 1 Furnace to Press Material Handling Upgrades	P	1.800	Base	

Line No.	GOCO AAP	PBS Project Title	Category	Project Value ($M)	Funding Type	2024 Mod. Plan
		Supports: Material Handling Operations; Artillery Production				
20	SCAAP	WB Hanover - Phosphate system Replacement ***Supports: Metal Parts Production; Artillery; Regulatory Compliance***	E,P	3.500	Base	
21	SCAAP	HVAC Replacement (Operations Center) ***Supports: Facility Operations***	P,S	7.345	Base	
22	SCAAP	Casting Platform Rebuild ***Supports: Casting and Forge Shop Subway Operations; Artillery Production***	S,P	3.920	Base	
23	SCAAP	Bliss 3 Press Line - Rebuild Phase III - Controls & MCC Replacement ***Supports: Artillery Production***	P	5.750	Base	
24	SCAAP	35K Capacity Expansion 1128 Upgrade ***Supports: 155mm Artillery Production***	P	60.000	Base	
25	SCAAP	Forge Press Line Capacity and Quality Upgrades ***Supports: Metal Parts Production Operations; 120mm Ammunition Production***	P	14.500	Base	
26	ALL	Planning & Design, MIF/LIF, Engineering Support, QWE, Emergent Maintenance	-	99.286	Base	
			FY31 Total	**672.535**		

11.3 APPENDIX C: PBS MODERNIZATION PROJECTS IN EXECUTION

The charts in Sections 11.3.1 through 11.3.5 provide cost, schedule, and performance metrics for PBS projects currently in execution (as of January 24, 2025). Projects are listed by GOCO AAP.

The following is a summary of all projects in execution:

GOCO Ammunition Facility	No. of Projects in Execution (As of January 24, 2025)	Total Cost of Projects ($M)
HSAAP	24	1,107.720
IAAAP	32	353.500
LCAAP	29	782.590
RFAAP	18	1055.900
SCAAP	19	1,117.800
TOTALS	**122**	**$4,417.510**

11.3.1 HSAAP Projects in Execution

Project Number	Project Title	Cost ($M)	Start Date	Completion Date
2018 ($M)	**60.15**			
1	New Fluid Energy Mill (FEM) Construction	$60.15	Sep 2018	Feb 2025
2019 ($M)	**74.80**			
2	G3 Recrystallization Construction	$74.80	Dec 2018	May 2025
2020 ($M)	**281.78**			
3	WAARP plus Tank Farm Construction	$145.60	Sep 2020	Oct 2025
4	Landfill Expansion	$31.48	Dec 2020	Nov 2025
5	Analytical Lab Construction	$69.86	Mar 2021	Dec 2025
6	Modernized Melt Cast Construction	$34.84	Mar 2021	Sept 2025
2021 ($M)	**370.85**			
7	Flashing Furnace Construction	$51.19	Aug 2021	Dec 2027
8	AAA Construction	$136.81	Oct 2021	Apr 2026
9	Kettle Drying Final Design/Construction	$55.26	Mar 2022	Mar 2026
10	Nitration Construction	$123.88	Dec 2021	Jan 2027
11	TATB Emissions Phase 1-A	$3.71	Feb 2022	Mar 2025
2022 ($M)	**261.74**			
12	TATB Emissions Phase 1-B Design	$3.36	Sep 2022	Jul 2025
13	Filtration and Wash Construction	$80.71	Nov 2022	Sep 2026
14	Critical Spare Parts Phase II	$6.07	Dec 2022	Mar 2025
15	ANSOL Thermal Destruction Facility 65% Design	$5.89	Feb 2023	Dec 2025
16	Explosive Decontamination Chamber Final Design	$165.72	Mar 2023	Apr 2027
2023 ($M)	**57.96**			
17	G-10 Toluene Emissions and Process Improvements	$19.10	Jan 2023	Nov 2025
18	RDX Wastewater Design, Upgrades, Construction	$11.51	Mar 2023	Apr 2025
19	Major Maintenance Additions (G-10 Tank Replacement)	$9.68	Apr 2023	Feb 2025
20	NAC/SAC Reactivation	$5.24	Apr 2023	Dec 2025
21	New River Water Intake Design	$5.34	Aug 2023	Feb 2025
22	Front Entrance Road Repairs	$3.35	Apr 2024	Jun 2025
23	Coal Pile Cleanup	$3.74	Oct 2024	May 2026
2024 ($M)	**0.43**			
24	Holston Jacks	$0.43	Oct 2024	Jun 2025
Total:	**$1,107.72**			

All projects in execution as of January 24, 2025.

11.3.2 IAAAP PROJECTS IN EXECUTION

Project Number	Project Title	Cost ($M)	Start Date	Estimated Completion Date
2018 ($M)	**$5.86**			
1	Isostatic Press	$5.9	Sep 2018	Jun 2025
2019 ($M)	**$57.11**			
2	Modernize Melt Pour Operations - Design	$48.1	May 2019	Mar 2025
3	Waste Reprocessing Center Phase I& II	$9.0	Jan 2019	Jan 2025
2020 ($M)	**$28.32**			
4	40mm Spitback Press Relocation	$5.1	Jun 2020	Sep 2025
5	Distributed Natural Gas Boilers Phase II	$13.3	Jul 2020	Dec 2025
6	Roadway Modernization Improvements	$6.3	Sep 2020	Feb 2025
7	M112 Bagging Automation	$1.9	Sep 2020	Dec 2025
2021 ($M)	**$37.78**			
8	OB/OD Alternatives Ph 1	$4.1	Feb 2021	Dec 2025
9	Rail Phase IV Yard D & Unit Train Track	$15.6	Dec 2021	Feb 2025
10	Metrology Lab	$9.5	Aug 2021	Mar 2026
11	Water Tower Phase III	$2.7	Sep 2021	Sep 2025
12	Design Hellfire & AMP (DARAD)	$1.6	Mar 2023	Jul 2026
13	Cyber Design	$1.0	Jul 2022	Jan 2025
14	HVAC 1-10 & 2-12	$3.4	Sep 2021	Aug 2025
2022 ($M)	**$78.30**			
15	OB/OD Alternatives Ph 2	$7.8	Feb 2023	Sep 2026
16	Roof Replacement Phase III	$10.4	Apr 2022	Jul 2025
17	Plant-Wide Electrical Design	$1.5	May 2022	Jul 2025
18	Tank Line Charrette	$3.9	Apr 2022	Feb 202
19	Primary Water Pump - Construction	$3.2	Jul 2022	Aug 2025
20	Building 1-11 Inert Storage	$0.5	Jul 2022	Jan 2025
21	Asbestos Abatement	$14.4	Apr 2023	Mar 2025
22	Line 3A Sustainment	$2.8	Sep 2022	Feb 2025
2023 ($M)	**$182.74**			
23	Energetics Facility - Modernization	$50.0	Feb 2023	May 2027
24	Line 3A Modernization Effort	$24.6	Aug 2023	Feb 2026
25	2nd X-Ray System for Large Caliber Ammunition	$8.1	Aug 2023	Aug 2025
26	Future Artillery Complex Site Prep	$35.9	Jul 2023	Mar 2025
27	New Tool Die Facility	$40.0	Feb 2023	Aug 2026
28	Roadway Modernization Improvements	$15.0	Feb 2023	Dec 2025
2024 ($M)	**$5.19**			
29	AI Equipment Implementation for M795	$0.2	Jul 2024	Jul 2026
30	Isostatic Press Phase III	$1.6	Jul 2024	Jan 2026
31	C43 Magazine Cat II Upgrades	$0.2	Sep 2024	Mar 2026
32	Main Sewage Treatment Plant	$16.2	Aug 2024	Feb 2027
Total:	**$353.5**			

All projects in execution as of January 24, 2025.

11.3.3 LCAAP Projects in Execution

Project Number	Project Title	Cost ($M)	Project Start Date	Estimated Completion Date
2020 ($M)	**$9.57**			
1	GP Projectile ECP and Supply Risk Reduction Effort	$9.57	Sep 2022	May 2025
2021 ($M)	**$32.07**			
2	B35 Roof and Structural Repairs (Design/Construct)	$16.95	Aug 2022	Jan 2026
3	Consolidated Chemistry Lab (Design)	$7.92	Sep 2023	Jan 2026
4	889 Compliance/Obsolescence	$7.20	Sep 2023	Mar 2026
2022 ($M)	**$59.21**			
5	Contaminated Waste Plant & 160C Lift Station	$7.36	Mar 2023	Feb 2025
6	Phase 3 Demolition [LIF]	$3.66	Sep 2022	Feb 2025
7	(NGSW) Interim Capability Equipping: Phase II	$42.40	Aug 2022	Mar 2025
8	Mix Houses & Neutralization (Design)	$4.33	Aug 2023	Feb 2025
9	External Natural Gas Distribution Upgrade	$0.17	Oct 2024	Apr 2025
10	Propellant Storage [replace existing 19 Series	$0.17	Nov 2023	Feb 2025
11	Pyrotechnics Operations Complex	$0.19	Dec 2023	Jan 2025
12	Primer Operations Complex [replace existing B35]	$0.19	Feb 2024	Jan 2025
13	Lead Free Primer (Planning Charrette)	$0.19	Apr 2024	Feb 2025
14	B2 HVAC: Loading & East Charging	$0.28	May 2024	Feb 2025
15	IT Infrastructure & Centralized Hub	$0.27	Sep 2024	Mar 2025
2023 ($M)	**$19.33**			
16	Explosive Waste Incinerator (Major Maintenance)	$2.80	May 2023	Feb 2025
17	7.62mm Reduced Range Ammo: Equip/Fac/Inf	$2.97	Apr 2024	Apr 2025
18	One Way Luminescence: Equip/Fac/Inf	$2.08	Apr 2024	Apr 2025
19	Roof Replacements (Critical Safety Project)	$1.38	Jul 2023	Jul 2026
20	Powder Delivery (B1) (Design)	$2.30	Sep 2023	Feb 2025
21	Automatic Transfer Switches/Back-Up Generators	$4.26	Sep 2023	Feb 2025
22	(NGSW) RACE Year 2	$3.54	Nov 2023	Mar 2025
2024 ($M)	**$653.79**			
23	Explosive Waste Incinerator (New Design)	$15.20	Jan 2024	Jan 2026
24	Cybersecurity Phase 2 Modernization (Design)	$17.08	May 2024	May 2025
25	(NGSW) Munro COTA Year 2	$5.98	Jan 2024	Jan 2025
26	NGSW-A Facility Construction	$607.20	Mar 2024	Jul 2027
27	(NGSW) Interim Capability Equipping: Phase III	$8.33	Mar 2024	Mar 2027
2025 ($M)	**$8.62**			
28	Bldg. 6 Roof MATOC	$5.50	Nov 2024	Mar 2025
29	Bridge 5 Replacement	$3.12	Dec 2024	Dec 2025
Total:	**$782.59**			

All projects in execution as of January 24, 2025.

11.3.4 RFAAP Projects in Execution

Project Number	Project Title	Cost ($M)	Project Start Date	Estimated Completion Date
2012 ($M)	**$399.4**			
1	Construct (NEW) Nitrocellulose Facility	$399.4	Jan 2012	May 2025
2019 ($M)	**$157.4**			
2	Combined EWI-CWP Facility - Construction	$157.4	Sep 2019	Apr 2026
2020 ($M)	**$12.0**			
3	Fire Protection System Upgrade (Deluge Valves)	$12.0	Dec 2021	Feb 2025
2022 ($M)	**$130.1**			
4	NAC/SAC Process Improvements (Design and Train 1 & Train 2 Construction)	$27.7	Sep 2022	Aug 2026
5	LIF Phase 9: Assess, Decontamination, Dispose & Demo Excess Facilities & Equipment	$31.1	Aug 2022	Dec 2026
6	Solvent Propellant Facility Sustainment, Phase 2	$16.3	May 2023	May 2025
7	Compressed Air House Control System Improvements	$2.6	Sep 2023	Mar 2025
8	Solventless/Rocket Area Sustainment Phase II	$12.3	Sep 2023	Apr 2026
9	NG-3 30% Design of Multi-Nitrate Ester Manufacturing Facility	$35.9	Oct 2023	Apr 2026
10	NG-2 Area Sustainment Phase II	$4.2	Jun 2024	June 2026
2023 ($M)	**$238.5**			
11	Chemical Grind (Includes $932 for Section 8121)	$92.8	Nov 2022	Oct 2026
12	Workplace Safety Upgrades	$46.4	Apr 2023	Aug 2026
13	Hammermill (New NC Facility)	$8.0	Jun 2023	June 2025
14	NC Propellant First Article Acceptance Testing (FAAT)	$3.4	Jun 2023	Dec 2025
15	Evenspeed Rocket Propellant Manufacturing Upgrades	$6.8	Aug 2023	Aug 2025
16	Advanced Ammunition (AA) Phase I Renovation	$3.4	Sep 2023	Jul 2025
17	Steam Distribution Phase I (90% of NTE Ceiling)	$77.7	Sep 2023	Dec 2026
2024 ($M)	**$118.5**			
18	Solvent Propellant Facility Sustainment Phase II	$118.5	Aug 2023	Aug 2026
Total:	**$1,055.9**			

All projects in execution as of January 24, 2025.

11.3.5 SCAAP Projects in Execution

Project Number	Project Title	Cost ($M)	Project Start Date	Estimated Completion Date
2020 ($M)	**$4.2**			
1	Infrastructure and Safety Upgrades	$4.2	Jun 2022	Feb 2025
2021 ($M)	**$0.7**			
2	Retaining Walls Repair	$0.3	Sep 2022	Jun 2025
3	Bathroom Upgrade	$0.4	Sep 2022	Feb 2025
2022 ($M)	**$356.2**			
4	M1128 Production Line	$43.1	Apr 2022	Aug 2025
5	XM1113/XM1210 Production Line	$42.3	Sep 2022	Oct 2025
6	155mm Capacity Improvement (Surge PWS)	$55.7	Sep 2022	Feb 2025
7	Universal Artillery Projectile Line 1	$203.1	Nov 2022	Jun 2025
8	Heat Treat Filter Tower Replacement	$1.8	Jul 2023	Jan 2025
9	Artillery Critical Spare Parts	$1.4	Sep 2023	Mar 2025
10	Bliss III Press Repair & Rebuild	$2.6	Feb 2024	Feb 2026
11	M795 Pangborn & A2 Shot Blast System Mod	$5.6	Mar 2024	Jun 2025
12	Steam & Water Piping Mod Upgrades	$0.5	Sep 2024	Mar 2026
2023 ($M)	**$747.6**			
13	Universal Artillery Projectile Line 2	$200.3	Feb 2023	Jul 2025
14	SCAAP 35k Capacity Expansion for 155mm Metal Parts	$347.5	Mar 2023	Sep 2025
15	Equipment Excess	$12.7	Mar 2023	Jun 2025
16	Universal Artillery Projectile Line 3	$187.1	Jun 2023	Jun 2026
2024 ($M)	**$9.1**			
17	Pit Pumps to Wastewater	$1.3	Jul 2024	Jul 2025
18	Electrical System Upgrades Phase I	$7.4	Jul 2024	Feb 2026
19	Forge Shop Emergency Water Valve Replacement	$0.4	Jul 2024	Jul 2025
Total:	**$1,117.8**			

All projects in execution as of January 24, 2025.

11.4 APPENDIX D: CAPACITY AND SCALABILITY DATA

11.4.1 HSAAP CAPACITY AND SCALABILITY DATA

Capability	Max Capacity (Pounds) [2]				Project Reference
	Current State	3-Years	5-Years	7-Years	
HMX	2,000,000	2,000,000	2,000,000	2,000,000	
RDX	9,500,000	12,000,000	12,000,000	12,000,000	RDX Expansion Projects
IMX-104[1]	3,000,000	5,000,000	13,000,000	13,000,000	IMX Expansion Projects

[1] Completion of the NAC/SAC reactivation project, the NTO production upgrade project, and the New Melt Cast facility, anticipated by the end of FY 2025, will allow HSAAP to achieve 5M lbs. IMX-104 capacity.

[2] Capacity is dependent on the overall product mix.

11.4.2 IAAAP CAPACITY AND SCALABILITY DATA

Capability	Max Capacity (Rounds)				Project Reference
	Current State	3-Years	5-Years	7-Years	
40mm Med Cal LAP (Tactical/Training) [1]	3,072,000	3,072,000	3,072,000	3,072,000	
Tank LAP (120mm Training) [2]	423,756	423,756	423,756	423,756	
Artillery 155mm M795[3]	324,000	300,300	700,000	700,000	Increased Capacity Online in FY 29/30
C4 Demo Block	283,500	283,500	283,500	283,500	
Hellfire	6,000	6,000	6,000	6,000	

[1] 40mm capacity is based on the overall product mix.

[2] Tank LAP capacity is dependent on the overall product mix.

[3] The FAC Site Preparation project, funded by the FY 2023 Supplemental, supports the increase in the 155mm LAP capacity to 700K.

11.4.3 LCAAP Capacity and Scalability Data

Capability	Max Capacity (Rounds)				Project Reference
	Current State	3-Years	5-Years	7-Years	
5.56mm	1,320,000,000	1,320,000,000	1,320,000,000	1,320,000,000	No production increase
7.62mm	230,000,000	230,000,000	230,000,000	230,000,000	No production increase
.50 caliber	75,000,000	75,000,000	75,000,000	75,000,000	No production increase
6.8mm	5,000,000	40,000,000	114,000,000	240,000,000	New Capability

11.4.4 QCCCF Capacity and Scalability Data

Capability	Max Capacity (Projectiles) [1]				Project Reference
	Current State	3-Years	5-Years	7-Years	
105mm Steel Cartridge Cases	60,000	60,000	60,000	60,000	
105mm Brass Cartridge Cases	85,000	85,000	85,000	85,000	
Navy 5” Steel Cartridge Cases	60,000	60,000	60,000	60,000	

[1] Concurrent production is limited to approximately 72,000 cases annually.

11.4.5 RFAAP Capacity and Scalability Data

Capability	Max Capacity (Pounds)				Project Reference
	Current State	3-Years	5-Years	7-Years	
Nitrocellulose (NC) [1,3]	20,000,000	28,000,000	28,000,000	28,000,000	Completion of new NC Facility and NAC SAC improvements.
Solvent SB [2]	4,300,000	6,000,000	6,000,000	6,000,000	Completion of the base requirements in the Solvent Sustainment Phase II UCA project.
Solvent MB [2]	1,300,000	2,500,0005	2,500,000	2,500,000	Completion of the base requirements in the Solvent Sustainment Phase II UCA project.
Solventless [2]	3,500,000	3,500,000	3,500,000	3,500,000	
Mk. 90	325,000	325,000	325,000	325,000	

[1] NC capacity is based on running non-hammermill product, i.e., Small Caliber, Medium Caliber, Tank, 155 Artillery.

[2] Capacity is dependent on the overall product mix.

[3] NAC/SAC process improvements, funded by the FY 2023 Supplemental funding, supports the NC increased capacity to 28M lbs.

11.4.6 SCAAP Capacity and Scalability Data

Capability	Max Capacity (Projectiles)				Project Reference
	Current State	3-Years	5-Years	7-Years	
155mm M795 Metal Parts [1]	192,000	360,000	360,000	360,000	Modernization Capacity Increase
155mm M1128	0	60,000	60,000	60,000	New Capability
155mm XM1113	0	15,000	15,000	15,000	New Capability

[1] The FY 2023 Supplemental funding supports the increased capacity for 155mm metal parts production as indicated above.